THE INDUSTRIAL GRAPHITE ENGINEERING HANDBOOK

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HIGH TECH HIGH TEMP

Chapter One

THE MANUFACTURE OF CARBON AND GRAPHITE

Industrial Graphite Engineering Handbook

HIGHTECH HIGHTEMP

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The Manufacture of Carbon and Graphite

Introduction

The use of graphite in many diverse areas of industrial activity is due directly to a unique combination of physical and chemical properties. Graphite is a good conductor of electricity, a superior refractory, has high thermal conductivity, high thermal shock resistance, excellent high temperature strength, excellent machinability, high sublimation temperature, and a low modulus of elasticity. In certain applications the superior hardness of carbon may be preferred over that of graphite. Therefore, there will be continual reference to both monolithic carbon and monolithic graphite as specialty industrial materials.

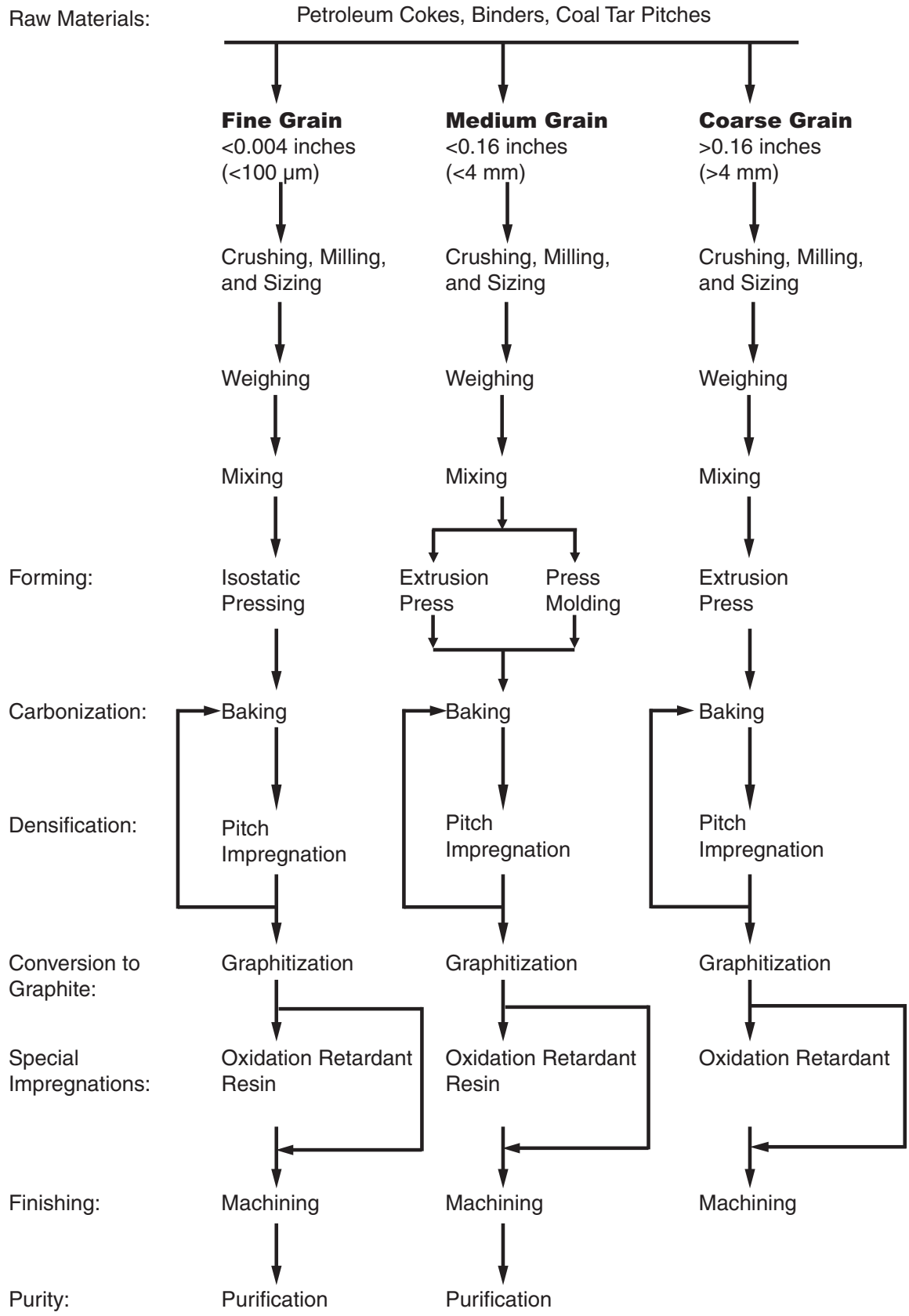
The graphite used in industry is manufactured from carbon-base materials, rather than mined as the natural substance. For this reason, it is frequently called “artificial graphite.” Manufactured graphite is superior to natural graphite, in that it does not have the disadvantages of mechanical weakness and high ash content that are commonly associated with natural graphite, and it can be made in a wide variety of monolithic forms.

Manufactured graphite is not one specific material, but a range of materials. Each is essentially pure carbon, but each varies from the other in such things as the orientation of the crystallites, the size and number of pore spaces, the degree of graphitization, etc. Consequently, there is a corresponding variation in physical properties. Specific graphite grades have, therefore, been established on the basis of these controlled characteristics. Each grade is adapted to particular areas of service, with quality and price limitations that are dictated by the requirements of the particular application. Proposed new uses of manufactured graphite must thus be considered from a knowledge of the many grades available, all the way from the lowest cost, coarse-grained, relatively weak materials to the ultrafine-grained, strongest and purest materials, which require many times the production effort.

This wide variation in graphite properties is due largely to controlled variation of the production process --- the choice of raw materials, filler particle sizing, and in the case of the fine- and ultrafine-grained material the degree of milling and classification, forming method, conditions of the first bake (or gas bake), densification, graphitizing, and purification. In addition, the manufacturing procedures essential to the production of tonnage quantities of many grades of graphite at a moderate cost introduce further property variations from piece-to-piece and within-a-piece.

The typical steps in processing coarse, medium, fine, superfine, and ultrafine grain carbon and graphite are outlined in Figure 1.1, and briefly discussed in the balance of this section.

Figure 1.1 Processing Specialty Graphites



Filler Effects

Graphite can be manufactured from almost any organic material that leaves a high carbon residue when heated in the absence of air. Petroleum coke, a petroleum refinery by-product of the delayed coking process, is a major raw material filler. The filler raw materials typically consisting about 70%, by weight, of the final graphite tend to dominate the properties of the final graphite, see Table 1.1. Delayed coke in large supply and relatively free from impurities is usually calcined as a preliminary to the carbon and graphite manufacture steps. Other important filler raw materials include calcined coal tar pitch, delayed cokes, carbon black, lamp black, raw petroleum coke, and recycled artificial graphite.

To illustrate the effect of raw material, a number of common room-temperature properties of a series of graphite lots is shown in Table 1.1. Each sample was identical in all respects -grade, manufacture, size - except for the petroleum cokes used in the raw materials. The wide range in property values is then directly attributable to the raw material. Varied filler raw materials permit the manufacturer to produce graphites with specific properties by appropriate raw material selection combinations.

Table 1.1 Typical Room-Temperature Physical Properties of Extruded Graphite

Property	Coke Source	With-Grain	Against-Grain
Specific Resistance, μΩm	A	8.4	11.7
	B	7.5	11.7
	C	8.8	12
	D	5.7	12
Thermal Conductivity, cal/cm°C sec	A	0.4	0.3
	B	0.4	0.32
	D	0.56	0.29
Thermal Expansion, 10^{-6}/°C (RT-60°C)	A	2.4	3.8
	B	1.5	3.6
	C	1.9	3.2
	D	1.1	2.7
Young's Modulus, 10^6 psi	A	1.43	0.95
	B	1.49	0.88
	C	1.54	1.07
	D	1.99	0.79

Particle Size

In manufacturing graphite, the raw delayed coke is first calcined in large rotary or shaft kilns to shrink it and to drive out the volatile content. The calcined coke is then broken down by crushers and sized into a series of carefully controlled fractions. The coarsest fraction in extruded graphites may be over 1" (25mm). The finest fraction termed "flour", in extruded and molded grades, typically has a maximum particle size of 0.0083" (210 μm). The flour is usually produced in a mill with a built-in classifier. In normal manufacture, selected size fractions are recombined with coke flour to produce a dry aggregate. Recycle artificial graphite is similarly prepared.

Isomolded grades are usually all-flour compositions. Flours are milled to controlled maximum and median sizes using built-in or stand-alone classifiers. Typical maximum particle size varies from about 75 μm to about 5 μm, depending on the grade produced. Recycle artificial graphite is similarly prepared.

The proportion of fractions and the fraction size is varied, within limits, to control the properties of the end product.

Forming Method

Extrusion and Molding

Prior to forming the pieces that are to be graphitized, the selected combination of dry aggregate and coal tar pitch binder is heated during mixing to assure homogeneity. The mixture is then cooled and the warm mix formed either by extrusion or molding. Forming is another variable that affects the final graphite properties. The magnitude of the effects is leveraged by the type of coke, be it needle, sponge, or isotropic. When the calcined needle (premium) coke is crushed or milled to obtain the dry aggregate, the individual particles, although irregular in shape, tend to have one dimension larger than the other two. In other words, particles shaped like splinters or needles, are obtained. The shapes are more distinct in the finer size fractions. The pre-graphitic layer planes are preferentially aligned along the major axes of these splinter and needle-shaped particles. At the other extreme, the isotropic cokes usually have no internal preferred orientation and have more of a rounded cube shape. The sponge cokes exhibit intermediate levels of preferred orientation and shape.

In forming, the long axes tend to take a preferred orientation - either with the direction of extrusion or perpendicular to the direction of molding. The final graphite product retains the same pattern of grain orientation. The with-grain direction is parallel to the direction of extrusion in extruded graphite and perpendicular to the direction of molding pressure in the molded piece (i.e. perpendicular to the major vertical axis). The against-grain direction is perpendicular to the with-grain. Thus, with the typical anisotropic lattice arrangement of graphite crystallites coupled with the level of preferred orientation in the starting coke there is a difference in the property values measured “with-grain” (WG) and “against-grain” (AG). These effects are most marked with the needle cokes which are used to make graphite electrodes for electric steel making. Intermediate effects are evidenced in sponge coke graphites, see Table 1.2. With-grain and against-grain effects are minimal when isotropic cokes are used.

Other factors influencing isotropy are the sizing of the coarser fractions in the dry aggregate and the ratio of the coarse fractions to the fine fractions. These effects are illustrated in Table 1.3 for three extruded products, all with a common base and manufactured under the same process conditions.

Table 1.2 WG and AG Physical Properties of Molded and Extruded Graphite Produced from a Sponge Coke

Property	Extruded Stock	Molded Stock
Density, g/cc	1.64	1.75
Specific Resistance, μΩm		
With-grain	8.6	9.6
Across-grain	16.2	13.2
Ratio	0.53	0.73
Thermal Expansion, $10^{-6}/^{o}C$		
With-grain	1.1	1.9
Across-grain	4.1	3.2
Ratio	0.27	0.6
Young's Modulus, 10^{6} psi		
With-grain	1.84	1.39
Across-grain	0.78	0.96
Ratio	2.4	1.45
Flexural Strength, psi		
With-grain	3420	3550
Across-grain	2370	2750
Ratio	1.44	1.29

Isomolding

In the case of isomolding, the hot mix of flour and coal tar binder pitch is cooled and milled into a molding powder. The molding powder is loaded into a rubber bag and shaped at an elevated pressure in an isostatic press.

Isostatic molding, where pressures during molding are equalized on all surfaces, tends to produce near isotropic properties irrespective of the starting coke type. However, the WG and AG jargon is carried over from molding so that the minor axis, usually horizontal, is referred to as WG, and the major axis (usually in the vertical) referred to as AG. The isotropy of isostatically molded materials is higher than with any other forming method, see Table 1.3.

Table 1.3 The Effect of Forming Method and Grain Size on Isotropy

	Maximum Grain Size, Extruded			Average Grain Size, Isomolded		
Property Ratio, WG/AG (Isotropy)	0.010 inches (0.25mm)	0.060 inches (1.52mm)	0.250 inches (6.35mm)	0.001 inches (25 µm)	0.0002 inches (2 µm)	0.0008 inches (2 µm)
Specific Resistance	0.51	0.79	0.85	0.9	1.0	1.0
Young's Modulus	2.91	1.71	1.4	1.0	1.0	1.0
Thermal Expansion	0.34	0.63	0.65	0.8	1.0	1.0

Baking and Densification

The next processing step is baking in which the pitch binder in the formed piece is carbonized to develop an infusible carbon bond between the filler particles. Baking is done by bringing the product to a temperature of 750°C to 900°C in oil or gas-fired furnaces. The pieces must be carefully arranged and spaced, and tightly surrounded by a granular carbonaceous packing material since the binder softens in the early stages of the bake and the product becomes plastic. Then, as gases are released from the binder, the piece expands. Therefore, the product must be well supported so that dimensional tolerances are retained. The carbonaceous packing material also provides oxidation protection.

Large furnaces are commonly used for baking. A high degree of control and temperature uniformity is necessary. These furnaces will range in size from 35,000 to 200,000 pounds product capacity. Since the load is large and has poor thermal conductivity, slow baking rates are necessary to keep temperature differences at a minimum. Baking cycles from 20 to 72 days, depending on furnace size and product mix, are not unusual.

In the course of baking the coal tar pitch binder is converted to binder coke. Approximately one-third of the pitch binder is driven off in this destructive distillation process. The baked product may show a loss in apparent density, and the development of porosity and permeability to liquids and gases.

The product after first baking is called carbon. It is characterized as being hard, abrasive, and difficult to machine (diamond tooling is required). Compared to graphite, it also has poor thermal properties, poor electrical characteristics, good mechanical properties, and is generally lower in cost.

Porosity and permeability can be reduced, and density and mechanical properties increased by pitch impregnation and rebaking prior to graphitization. Impregnation is generally accomplished by vacuum/pressure techniques, assuring complete penetration of the liquid pitch through the cross-section of all pieces. The impregnant is converted to carbon in another baking step often referred to as rebaking. This rebaking is typically accomplished in large furnaces similar in size and design to those used for the initial baking. However, baking rates for rebaking are significantly faster because the artifact is already rigid, and it is a minor carbonaceous component within the porosity which is converted to carbon. Whilst the impregnated bulk artifact does not require support during the rebaking, protection from burning is essential. The cycle of impregnation and rebaking is repeated until the desired properties are achieved.

Graphitization

The final step in graphite manufacture is the conversion of the carbon to graphite. This is known as graphitizing and requires a temperature in the 2600°C to 3300°C range. In graphitization, the tiny two-dimensional crystallites already present in random arrangement in the baked carbon piece, rearrange to very large three-dimensional graphite crystallites. The development of three dimensional order which heralds graphitization is accompanied by abrupt changes in physical properties.

Tests have shown that the properties of any piece of graphite are directly dependent upon the highest temperature reached in graphitization. In Figures 1.2, 1.3, and 1.4, the changes in electrical resistivity, flexural strength, and thermal expansion that occur in a fine-grain isomolded material as a function of final temperatures are shown.

Figure 1.2 Electrical Resistivity As a Function of Final Graphite Temperature

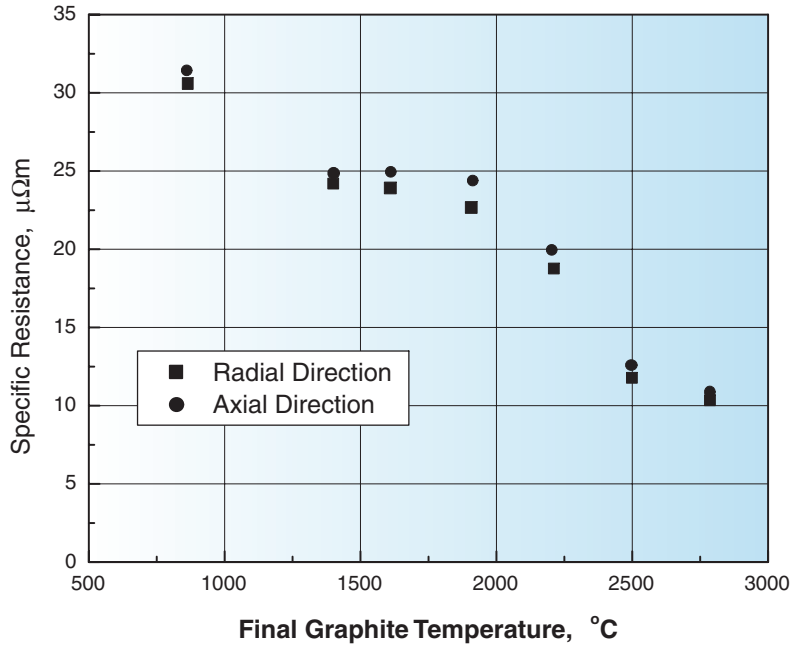


Figure 1.3 Flexural Strength As a Function of Final Graphite Temperature

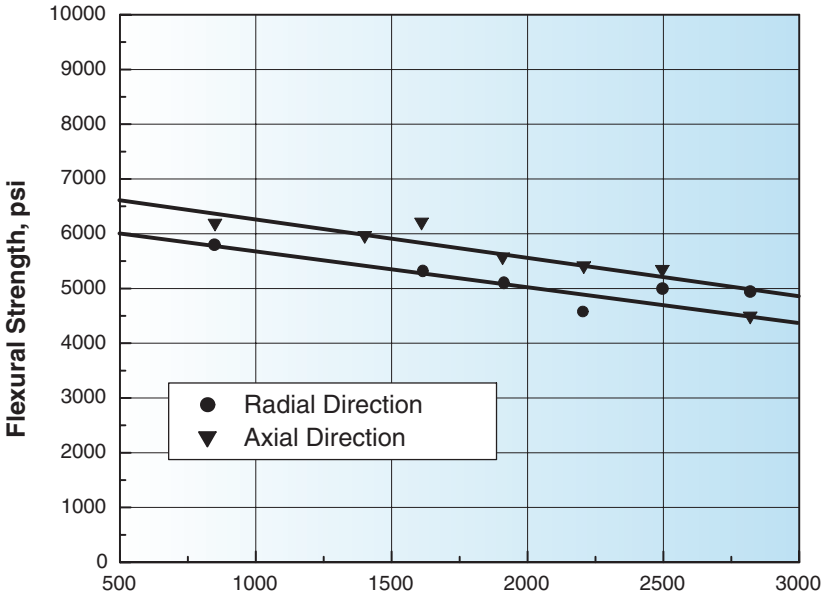
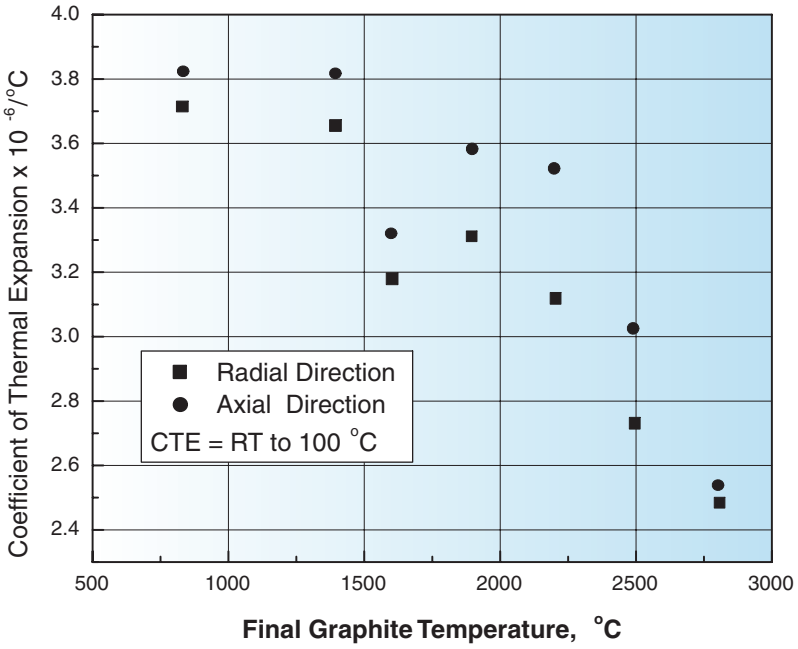


Figure 1.4 Thermal Expansion As a Function of Final Temperature



Within practical limits, higher temperatures and greater temperature uniformity throughout the furnace can be obtained by increasing the electrical power input in the graphitizing operation. However, heat losses increase exponentially with increasing temperatures, which means that each temperature increment used becomes increasingly expensive. So, again, it is a question of economics with regard to product application in choosing a graphite for particular uses.

Purification

Even after graphitization at around 3000°C most bulk commercial graphites contain small amounts of metallic impurities. These impurities originate in the raw material calcined cokes, recycled graphites, and coal tar and petroleum pitches. The total impurities may be up to 2000 ppm as measured by combustion ash, see Table 1.4. The ash level will be higher if inorganic compounds were purposely added at the mixing stage to control puffing.

In the carbon-rich environment many of the metal species are likely present as carbides, many of which are extremely stable, see Table 1.5. Scanning Electron Microscopy reveals that high concentrations of some metals can be present in the final graphite as micron-sized pools of apparently complex alloys.

The use of graphite as a moderator for thermal neutrons in water- and gas-cooled nuclear fission reactors require high purity levels. Some impurities require removal because of their ability to absorb neutrons, whilst others because of their undesirable radioactive derivatives. Amongst the former elements, boron is noteworthy because it substitutes in the carbon lattice. There is a concomitant difficulty in expelling boron from the lattice.

Table 1.4 Typical Room Temperature Properties of UCAR® Grades

Group	UCAR® Grade	Largest Cross Section inches	Largest Cross Section mm	Grain Size,(1) inches	Grain Size,(1) mm	Ash ppm
Extruded Medium Grain						
	CBY	24	600	0.03	0.8	800
	CS	30	600	0.03	0.8	800
	SLWX	24	600	0.03	0.8	<5
	CHV	24 x 48	600 x 1200	0.06	1.7	1.50%
Molded Medium Grain						
	PGW	52	1300	0.03	0.8	1000
	PGX	73	1825	0.03	0.8	700
Extruded Fine Grain						
	FGE	8.5 x 8.5	215 x 215	0.006	0.150	1000
	FGEX	8.5 x 8.5	215 x 215	0.006	0.150	1000
Isomolded Fine Grain						
	ATJ	24	600	0.001	0.025	900
	CGW	16 x 16	400 x 400	0.001	0.025	2000
Isomolded Superfine Grain						
	DM306	8 x 16	200 x 400	0.0002	0.005	2000
	DM300	8 x 16	200 x 400	0.0002	0.005	1600
	DM301	8 x 16	200 x 400	0.0002	0.005	1300
	DM302	8 x 16	200 x 400	0.0002	0.005	1100
	DM303	8 x 16	200 x 400	0.0002	0.005	1500
	DM342	8 x 16	200 x 400	0.0002	0.005	1500
	DM332	24	600	0.0002	0.005	600
Isomolded Ultrafine Grain						
	DM362	6 x 12	150 x 300	0.00008	0.002	1500

(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.

Table 1.5 Molecular Weight and Boiling Points of Carbides

Metal	Formula	MW	M.Pt., °C	B.Pt., °C	Name
Boron	B_4C	55.29	2350	>3500	Boron carbide
Sodium	Na_2C_2	70	~700		Sodium carbide
Magnesium					
Aluminum	Al_4C_3	143.91	2100	decomp > 2200	Aluminum carbide
Silicon	SiC	40.07	sublimes ~2700		Silicon carbide
Potassiuim					
Calcium	CaC_2	64.1	2300		Calcium carbide
Titanium	TiC	59.89	3140 +/- 90	4820	Titanium carbide
Vanadium	VC	62.95	2810	3900	Vanadium carbide
Chromium	Cr_3C_2	180.01	1980	3800	Trichromium dicarbide
	CrC	64.02	1550		Chromium monocarbide
Manganese	Mn_3C	176.83	1520		Manganese carbide
Iron	Fe_3C	179.55	1227		Iron carbide
Nickel	Ni_3C	188.08			Nickel carbide
Barium	BaC_2	161.35	1780		Barium carbide
Strontium	SrC_2	111.64	1950		Strontium carbide
Zirconium	ZrC	103.23	3540	5100	Zirconium carbide
Molybdenum	MoC	107.95	2692		Molybdenum carbide
	Mo_2C	203.89	2687		Molybdenum dicarbide

References:
Merck Index, 11th Edition, 1989
CRC Handbook of Chemistry and Physics, 75th ed., 1994
PlenumPress Handbook of High-Temperature Materials, No. 1: Materials Index by P.T.B. Shaffer, 1964

Purification Process

Purification processes involving halogen gas purges either during graphitization, or as a subsequent step, developed for the production of spectroscopic-grade electrodes for chemical analysis, were enhanced in the 1940's to cope with monolithic nuclear blocks. Chlorine is an effective agent except for boron. Boron removal to below 1 ppm requires fluorine which is generated in-situ by the reduction of either carbon tetrafluoride, or chlorofluorocarbons. The reaction between graphite and a stream of neat fluorine is too virulent.

In considering the likely effectiveness of purification, it is helpful to be mindful of the volatility of chlorides and fluorides, see Table 1.6 and 1.7, and to run the appropriate thermodynamic calculations.

Although the demand for the purification of monolithic graphite moderator blocks has waned, the 1990's saw a rising need both quantitatively, and qualitatively, from the semiconductor sector. A current typical limit of 10 ppm ash is met with high temperature chlorine flushes of machined parts. Interestingly, the machining of nuclear reactor graphite components is commonly done in an appropriately "clean" environment after purification.

Continuous improvement and new semiconductor materials pushing towards impurity levels a thousand times lower (to 10 ppb) require more effective purification processes as well as the development of enhanced analytical techniques.

Table 1.6 Molecular Weight and Boiling Points of Metal Chlorides

Metal	Formula	MW	M.Pt., °C	B.Pt., °C	Name
Boron	B_2Cl_4	163.47	-92.6	various	Diboron tetrachloride
	BCl_3	117.19	-107	12.5	Boron trichloride
Sodium	NaCl	58.45	804	1413	Sodium chloride
Magnesium	$MgCl_2$	95.23	712	1412	Magnesium chloride
Aluminum	$AlCl_3$	133.34	(Note 1)	decomposes 262	Aluminum chloride
Silicon	$SiCl_4$	169.89	-70	59	Silicon tetrachloride
Potassium	KCl	74.55	773	sublimes 1500	Potassium chloride
Calcium	$CaCl_2$	110.99	772	>1600	Calcium chloride
Titanium	$TiCl_2$	118.81	1035		Titanium dichloride
	$TiCl_3$	154.27	decomp. > 500		Titanium trichloride
	$TiCl_4$	189.73	-24.1	136.4	Titanium tetrachloride
Vanadium	VCl_2	121.85			Vanadium dichloride
	VCl_3	157.3	decomposes		Vanadium trichloride
	VCl_4	192.75	-28 +/- 2	148.5	Vanadium tetrachloride
Chromium	$CrCl_2$	122.92	824		Chromous chloride
	$CrCl_3$	158.38	1152	dissociates >1300	Chromic chloride
Manganese	$MnCl_2$	125.84	58	(differing values)	Maganese chloride
			650	1190 (differing values)	
	$MnCl_3$	161.3	slight decomp.		Maganese trichloride
Iron	$FeCl_2$	126.76	674	1023	Ferrous chloride
	$FeCl_3$	162.22	300	316	Ferric chloride
Nickel	$NiCl_2$	129.61	1001	sublimes 973	Nickel chloride
Copper	CuCl	99	430	1490	Cuprous chloride
	$CuCl_2$	134.45	(Note 2)		Cupric chloride
Zinc	$ZrCl_2$	136.29	290	732	Zinc chloride
Barium	$BaCl_2$	208.27	963	1560	Barium chloride
Strontium	$SrCl_2$	158.52	875	1250	Strontium chloride
Zirconium	$ZrCl_2$	162.13	decomp. 350		Zirconium dichloride
	$ZrCl_3$	197.58	decomp. 350		Zirconium trichloride
	$ZrCl_4$	233.05	437	sublimes 331	Zirconium tetrachloride
Molybdenum	$MoCl_2$	166.85	decomposes		Molybdenum dichloride
	$MoCl_3$	202.3	decomposes		Molybdenum trichloride
	$MoCl_4$	237.75	decomposes	volatile	Molybdenum tetrachloride
	$MoCl_5$	273.2	194	268	Molybdenum pentachloride

References:
Merck Index, 11th Edition, 1989
CRC Handbook of Chemistry and Physics, 75th ed., 1994
PlenumPress Handbook of High-Temperature Materials, No. 1: Materials Index by P.T.B. Shaffer, 1964
Note1: Fumes in air; volatilizes without melting when heated in small quantities.
Note2: Partial decomposition >300°C to CuCl + Cl M Pt. (extrapolated) is 630°C. Melting point of 498°C usually reported for this compound is the melting point of the CuCl + Cl_2 mixture.

Table 1.7 Molecular Weight and Boiling Points of Metal Fluorides

Metal	Formula	MW	M.Pt., °C	B.Pt., °C	Name
Boron	BF_3	67.82		-127.1 (differing values) -100.4 (differing values)	Boron trifluoride
Sodium	NaF	42	993	1704	Sodium fluoride
Magnesium	MgF_2	62.32	1248	2260	Magnesium fluoride
Aluminum	AlF_3	83.89		sublimes 1272	Aluminum fluoride
Silicon	SiF_4	104.06	-90.2	sublimes -95.7	Silicon tetrafluoride
	Si_2F_6	170.16	-18.7	-18.5	Hexafluorodisilane
Potassiuim	KF	58.1	859.9	1505	Potassium fluoride
Calcium	CaF_2	78.08	1403	2500	Calcium fluoride
Titanium	TiF_3	104.88	1200	1400	Titanium trifluoride
	TiF_4	123.9	>400	sublimes > 284	Titanium tetrafluoride
Vanadium	VF_3	107.95	~1406 (differing values) >800 (differing values)	sublimes	Vanadium trifluoride
	VF_4	126.95		decomposes 325	Vanadium tetrafluoride
	VF_5	145.95	19.5	47.9 (differing values) 111.2 differing values)	Vanadium pentafluoride
Chromium	CrF_2	90.01	894		Chromous fluoride
	CrF_3	109.01	1100	sublimes 1100-1200	Chromic fluoride
	CrF_4	128.01	estimate 277	~400	Chromic tetrafluoride
Manganese	MnF_2	92.93	856		Manganese difluoride
	MnF_3	111.93	stable to 600		Manganese trifluoride
Iron	FeF_2	93.85		sublimes >1100	Ferrous fluoride
	FeF_3	112.85		sublimes >1000	Ferric fluoride
Nickel	NiF_2	96.69		sublimes >1000 (in HF)	Nickel fluoride
Copper	CuF	82.54	980	sublimes > 1100	Copper (I) fluoride
	CuF_2	101.54	decomp.@950		Cupric fluoride
Zinc	ZnF_2	103.38	872	1500	Zinc fluoride
Barium	BaF_2	175.36	1353	2260	Barium fluoride
Strontium	SrF_2	125.63	~1400	2460	Stronium fluoride
Zirconium	ZrF_4	167.22		sublimes > 600	Zirconium fluoride
Molybdenum	MoF_6	209.95	17.5	35	Molybdenum hexafluoride

References:
Merck Index, 11th Edition, 1989
CRC Handbook of Chemistry and Physics, 75th ed., 1994

Analytical Methods

The chemical analysis of nuclear graphite traditionally consisted of combustion ash and wet chemical analysis of selected elements, especially boron (see ASTM C560). Improvements in analysis techniques have allowed for more completed analyses in a shorter period of time.

In a recent comparison of five modern instrumental methods for the analysis of halogen-purified graphite, three suitable techniques were reported[1]. Each technique has multi-element capability and detection limits (DL) which vary by element. Direct current arc-optical emission spectroscopy (DCA-OES or “DC-arc”) provides timely and cost effective analyses, but has an average (DL) of 0.2 ppm (200 ppb), the poorest DL of the three techniques. Inductively coupled plasma-atomic emission spectroscopy (ICP-AES) has the highest accuracy and acceptable DL of 0.04-3.0 ppm (40-300 ppb), but cannot analyze volatile elements such as boron, phosphorus, or sulfur, as they are lost upon sample preparation. The most capable technique is glow discharge-mass spectrometry (GD-MS) which can detect 75 elements with DL of 0.005-0.05 ppm (5-50 ppb).

Concluding Comments

Thus, it is seen that properties and uniformity of the final graphite product are greatly influenced by material and manufacturing variables. Where product uniformity is important, and special properties are sought, the manufacturing variables can be controlled, and product meeting certain specifications selected. The important thing is to determine the graphite best suited to the application from the standpoint of properties and economics.

The manufacture of artificial graphite is characterized by the following:

- Long processing cycles;
- Energy intensive with respect to raw materials and processes;
- Wide range of grades and property combination through appropriate selection of raw materials and processing.

References

1. Mayer, H.K., Proceedings of the 24th Conference on Carbon, 1999, Vol. 2, p. 644-645.

Chapter Two

TYPICAL GRADES AND ROOM TEMPERATURE PROPERTIES

Industrial Graphite Engineering Handbook

HIGHTECH HIGHTEMP

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Typical Grades and Room Temperature Properties

Introduction

The worldwide consumption of about 150,000 tons of specialty bulk carbons and graphites is satisfied by a wide variety of materials differing in billet size, grain size, property combination, and other characteristics. In selecting a grade for a particular end-use many factors are considered to ensure the choice of the most viable material for the application.

In this chapter, a number of frequently used carbon and graphite grades are briefly reviewed in general terms in the following groupings:

- Extruded Medium Grain
- Molded Medium Grain
- Extruded Fine Grain
- Isomolded Fine Grain
- Isomolded Superfine Grain
- Isomolded Ultrafine Grain

The grain size of Specialty Carbons and Graphites is always important. For instance, there are numerous applications where the machined graphite, in use as a die, directly affects the surface finish and value of the manufactured product. Then again, the strength properties of carbons and graphites are directly related to grain size.

Unfortunately, there is no agreed industry standard definition of grain size so the approach here is to use the National Electrical Manufacturer's Association (NEMA) definitions as set out in Table 2.1. The NEMA definitions focus on the size of the largest particle in the starting ingredients. Hence, the extruded and pressure molded graphites with the largest particles greater than 100 μm (0.004 inches) are appropriately described in this Handbook as medium grain. The fine grain extruded graphites with a maximum grain size of 150 μm (0.006 inches) should strictly be defined as medium grain, but are so distinct from the relative coarseness of the medium grain grades that the NEMA definition has been stretched.

The isomolded materials are well-covered by the NEMA definitions for fine (maximum particle is 50 μm-100 μm, 0.002 inches - 0.004 inches), superfine (maximum particle is 10 μm-50 μm, 0.0004 inches - 0.002 inches), and ultrafine (maximum particle is less than 10 μm, 0.0004 inches).

Table 2.1 Classification of Grain Size

Graphite material can be classified by grain size as follows:
a. Ultrafine (or micrograin graphite) - less than 10 microns maximum grain size;
b. Superfine - between 10 and 50 microns maximum grain size;
c. Fine - greater than 50 but less than or equal to 100 micron maximum grain size; or
d. Medium -greater than 100 microns maximum grain size.
The above classifications are based on size measuring techniques of the initial raw materials. It should be noted that the measuring techniques involved are not applicable to the finished product.

"Semiconductor Graphite"" CB-4-1989, National Electrical Manufacturers Association, Washington, DC, 1989"
For more detail see: Norley, J., Proceedings of the 24th Conference on Carbon, 1999, Vol. 2, p. 646.

For convenience, typical properties of extruded and molded medium grain, extruded fine grain, and isomolded fine, superfine, and ultrafine grain grades are summarized in English units in Table 2.2 and in SI units in Table 2.3. An indication of billet size in terms of largest cross-section is also given in the tables. The structure of selected grades is highlighted using photomicrographs, see Figures 2.1 to 2.5. Please note that this chapter is not a comprehensive compilation. UCAR Carbon Company offers over one thousand combinations of grade, size, machined shape, purity, oxidation-retardant, and other post-treatments. Please contact your UCAR Carbon Company representative for detailed information.

Extruded Medium Grain Graphites

UCAR® Grades CBY, CS, SLWX, CHV

Grades CBY, CS, and SLWX are extruded graphites of good structural quality, having uniform medium grain size throughout the range of sizes in which they are made. The maximum particle size is 0.03" (0.8 mm). Grade CBY is not pitch-impregnated, while CS is pitch-impregnated. Grades CBY and CS are frequently referred to as "Mold Grades" because they impart a smooth finish to castings. They are available in a variety of rounds with diameters up to 31" (775 mm) and in several rectangular configurations. CBY and CS are extensively used for handling, containment, and shaping hot metals, ceramics, and refractories.

Grade SLWX is a 24" extruded round, densified, and purified grade developed for use in silicon crystal growing applications.

Grade CHV is a slightly coarser grade with a maximum particle size of 0.06 inches (1.5 mm) produced as a large rectangle 24" x 48" (600 mm x 1200 mm) in cross-section for applications such as pressure casting of steel.

For typical room temperature properties of these grades, see Tables 2.2 and 2.3.

Molded Medium Grain Graphites

UCAR® Grades PGW and PGX

The grades PGW and PGX are uniaxially pressure molded graphites having a uniform medium-size grain (0.03 inches, 0.8mm, maximum) throughout the range of sizes in which they are made.

Grade PGW, a molded version of grade CBY, is available in diameters up to 50.8" (1270 mm). This material finds application as a casting mold, crucible, heater, and susceptor.

Grade PGX molded to diameters of 36" to 74" (900 mm to 1850 mm) is used whenever large rounds are required such as chemical reactors, induction furnaces, and heat exchangers.

Typical room temperature properties of these grades are shown in Tables 2.2 and 2.3.

Extruded Fine Grain Graphites

UCAR® Grades FGE and FGEX

The grades FGE and FGEX are economically priced extruded graphites in 8.6 inch (215 mm) squares with a fine grain structure. The maximum particle size is 0.006 inches (150 µm). Grade FGEX is a denser version of FGE. These grades are used for handling, containing, and shaping hot metals, ceramics, refractories, and semi-conductor materials.

Typical room temperature properties are summarized in Tables 2.2 and 2.3. The fineness of the grain of the FGEX is shown in the photomicrographs of Figure 2.1.

Isomolded Fine Grain

UCAR® Grades ATJ and CGW

These isostatically molded graphites are produced using fine powders with a maximum particle size of 0.003" (75 μm).

Grade ATJ is a high strength fine grain flagship graphite first introduced in the 1940's that can be machined to very close tolerances and sharp detail with a fine surface finish. It is available in a variety of sizes of rounds up to 24" (600 mm), and in rectangles up to 18 inch x 18 inch (460 mm x 460 mm). Applications include dies and molds, rocket motor nozzle and thrust inserts, and tooling for forming composite parts.

Grade CGW is a densified version of ATJ and is used where high strength, low permeability, and erosion resistance are required.

Typical room temperature properties of ATJ and CGW are shown in Tables 2.2 and 2.3. The fine grain structure of ATJ is shown in the photomicrographs in Figure 2.2.

Isomolded Superfine Grain

UCAR® Grades DM306, 300, 301, 302, 303, 342, and 332

This family of isostatically molded carbons and graphites is produced using superfine powders with a maximum particle size below 0.002" (50 μm) and an average particle size of 0.0002" (5 μm). All are capable of an excellent surface finish. These grades are available in a variety of sizes up to 11 inch x 14 inch (275 mm x 600 mm) in cross sections.

Grade DM306 is a hard carbon of high strength for use in aggressive and demanding applications such as hot glass bottle handling, permanent molds, cooling plates, and mechanical carbons. Grades DM300, 301, 302, and 303 are a group of general purpose graphites representing a continuum of properties. Applications include continuous casting molds, silicon crystal growing, and fiber optic furnace parts.

Graphite grade DM342 is suitable where strength is a major requisite.

Grade DM332 is a graphite having reduced CTE which is ideal for silicon carbide coating. This grade is available in rounds up to 24 inches (600 mm) and rectangles up to 11 inch x 24 inch (275 mm x 600 mm).

Typical room temperature properties of all these superfine grades are summarized in Tables 2.2 and 2.3. The typical microscopic structure of grade DM302 is shown in Figure 2.3, and grade DM342 in Figure 2.4.

Ultrafine Grain

UCAR® Grade DM362

This isostatically molded graphite is produced using ultrafine powders with a maximum particle size below 0.0004" (10 um). The average particle size is 0.00008" (2 um). This graphite represents the high end of surface finish and can be machined to thicknesses below 0.005". Applications include fine detail finish EDM, and aerospace.

Typical properties and microscopic structure are shown in Tables 2.2 and 2.3, and Figure 2.5, respectively.

Table 2.2 Typical Room Temperature Properties for UCAR® Grades, English Units

Group	UCAR® Grade	Largest Cross Section inches	Grain Size,(1) inches	Bulk Density, lb/ft^3	Specific Resistance 10^{-4} Ω inches
Extruded Medium Grain					
	CBY	24	0.03	103	3.2
	CS	30	0.03	107	3.0
	SLWX	24	0.03	112	2.8
	CHV	24 x 48	0.06	110	3.2
Molded Medium Grain					
	PGW	52	0.03	102	3.7
	PGX	73	0.03	107	4.0
Extruded Fine Grain					
	FGE	8.5 x 8.5	0.006	105	2.8
	FGEX	8.5 x 8.5	0.006	108	3.2
Isomolded Fine Grain					
	ATJ	24	0.001	110	4.6
	CGW	16 x 16	0.001	113	4.6
Isomolded Superfine Grain					
	DM306	8 x 16	0.0002	108	14.2
	DM300	8 x 16	0.0002	106	5.7
	DM301	8 x 16	0.0002	112	4.9
	DM302	8 x 16	0.0002	116	4.5
	DM303	8 x 16	0.0002	115	5.5
	DM342	8 x 16	0.0002	117	4.7
	DM332	24	0.0002	113	4.4
Isomolded Ultrafine Grain					
	DB362	6 x 12	0.00008	117	7.3

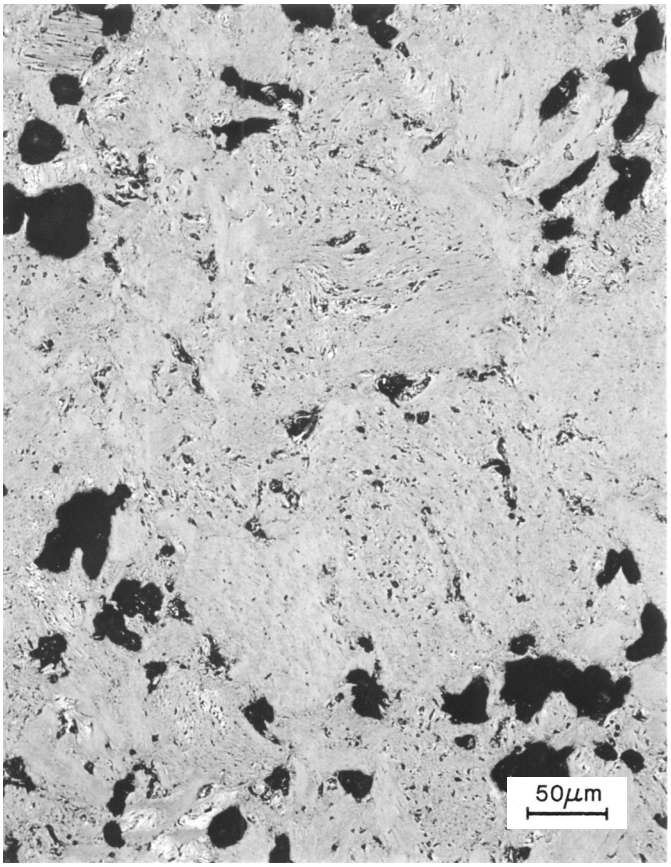
(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.

Table 2.3 Typical Room Temperature Properties for UCAR® Grades, SI Units

Group	UCAR® Grade	Largest Cross Section mm	Grain Size(1) mm	Bulk Density g/cm³	Specific Resistance μΩm
Extruded Medium Grain					
	CBY	600	0.8	1.64	8.5
	CS	600	0.8	1.74	7.5
	SLWX	600	0.8	1.80	7.0
	CHV	600 x 1200	1.7	1.76	8.0
Molded Medium Grain					
	PGW	1300	0.8	1.63	9.6
	PGX	1825	0.8 0.150	1.73	10.2
Extruded Fine Grain					
	FGE	215 x 215	0.150	1.68	7.0
	FGEX	215 x 215	0.150	1.74	8.0
Isomolded Fine Grain					
	ATJ	600	0.025	1.76	11.7
	CGW	400 x 400	0.025	1.80	11.7
Isomolded Superfine Grain					
	DM306	200 x 400	0.005	1.73	36
	DM300	200 x 400	0.005	1.70	14.6
	DM301	200 x 400	0.005	1.79	12.4
	DM302	200 x 400	0.005	1.86	11.4
	DM303	200 x 400	0.005	1.85	14.0
	DM342	200 x 400	0.005	1.87	12.0
	DM332	600	0.005	1.81	11.2
Isomolded Ultrafine Grain					
	DM362	150 x 300	0.002	1.88	15.5

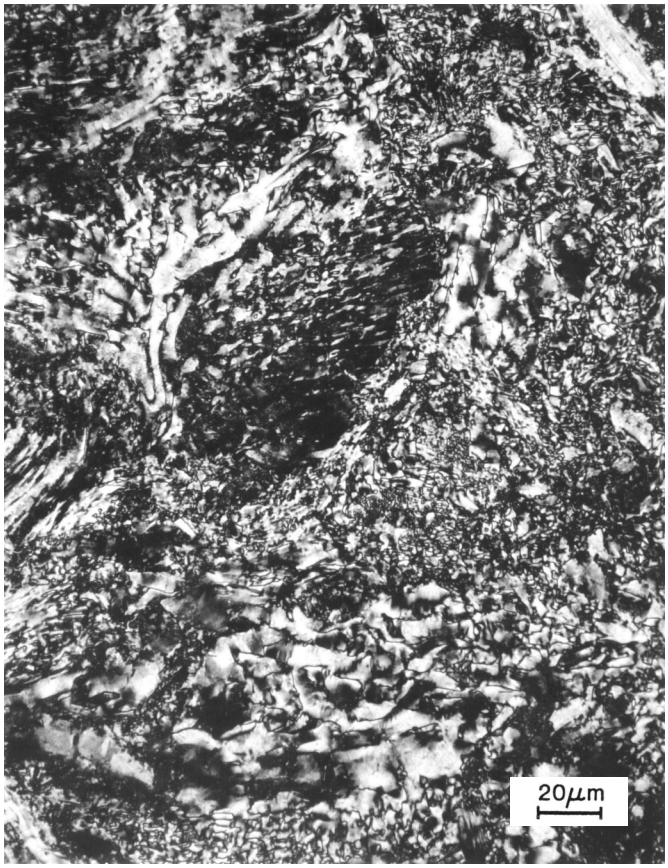
(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.

Figure 2.1 Optical structures of UCAR® Grade FGEX



200x Bright Field

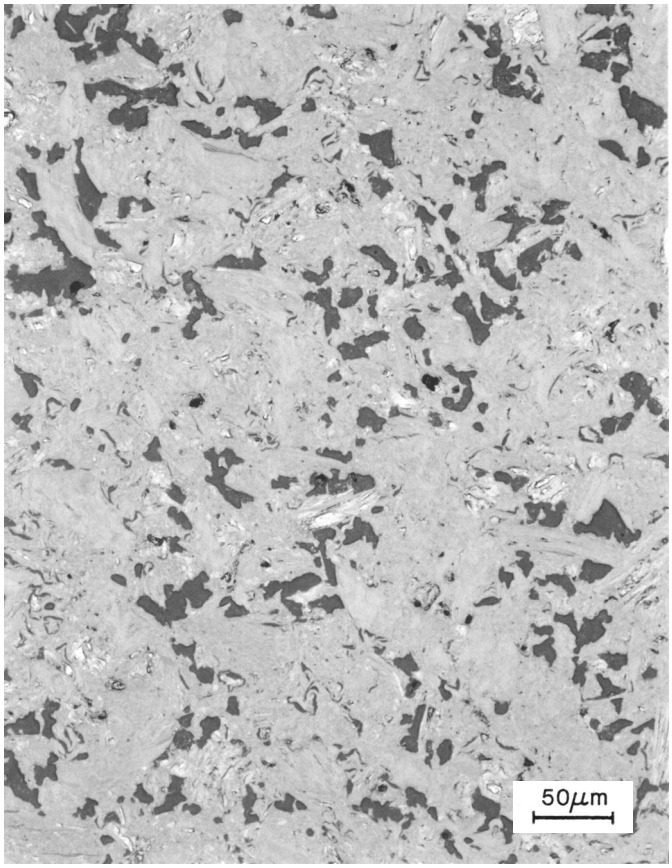
Here is a visual of the porosity. The black area is porosity. (The gray area is the solid). Notice the size and distribution of the porosity.



400x Polarized Light

Here is a visual of the solid. Focus on the texture of the white and gray areas, and note the degree of fineness. The largest solid particles can usually be distinguished.

Figure 2.2 Optical Structures of UCAR® Grade Isomolded Fine Grain ATJ



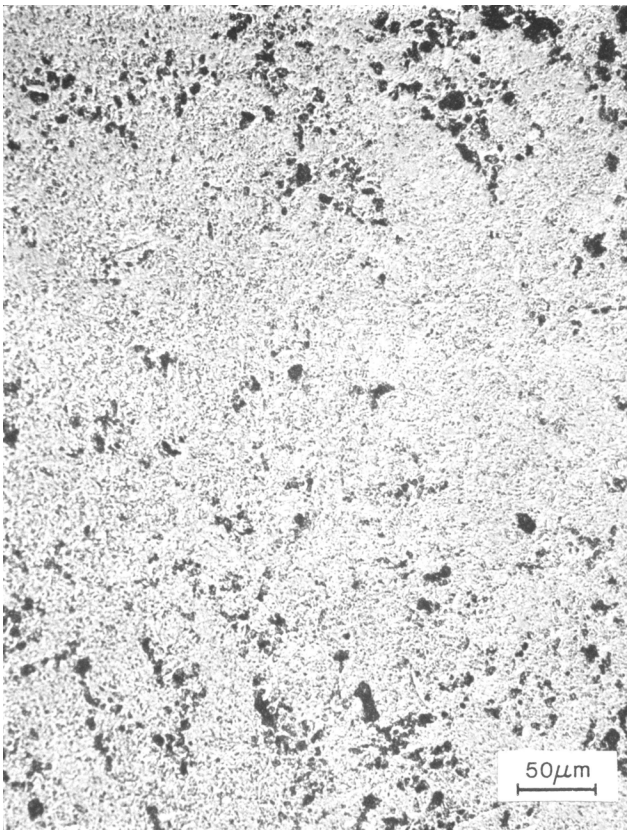
200x Bright Field

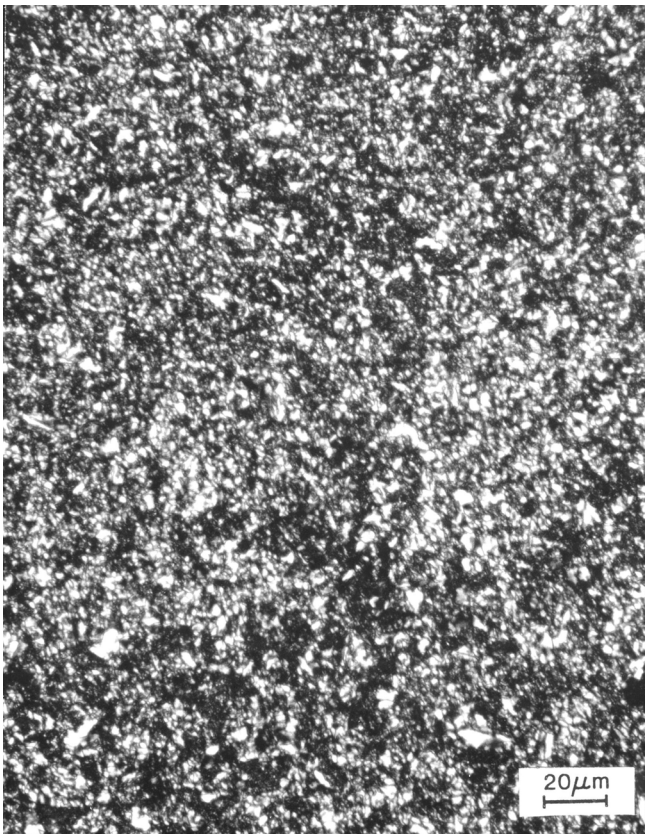
Here is a visual of the porosity. The black area is porosity. (The gray area is the solid). Notice the size and distribution of the porosity.

400x Polarized Light

Here is a visual of the solid. Focus on the texture of the white and gray areas, and note the degree of fineness. The largest solid particles can usually be distinguished.

Figure 2.3 Optical Structures of UCAR® Grade DM-302





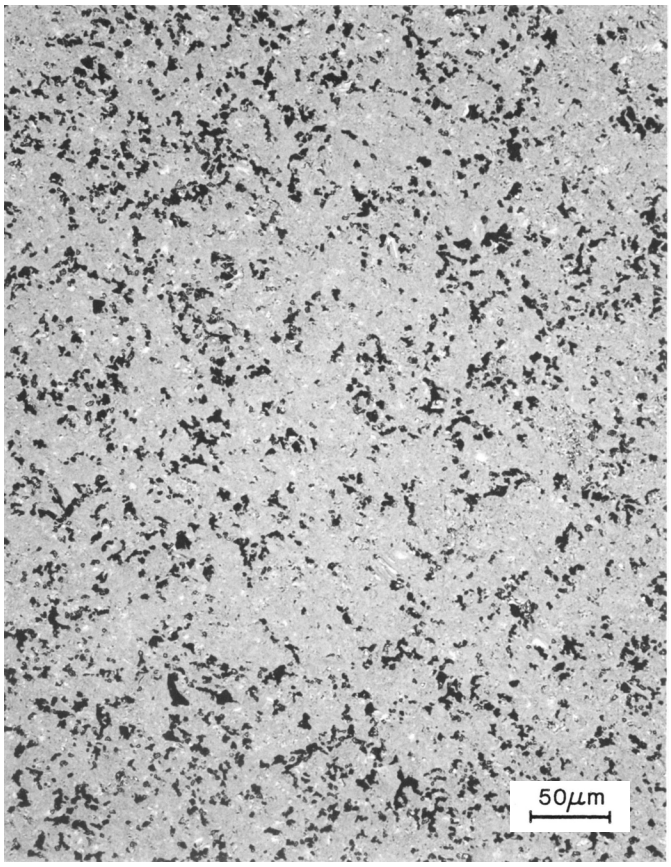
200x Bright Field

Here is a visual of the porosity. The black area is porosity. (The gray area is the solid). Notice the size and distribution of the porosity.

400x Polarized Light

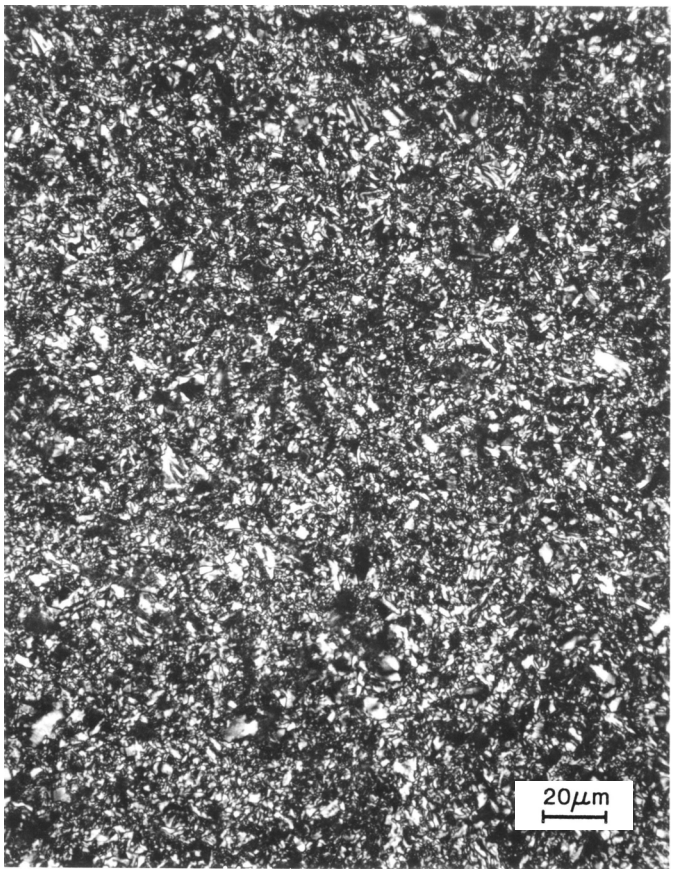
Here is a visual of the solid. Focus on the texture of the white and gray areas, and note the degree of fineness. The largest solid particles can usually be distinguished.

Figure 2.4 Optical Structures of UCAR® Grade DM-342



200x Bright Field

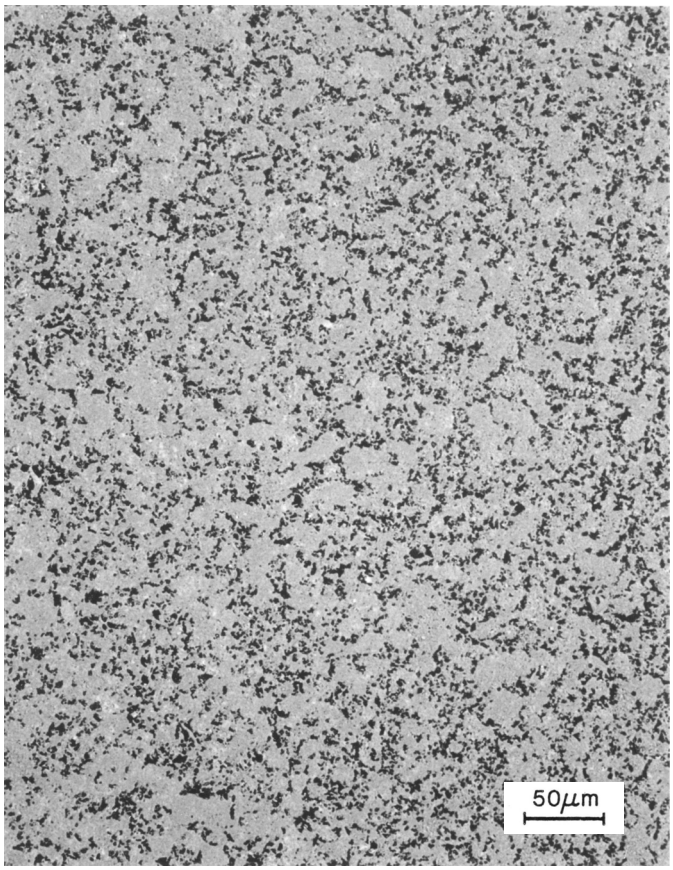
Here is a visual of the porosity. The black area is porosity. (The gray area is the solid). Notice the size and distribution of the porosity.



400x Polarized Light

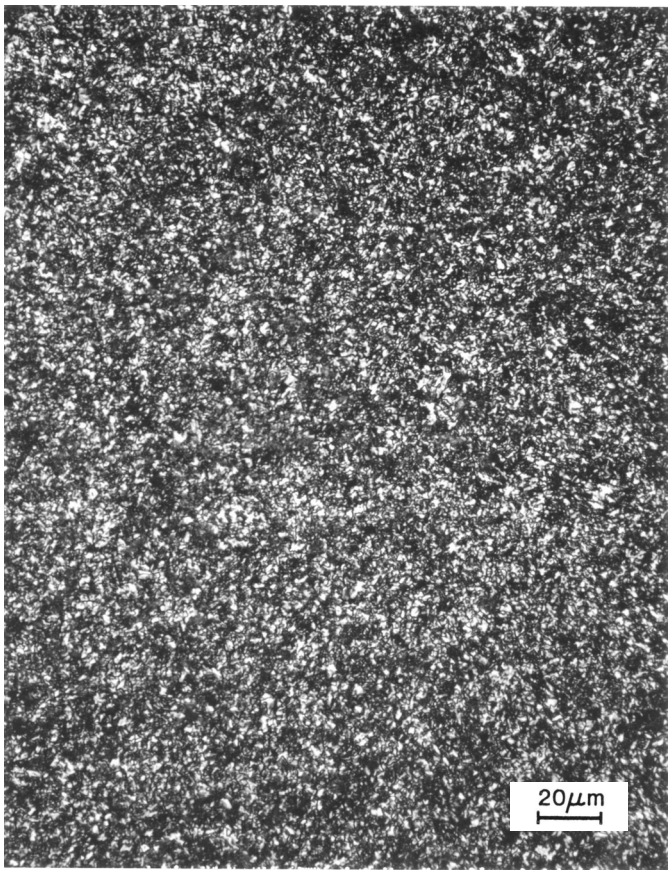
Here is a visual of the solid. Focus on the texture of the white and gray areas, and note the degree of fineness. The largest solid particles can usually be distinguished.

Figure 2.5 Optical Structures of UCAR® Grade DM-362



200x Bright Field

Here is a visual of the porosity. The black area is porosity. (The gray area is the solid). Notice the size and distribution of the porosity.



400x Polarized Light

Here is a visual of the solid. Focus on the texture of the white and gray areas, and note the degree of fineness. The largest solid particles can usually be distinguished.

Chapter Three

MECHANICAL PROPERTIES

Industrial Graphite Engineering Handbook

HIGH TECH HIGH TEMP

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Mechanical Properties

Bulk Density

Bulk density (or apparent density) is defined as weight per unit volume. The methodology for bulk carbon and graphite is described in ASTM C559.

Average values for several typical manufactured carbon and graphite grades are listed in Tables 3.1 and 3.2.

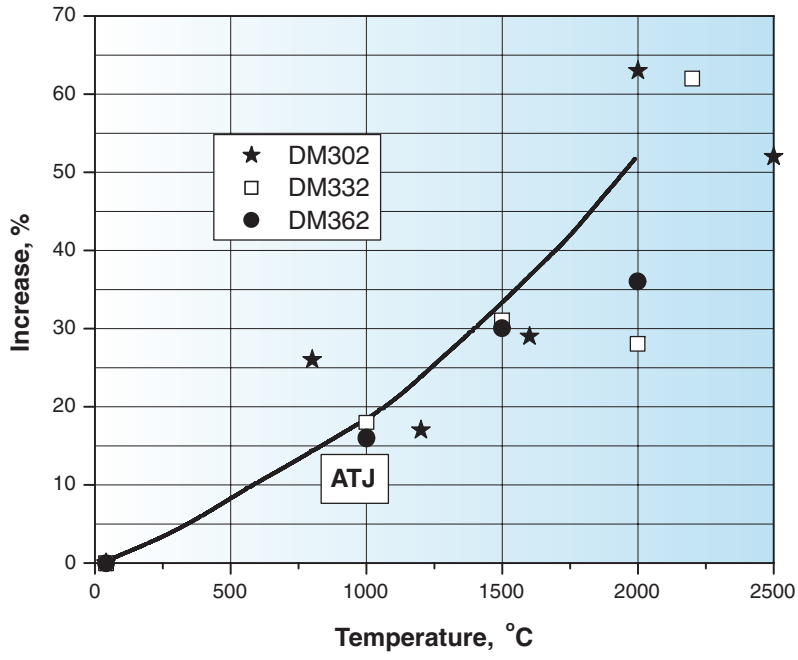
In the graphite family itself, the level of bulk density has some significance other than being purely descriptive. For example, with major material and manufacturing factors remaining reasonably the same, a higher apparent density will be reflected in improved mechanical strengths and higher thermal conductivity. Such properties as electrical resistance and permeability will be lowered. In nuclear applications, a shorter diffusion length will be realized. The above effects are not necessarily directly proportional to bulk density changes.

Tensile Strength

Tensile strength is defined as the pure tensile force per unit area required to rupture a material by pulling it apart. The room-temperature measurement is described in ASTM C565; measurement at elevated temperatures up to 2400°C is covered in ASTM C749. Average room-temperature values of several typical manufactured carbon and graphite grades are indicated in Tables 3.1 and 3.2.

The variation in tensile strength with temperature of isomolded fine grain UCAR® grades ATJ, superfine grades DM302, DM332, and ultrafine grade DM362 has been measured to give the relative value shown in Figure 3.1. Note the increase in tensile strength as temperatures increase. This is generally true of most graphites. At about 2500°C, strength can be expected to increase 80 to 100 percent from room temperature values.

Figure 3.1 % Increase in Tensile Strength vs. Temperature for UCAR® Grades ATJ, DM302, DM332, and DM362



In a similar situation, good correlation has been found between the apparent density and tensile strength, as shown by Figure 3.2[2]. Conversion of data to a constant density results in a “normalized” curve, and this is the one reported for UCAR® Grade ATJ. This correlation, which holds for graphite derived from a common base stock, is not necessarily valid when graphites of different base compositions are compared.

Figure 3.2 Bulk Density (at Room Temperature) vs. Tensile Strength (at 1200°C) for UCAR® Grades ATA, ATR, ATJ

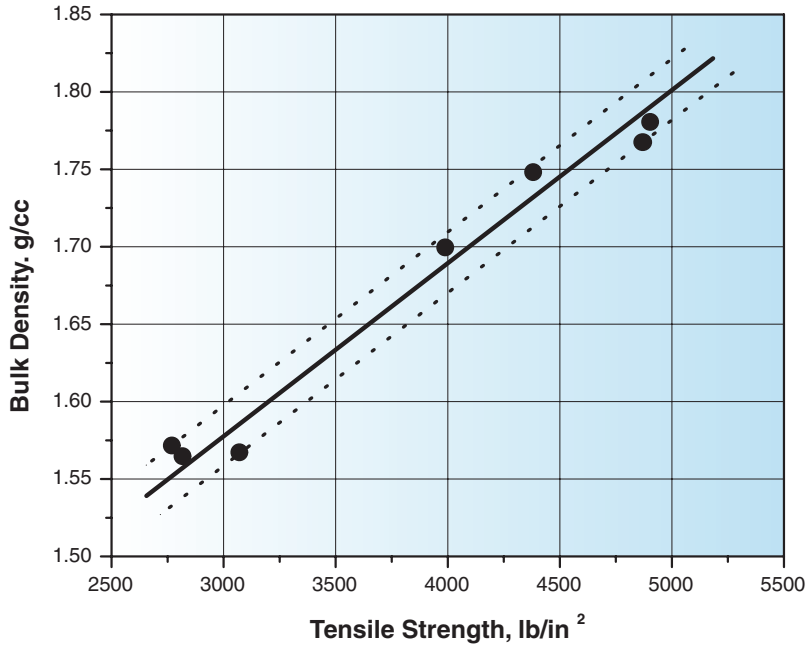


Figure 3.3[1] compares the strength-to-weight ratio of graphite at high temperatures with that of alloys and ceramics that are conventionally used in high-temperature applications.

Figure 3.3 Tensile Strength-to-Weight Ratios vs. Temperature for High-Temperature Materials (including UCAR® Grade ATJ)

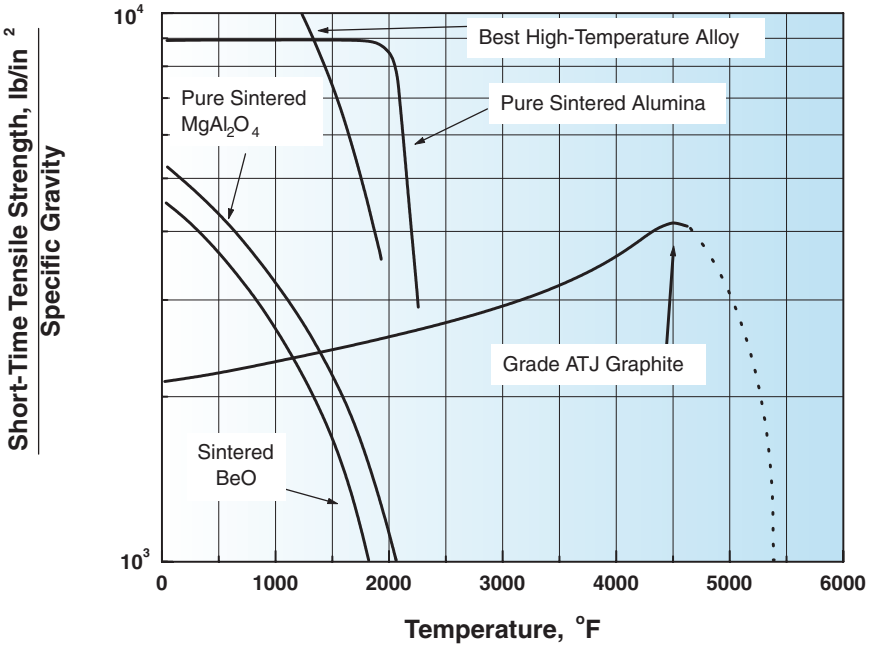


Table 3.1 Typical Room Temperature Properties UCAR® Grades, English Units

Group	UCAR® Grade	Largest Cross Section inches	Grain Size(1) inches	Bulk Density lb/ft3	Flexural Strength 4-Point psi		Tensile Strength psi		Compressive Strrength psi	Young's Modulus 10^6 psi		Permeability Darcy		Hardness Rockwell Shore	
					WG	AG	WG	AG	WG	WG	AG	WG	AG		
Extruded Medium Grain															
	CBY	24	0.03	103	2400	1600	1700	1150	5000	1.32	0.96	0.07	0.05	70R	---
	CS	30	0.03	107	3500	2300	2600	1700	7250	1.74	1.2	0.03	0.03	83R	---
	SLWX	24	0.03	112	3300	3000	2300	2200	8500	1.48	1.35	0.01	0.01	85R	---
	CHV	24 x 48	0.06	110	2600	1900	1650	1150	5200	1.9	0.97	0.02	0.02	---	---
Molded Medium Grain															
	PGW	52	0.03	102	---	1960	---	1255	4500	---	---	---	0.17	---	32
	PGX	73	0.03	107	1600	1300	1600	1300	4350	1	0.7	0.04	0.05	---	---
Extruded Fine Grain															
	FGE	8.5 x 8.5	0.006	105	3190	1740	2540	1015	6525	1.86	0.96	0.004	0.003	---	---
	FGEX	8.5 x 8.5	0.006	108	3625	2465	2900	1850	8410	1.75	1.12	0.002	0.001	---	---
Isomolded Fine Grain															
	ATJ	24	0.001	110	4250		3740		9700	1.4		0.002		60L	---
	CGW	16 x 16	0.001	113	4900		4500		11,200	1.55		0.0001		90L	---
Isomolded Superfine Grain															
	DM306	8 x 16	0.0002	108	7100		6000		22,000	1.6		<0.002		105H	---
	DM300	8 x 16	0.0002	106	4450		3400		9,500	1.45		<0.006		37H	43
	DM301	8 x 16	0.0002	112	6000		4400		12,500	1.6		<0.001		55H	50
	DM302	8 x 16	0.0002	116	7500		6000		14,000	1.7		<0.0003		75H	60
	DM303	8 x 16	0.0002	115	9100		7900		18,000	1.7		<0.0006		90H	75
	DM342	8 x 16	0.0002	117	9200		8000		18,000	1.8		<0.0001		---	68
	DM332	24	0.0002	113	6200		4900		13,000	1.7		<0.0007		53H	---
Isomolded Ultrafine Grain															
	DM362	6 x 12	0.00008	117	12,000		9000		24,000	1.9				---	75

(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.

Table 3.2 Typical Room Temperature Properties UCAR® Grades, SI Units

Group	UCAR® Grade	Largest Cross Section, mm	Grain Size,(1) mm	Bulk Density, g/cm³	Flexural Strength, 4-Point MPa		Tensile Strength, MPa		Compressive Strength MPa		Young's Modulus GPa		Permeability Darcy		Hardness Linear Scale MPa
					WG	AG	WG	AG	WG	AG	WG	AG	WG	AG	
Extruded Medium Grain															
	CBY	600	0.8	1.64	16.5	11	12	8.0	35	---	9.1	6.6	0.07	0.05	125
	CS	600	0.8	1.74	24.0	16	18	12.0	50	---	10.3	8.0	0.03	0.04	200
	SLWX	600	0.8	1.80	22.8	20.7	15.9	15.2	58.6	---	10.2	9.3	0.01	0.01	
	CHV	600 x 1200	1.7	1.76	18.0	13.0	11.5	8.0	36	---	10.1	6.7	0.02	0.02	130
Molded Medium Grain															
	PGW	1300	0.8	1.63	---	11	---	8.7	31	---	---	---	---	0.17	43
	PGX	1825	0.8	1.73	11.7	10.3	11.6	9.5	30	---	7.6	5.5	0.04	0.05	---
Extruded Fine Grain															
	FGE	215 x 215	0.150	1.68	22	12	17.5	7	45	---	12.8	6.6	0.004	0.003	200
	FGEX	215 x 215	0.150	1.74	25	17	20	12.5	58	---	12.1	7.7	0.002	0.001	310
Isomolded Fine Grain															
	ATJ	600	0.025	1.76	29.5		25.5		67		9.6		0.002		60
	CGW	400 x 400	0.025	1.80	34		31		78		10.7		0.001		90
Isomolded Superfine Grain															
	DM306	200 x 400	0.005	1.73	49		41.4		151.7		11		<0.002		105
	DM300	200 x 400	0.005	1.70	30.7		23.4		65.5		10		<0.006		35
	DM301	200 x 400	0.005	1.79	41.4		30.3		86.2		11		<0.001		55
	DM302	200 x 400	0.005	1.86	51.7		41.4		96.5		11.7		<0.0003		75
	DM303	200 x 400	0.005	1.85	62.7		54.9		124.1		11.7		<0.0006		90
	DM342	200 x 400	0.005	1.87	63.4		55.2		124.1		12.4		<0.0001		
	DM332	600	0.005	1.81	42.7		33.8		89.6		11.7		<0.0007		
Isomolded Ultrafine Grain															
	DM362	150 x 300	0.002	1.88	82.7		---		---		13.1		---		

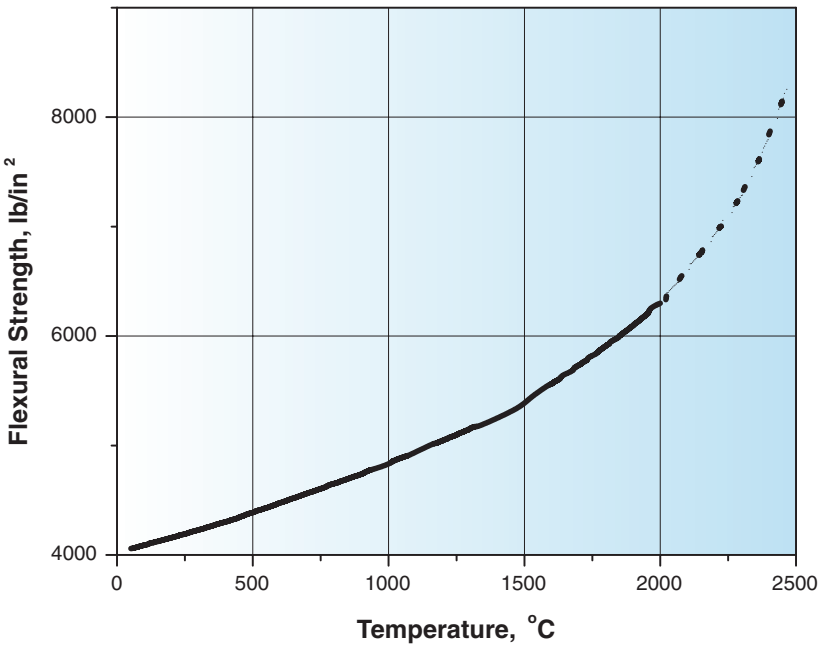
(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.

Flexural Strength

Flexural strength (modulus of rupture) is the maximum fiber stress that a material will withstand before rupture in bending. The room-temperature measurement using four-point loading is described in ASTM C1025. The alternative three-point loading (see ASTM C1161) yields values for flexural strength about 25% higher than the four-point loading.[17] However, the three-point loading has the higher variance[17]. Both loadings are affected by specimen size, shape, and surface finish.[17] For elevated temperature methodology, see ASTM C1366. Average room-temperature flexural strengths for several typical carbon and graphite grades using four-point loading are listed in Tables 3.1 and 3.2.

The flexural strength of UCAR® Grade ATJ graphite has been measured at elevated temperatures[3], to show, as expected, that this increases markedly with increasing temperature, just as do the other strength characteristics of graphite. The results are plotted in Figure 3.4.

Figure 3.4 Modulus of Rupture vs. Temperature for UCAR® Grade ATJ Graphite Stressed Against-the-Grain



These results were measured on 0.5 inch x 0.5 inch x 8.0 inch samples, mounted horizontally in the center of a 40 inch I.D. graphite tube furnace. A four-point loading method was employed, the piece resting on supports six inches apart, with the stress applied from above on two rounded knife edges 1 inch apart. With graphite, this has been found to give results more consistent than with single-point loading, although the values are always of lower magnitude, by about 25 percent. The upper portion of the curve is shown as a dashed line, since plastic flow is quite significant at 2500°C, and a more precise control of the rate of loading is needed to improve the reliability of this last point.

Creep

Creep is defined as plastic flow under constant stress. Graphite displays extremely low creep at room temperature, its flow characteristics being comparable to those of concrete. Graphite is not perfectly elastic, even at room temperature, and yields plastically under stress as the stress is increased. The stress-strain curves for graphite are not linear, and there is a residual strain in the material after stressing, which increases as the maximum stress is increased. However, this plastic straining is not generally considered to be creep. Genuine creep over periods up to several hours has been observed in graphite only at temperatures of approximately 1700°C and above.

Carbon products that have been processed to temperatures of about 900°C show much more creep than graphite, and at much lower temperatures. They are much stronger than graphite and have higher Young's Moduli at room temperature, but creep more easily at high temperatures. Creep in carbon has been observed at a temperature as low as 1000°C. As the processing temperature of carbon is raised, its creep tends to decrease. In general, a graphite displays less creep as the crystal habit becomes more highly ordered.

Typical tensile creep curves for UCAR® grade ATJ graphite are shown in Figure 3.5. The curves are characterized by an initial elastic strain, which is followed by a region of decreasing creep rate. In the relatively short-time creep tests of 90 minutes, there is a region of steady-state creep in which the creep rate is constant over approximately the last 30 minutes. In some graphites, however, the creep rate has been observed to continue to decrease over a testing period of 6 to 7 hours. When the stress is removed from a specimen that has been creeping for some time, there is an immediate recovery of the elastic part of the strain and a gradual recovery of part of the creep strain which can extend for several hours. Mainly because of its behavior in recovery, graphite can be considered to be a viscoelastic material at high temperatures.

Creep in graphite is strongly dependent on the grain orientation. As Figure 3.5 shows, the creep of UCAR® grade ATJ graphite is greater in the "against-the-grain" direction than in the "with-the-grain" direction. Figure 3.6 compares the tensile creep of two other graphites. For UCAR® grade ZTA graphite, which has a higher density and a much higher degree of anisotropy than UCAR® grade ATJ graphite, the creep is very low in the "with-the-grain" direction and relatively high "against-the-grain." The creep of an isotropic, molded lampblack-base graphite (UCAR® grade CEP), having a relatively low density, is almost independent of the orientation of the specimen relative to the direction of the applied force in molding. Lampblack-base graphite displays much less creep than UCAR® grade ATJ graphite under the same conditions of stress and temperature.

Figure 3.5 Tensile Creep of UCAR® Grade ATJ Graphite

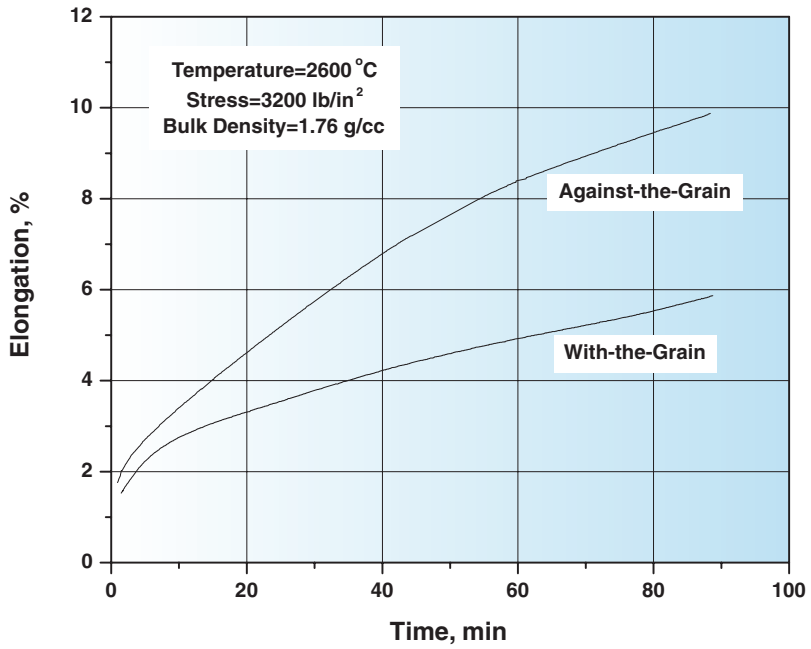
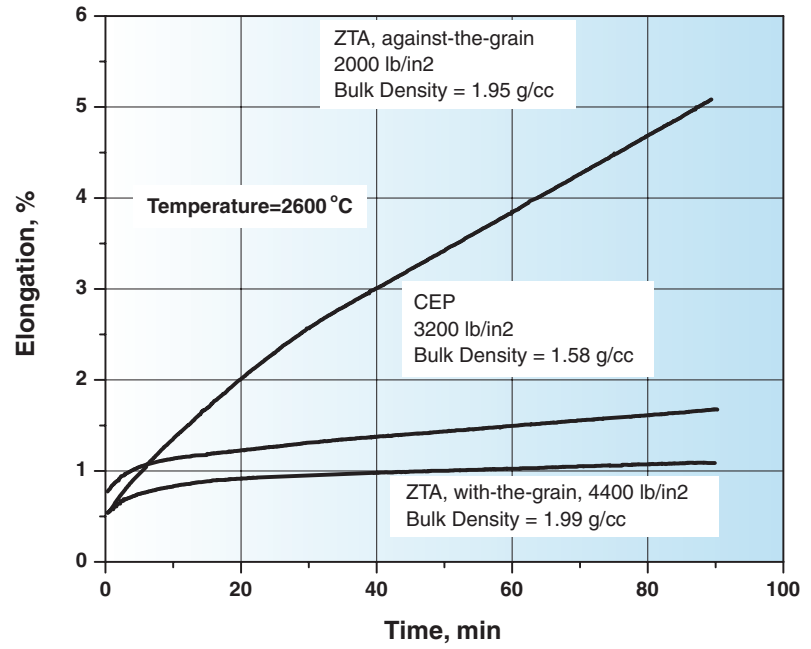


Figure 3.6 Tensile Creep of UCAR® Grade ZTA Graphite and a Lampblack-Base Graphite



The stress dependence of the steady-state creep rate of UCAR® grade ATJ specimens oriented “with-the-grain” is shown in Figure 3.7. At constant temperature, the steady-state creep rate varies with the fourth power of the stress. The same stress dependence was found for specimens oriented “against-the-grain.” Temperature dependence of the steady-state creep rate of UCAR® grade ATJ specimens oriented “with-the-grain” is shown in Figure 3.8, and an activation energy for the creep process of 124 kcal/mole was calculated from these results. “Against-the-grain” specimens yielded the same activation energy, even though under identical conditions of temperature and stress the steady-state creep rate was 1.8 times as great as that of the “with-the-grain” specimen. Tensile creep data of UCAR® grade ZTA graphite and lampblack-base graphite yielded the same activation energy.

Figure 3.7 Tensile Creep of UCAR® Grade ATJ Oriented “With-the-Grain”

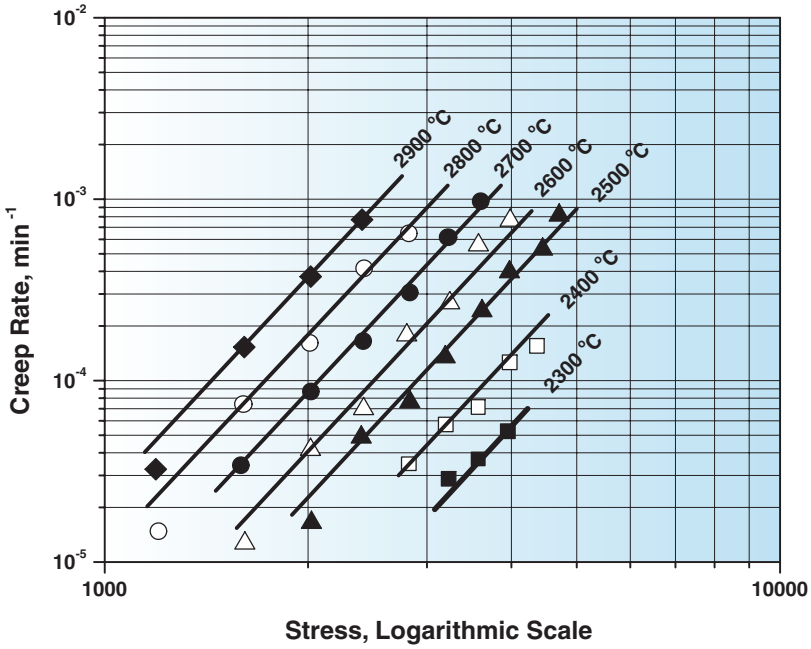
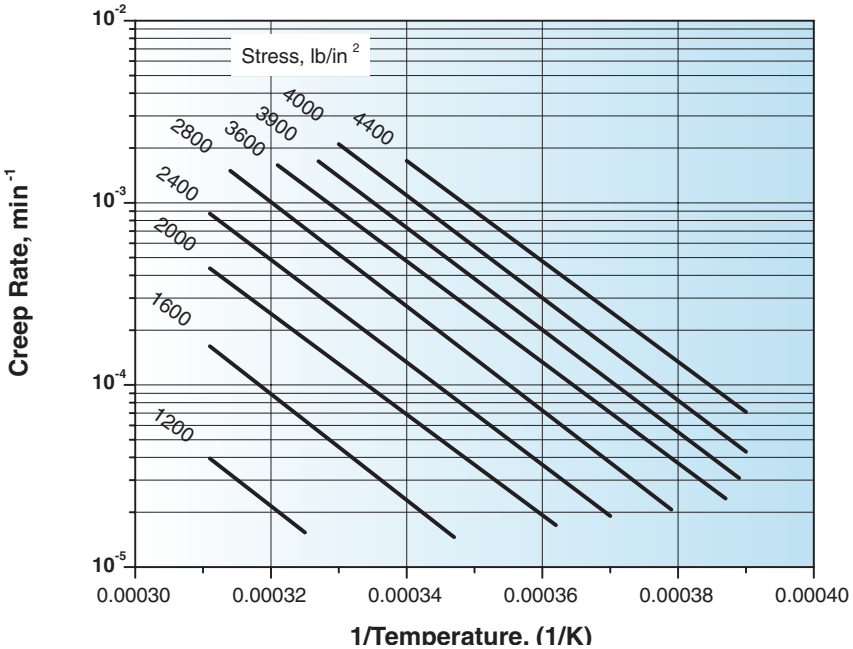


Figure 3.8 Tensile Creep of UCAR® Grade ATJ Oriented “With-the-Grain”



The mechanisms for creep in graphite have not been definitely established. For practical purposes, creep of a coke-base graphite decreases in the direction “with-the-grain” as the density increases and as the anisotropy of the graphite increases. There is a corresponding increase in creep in the direction “against-the-grain.” A lampblack-base graphite has relatively low creep in all directions.

Compressive Strength

Compressive strength is defined as the maximum strength that can be developed under a compressive load, the load being referred to is the original cross-sectional area and expressed as load per unit area. Compressive strength is not a fundamental constant of a material. Different unit values result from different sizes and shapes of sample, different rates of loading, etc. Average compressive strengths measured using ASTM C695 for several typical carbon and graphite grades are shown in Table 3.1 and 3.2.

The temperature variation of compressive strength of a medium grain graphite UCAR® grade AGSR has been measured using in-house methodology on samples one-inch in diameter to two inches long, cut from a 20-inch diameter piece with the length taken both from the diameter of the piece (transverse) and from the length of the piece (longitudinal)[4]. The results are shown in Table 3.3, where the compressive strength of graphite is seen to increase with increasing temperature.

The following relationship has been found to hold between crushing strength and the more commonly measured flexural strength for a variety of miscellaneous graphite samples[5].

$$\frac{\text{Crushing Strength}}{\text{Flexural Strength}} = 2.07 \text{ (Range 1.61 to 2.90)}$$

Table 3.3 Compressive Strength of 20-inch Extruded Graphite

	Compressive Strength, psi	
	With-the-Grain	Across-the-Grain
Direction	at 25°C	at 1600°C
Longitudinal	2840	3260
Transverse	2450	3050

Impact Strength

The impact strength of a material is a measure of its ability to withstand a sudden blow, a property conventionally measured with a pendulum device, such as that described in ASTM test method D256. However, with rigid plastics, there are many instances where the strengths so measured are not correlated with a practical experience[6]. A comparison of dynamic with static stress-strain behavior using oscillographic techniques has shown that rupture (failure) under dynamic stress-strain conditions results from the same basic considerations as those which govern static or flexural failure. Therefore, the conventional static flexural test technique can be utilized in evaluating the energy required for fracture and the notch resistivity. These values are a much better measure of the service characteristics of brittle materials under impact loading than is given by the pendulum test methods.

For graphite, the static test data are found to give a much improved correlation of the impact behavior as compared with pendulum test results[7]. The work done to failure (the recommended measure of impact strength) is given by the area under the stress-strain diagram. Since, with graphite, this curve approximates a straight line, it can be shown that:

$$\text{Impact Strength (ft-lb)} = 0.00463\frac{F^2}{E}$$

where:

F = Flexural Strength, psi
E = Young's Modulus, psi

Impact strength values calculated using average F and E from the room-temperature property Tables 3.1 and 3.2 are given in Table 3.4 for UCAR® grades CS and ATJ.

Another conventional measure of the static and impact behavior of brittle materials is "notch sensitivity," which is defined as the ratio of the strength of a 0.400 inch thick test bar to that of a 0.500 inch thick bar with a 0.100 inch deep notch cut into it. Because of the inherent high modulus of elasticity of carbon and graphite, the impact strength is "notch" and shape sensitive. For this reason, the test values apply directly only to specimens of identical geometry.

Table 3.4 Impact Strengths of UCAR® Grades

	Impact Strength, ft-lb	
Grade and Size	With-the-Grain	Across-the-Grain
CS		
3 - 9 inch, Diameter	0.018	0.016
10 - 24 inch, Diameter	0.018	0.014
ATJ		
Rectangles	0.036	0.045

Young's Modulus, Modulus of Rigidity, and Poisson's Ratio

Young's modulus of elasticity is a measure of the resistance to change in length per unit length of a sample subjected to tensile or compressive stress. It is expressed as the ratio of the stress (load per unit area) to the resulting strain (change in length per unit length). Typically, Young's Modulus is estimated from measurement of the sonic resonance in a specimen, see ASTM C747. An approximate value for the Young's Modulus (sonic modulus) is estimated from measurement of the sonic velocity in a specimen, ASTM C769.

The modulus of rigidity (or shear modulus) of a material is a measure of the resistance to angular deformation which occurs when the material is subject to pure shear stress. It is expressed as the ratio of shearing stress (load per unit area) to the resulting shear strain.

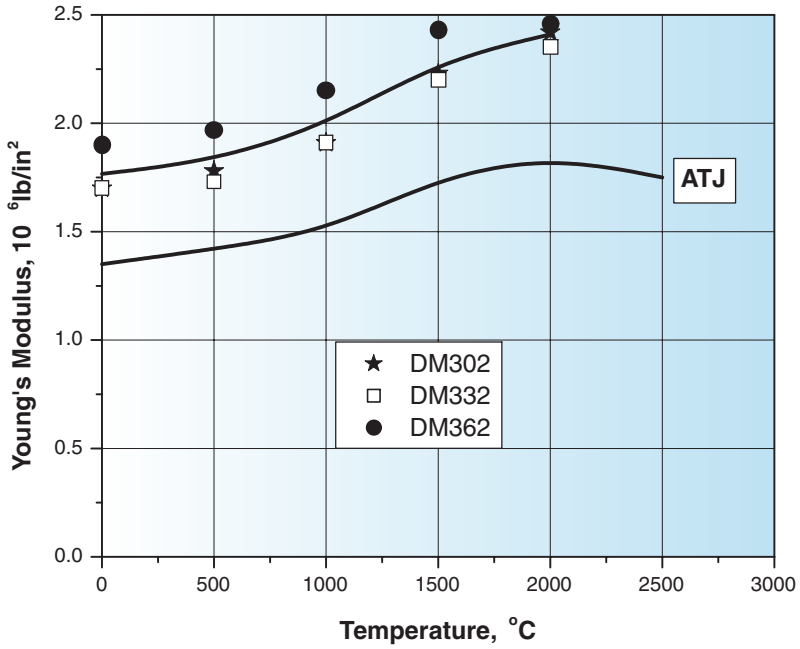
When a material is subjected to a tensile (or compressive) stress, there is a transverse contraction (or expansion) which accompanies the longitudinal expansion (or contraction). The ratio of the transverse strain to the longitudinal strain is a dimensionless number known as Poisson's ratio. For graphite, which has a nonlinear stress-strain relationship, Poisson's ratio is taken as the ratio of the transverse to longitudinal strains in the limit of zero strain. Three Poisson's ratios may be given for graphite. When the stress is in the with-the-grain direction, ν_{ww} is the ratio of the transverse strain in the across-the-grain direction to the longitudinal strain in the with-the-grain direction. When the stress is in the across-the-grain direction, ν_{aw} is the ratio of the transverse strain in the with-the-grain direction to the longitudinal strain in the across-the-grain direction. Table 3.5 lists the Young's moduli, modulus of rigidity, and Poisson's ratios at room temperature of several grades of molded graphites[13]. Room temperature Young's moduli for typical carbon and graphites are reported in Table 3.1 and 3.2.

Table 3.5 Young's Moduli, Modulus of Rigidity, and Poisson's Ratios of UCAR® Molded Graphites at Room Temperature

	Young's Modulus, 10^{-6} psi		Modulus of Rigidity, 10^{6} psi	Poisson's Ratios		
	With-the-Grain	Across-theGrain				
Grade	E_w	E_a	G	ν_{ww}	ν_{wa}	ν_{aw}
ATJ	1.40	1.40	0.57	0.10	0.16	0.11
ZTA	2.85	0.70	0.50	0.07	0.25	0.05

The Young's moduli of isomolded UCAR® grades ATJ, DM302, DM332, and DM362 have been measured as a function of temperature in the range from 25 to 2700°C, see Figure 3.9. Young's modulus was determined by measuring the resonant frequency of the fundamental flexural mode of vibration of a graphite bar or rod.

Figure 3.9 Young's Modulus vs Temperature for UCAR® Isomolded Graphites



Single Crystal Elastic Constants

The elastic constants of the graphite crystal may be given either by a set of stiffnesses, c_{ij}, or a set of compliances, s_{ij}. The components of stress σ_j are related to the components of strain ε_j through the relationships:

$$\sigma_i = \sum_{j=1}^{6} c_{ij}\,\varepsilon_j \quad \text{and} \quad \varepsilon_j = \sum_{i=j}^{6} s_{ij}\sigma_i$$

The values of the elastic constants, with their estimated uncertainties, are given in Table 3.6 [14]. The Young's moduli of the single crystal are $E_w = 1/S_{11}$ and $E_a = I/S_{33}$. The modulus of rigidity of the graphite crystal, corresponding to shear parallel to the basal planes, is $G = I/S_{44} = C_{44}$.

Table 3.6 Elastic Constants of Graphite

(Units: C_{ij} and 1 /S_{ij} in 10^{11} dynes/cm^2, S_{ij} in 10^{-11} cm^2/dyne)		
C_{11} = 106 ± 2	S_{11}=	0.0098 ± 0.0003
C_{12} = 18 ± 2	S_{12} =	-0.0016 ± 0.0006
C_{13} = 1.5 ± 0.5	S_{13} =	-0.0033 ± 0.0008
C_{33} = 3.65 ± 0.10	S_{33} =	0.275 ± 0.010
C_{44} = 0.40 ± 0.04	S_{44} =	2.50 ± 0.25
C_{66} = 44 ± 2	S_{66} =	0.023 ± 0.002
	$1/S_{11}$ =	102± 3
	1 /S_{33} =	3.65 ± 0.10
Poisson's Ratios:		
$\nu_{ww} = S_{12}/S_{11}$ = 0.16 ± 0.06		
$\nu_{wa} = S_{13}/S_{11}$ = 0.34 ± 0.08		
$\nu_{aw} = S_{13}/S_{33}$ = 0.012 ± 0.003		

Permeability

Liquid and gas permeabilities of unconsolidated media are well understood and remain simple in consolidated media of high permeability below the velocity range in which critical or turbulent flow come into play[8]. However, as pore size becomes smaller, details of pore shape become increasingly important and gas flow and liquid flow are not simply interrelated. In low-permeability graphites, liquid and gas flow should be discussed separately. Gas flow in manufactured carbon and graphite is determined by ASTM C577 and adequately described by Darcy's Law, in the form:

Equation (1)

$$Q_v = K \frac{A}{L} \frac{<p>}{\eta} \Delta p$$

where K is Darcy's constant, A and L are, respectively, the sample area (cm^2) and length (cm), η the viscosity in centipoises, <p> the average pressure and Δp the pressure differential in atmospheres. K is thus defined to be in Darcy units and the volume flow, Q_v, is expressed in ml/sec at one atmosphere pressure. The room temperature permeabilities for some commonly used graphites have been measured using nitrogen as the fluid, (helium for the finer isomolded grades), and at small pressure gradients (Tables 3.1 and 3.2).

Graphite permeability is affected by temperature as a result of the change in viscosity of the test gas. Gas viscosities increase as the temperature increases and, as indicated in Equation (1), there is a corresponding reduction in the volume flow rate. Darcy's Law is valid at elevated temperatures so long as the flow of the test gas is predominantly viscous in nature. The effective pore size of the solid must be large compared to the mean-free path of the gas molecules. Since the mean-free path increases with temperature, it is possible to encounter diffusive flow at elevated temperatures. In such cases, Equation (1) is no longer adequate. The pressure dependence of the flow rate is a reasonable indication that the test conditions lie in the region where Darcy's Law applies.

Gas flow in low permeability graphites is more complex than that defined by Darcy's equation and may be described as viscous flow with Slip[9]. This type of flow is given by an equation of the form:

$$Q_{pv} = F \frac{A}{L} \Delta p$$

Where Q_{pv} is the pressure volume flow in milliliter-atmospheres per minute, F is an admittance (ml-cm/min), A and L are, respectively, the sample area (cm^2) and length (cm) and Δp the pressure differential (atm). The admittance, F, in turn consists of two terms in which K_0 and K_1 are[10]:

$$F = F_0 + F_1 <p> = K_0 \sqrt{\frac{T}{M}} + K_1 \frac{<p>}{\eta}$$

constants characteristic of the material, (strictly speaking, K_0 is a weak function of pressure in the transition region from viscous to diffusive flow.), T is the absolute temperature (K), M the molecular weight of the gas, <p> the average pressure (atm), and η the viscosity (poise). The coefficient, K_1, is similar to that appearing in Darcy's equation above. This relationship has been carefully verified both for various gases and at temperatures up to 1000°C[11].

The constants $K_0$and K_1 may be expressed in terms of effective capillary dimensions:

$$K_0 = 1.48 \times 10^6 <\frac{Nr^3}{x}> \ ml \cdot min^{-1} \cdot cm^{-1} \cdot (K)^{-1/2}$$

$$K_1 = 23.9 \times 10^6 <\frac{Nr^4}{x}> \ ml \cdot poise \cdot min^{-1} \cdot cm^{-1} \ atm^{-1}$$

in which N is the number of pores of radius r and effective length xL, x being a tortuosity factor and L the sample length as above. The angular brackets indicate that appropriate averages over the pore distribution are to be taken. From the ratio K_0/K_1, an effective pore radius for flow may be computed, and hence an effective factor N/x. Without more exact calculation the values are only qualitative, but give some idea of the orders of magnitude involved. Parameters are given in Table 3.7 for two graphites. The fact that the effective pore radius occurs at high exponents in K_0 and K_1 indicates the sensitivity of the admittance factor to small changes in pore radius. Finally, it should be mentioned that BET surface area measurements in conjunction with the porosimetry results indicate the tortuosity factor "x" is of the order two to ten and increases as the effective pore radius decreases.

Table 3.7 Permeability Coefficients

	Unimpregnated Stock	Impregnated Stock	Units
F_o (Helium)	25.5	1.53	$ml \cdot min^{-1} \cdot cm^{-1}$
F_1 (Helium)	78.1	0.62	$ml \cdot min^{-1} \cdot cm^{-1} \cdot atm^{-1}$
Effective Pore Radius	3.2	0.42	Microns
Effective N/x	6×10^4	1.6×10^6	cm^{-2}

Heterogeneous liquid flow is related in a complex way to both static and dynamic phenomena. If there were purely liquid flow, the Darcy equation would hold:

$$Qv = K \frac{A}{L} \frac{\langle p \rangle}{\eta} \Delta p$$

in which Q_v, is the volume flow, h the viscosity, K' a permeability coefficient similar to the K_1 above for gas flow. With liquid moving against a gas, however, it is necessary for the applied pressure to overcome the capillary force arising from an interface arrested at sharp discontinuities in pore diameter. This defines a critical radius r_c for each pore and, until the pressure difference Δp exceeds the value given by:

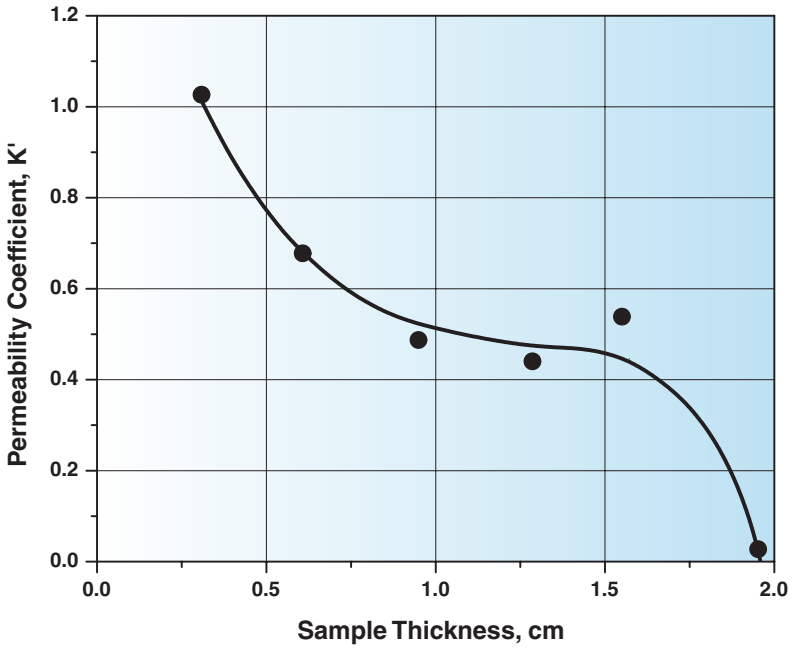
$$\Delta p = \frac{2\gamma \cos \Theta}{r_c}$$

where γ is the surface tension, the pore cannot support flow. As a result, liquid permeability through the material will vary sharply with the sample thickness, attaining a cutoff at which no flow will occur at reasonable Δp. This behavior is difficult to measure due to the tendency toward bubble formation, but the effect is shown in Figure 3.10 for water through an impregnated graphite. The S-shape of the curve correlates with the transition from macro- to micro-pore structure. For water ($\theta = 0$, $\gamma = 72.8$ dynes/cm), the critical radius is given below, where Δp is in atmospheres:

$$r_c = \frac{1.5}{\Delta p} \text{ microns}$$

which gives a value of $r_c = 0.15\ \mu$ at the cutoff with $\Delta p = 150$ psi. Since the minimum pore radius decreases sharply as the length increases, the sharp drop-off in K' results.

Figure 3.10 Water Permeability vs. Sample Thickness at Δp= 150 psi



At very small pore radii, wall friction may be more important than viscosity in determining the liquid flow characteristics. In a practical sense, complete impermeability to liquid flow can be supplied by a sufficient thickness of graphite still permeable to gases.

To achieve low-permeability materials, selected base materials are subjected to additional densifications. The conventional pitch-type impregnation has the effect of increasing density while reducing permeability.

The effect of pitch impregnation on the mercury intrusion pore spectrum of graphites are shown in Figures 3.11 for UCAR® grade CBY, and in Figure 3.12 for UCAR® grade CS. The UCAR® grade ATJ is shown in Figure 3.13. Table 3.8 summarizes these investigations.

Table 3.8 Porosimetry Analysis of UCAR® Grades[16]

	ATJ	CS	CBY
Maximum Particle Size, inches.	0.001[1]	0.03	0.03
Bulk Density, g/cc	1.76	1.73	1.66
Median Pore Diameter, μm			
Volume	1.714	4.347	4.866
Area	0.012	0.164	0.183
Average Pore Diameter, μm	0.040	0.953	1.238

[1]Average Grain Size

Figure 3.11 Porosimetry - UCAR® Grade CBY

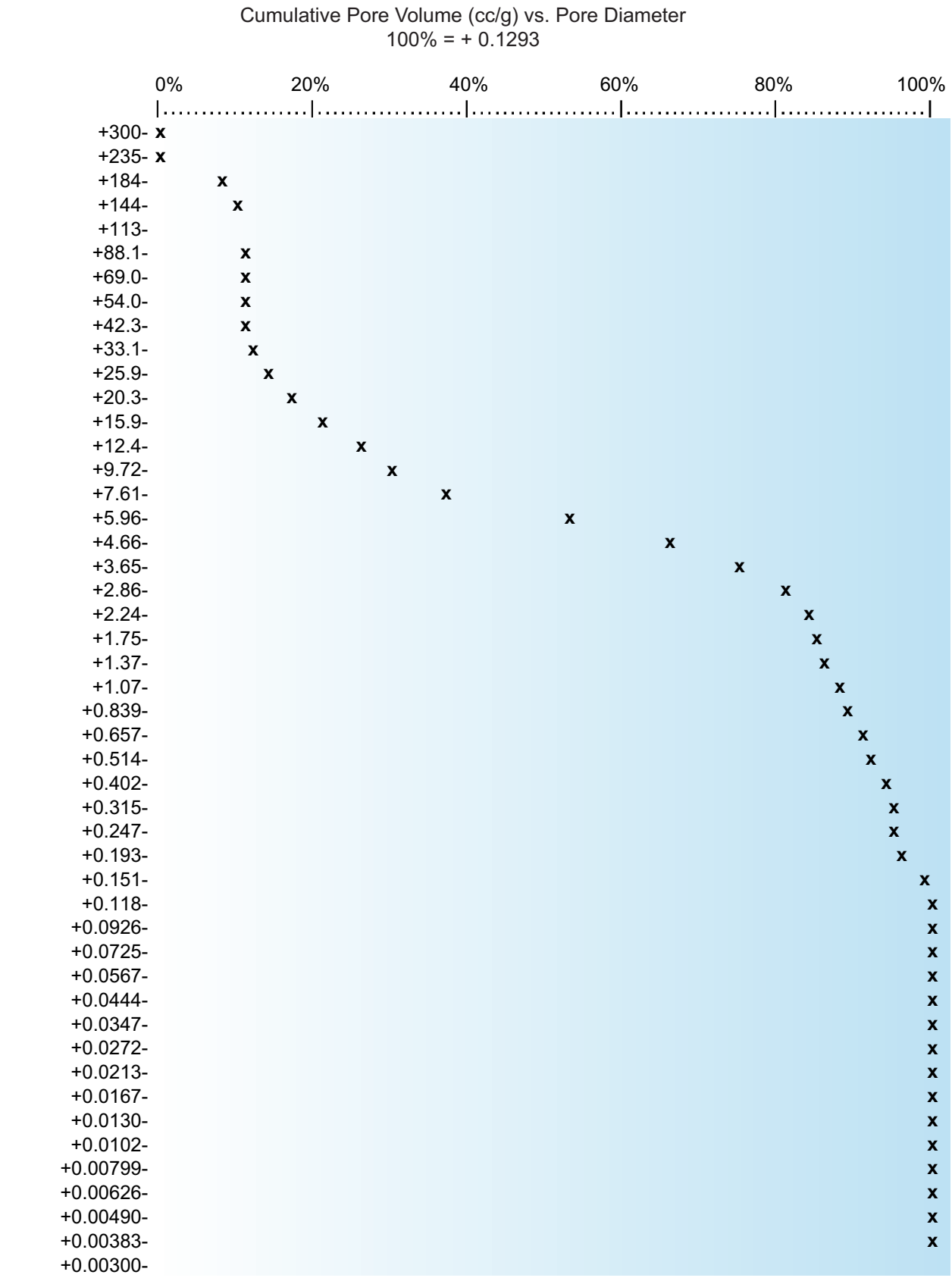


Figure 3.12 Porosimetry - UCAR® Grade CS

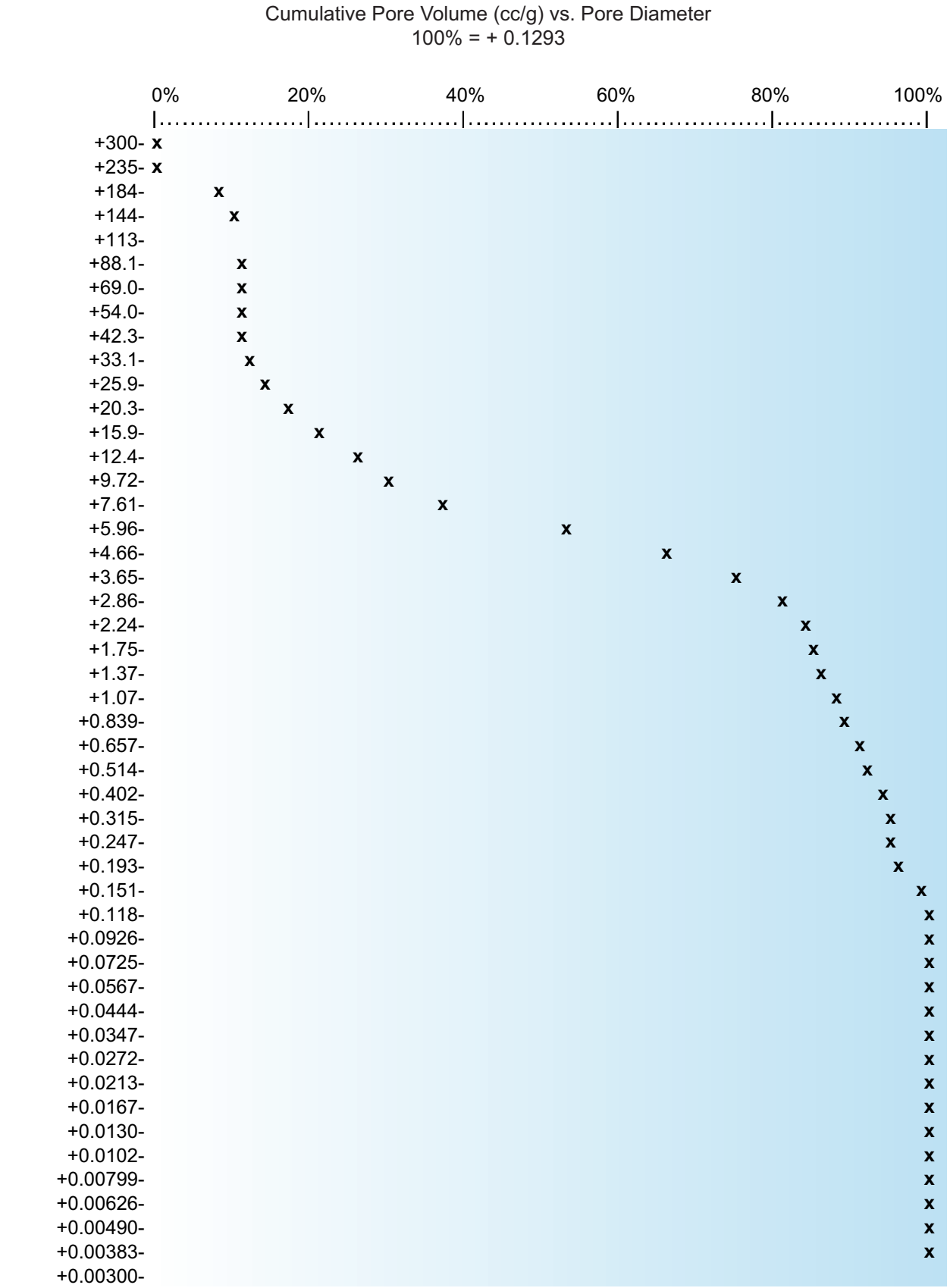
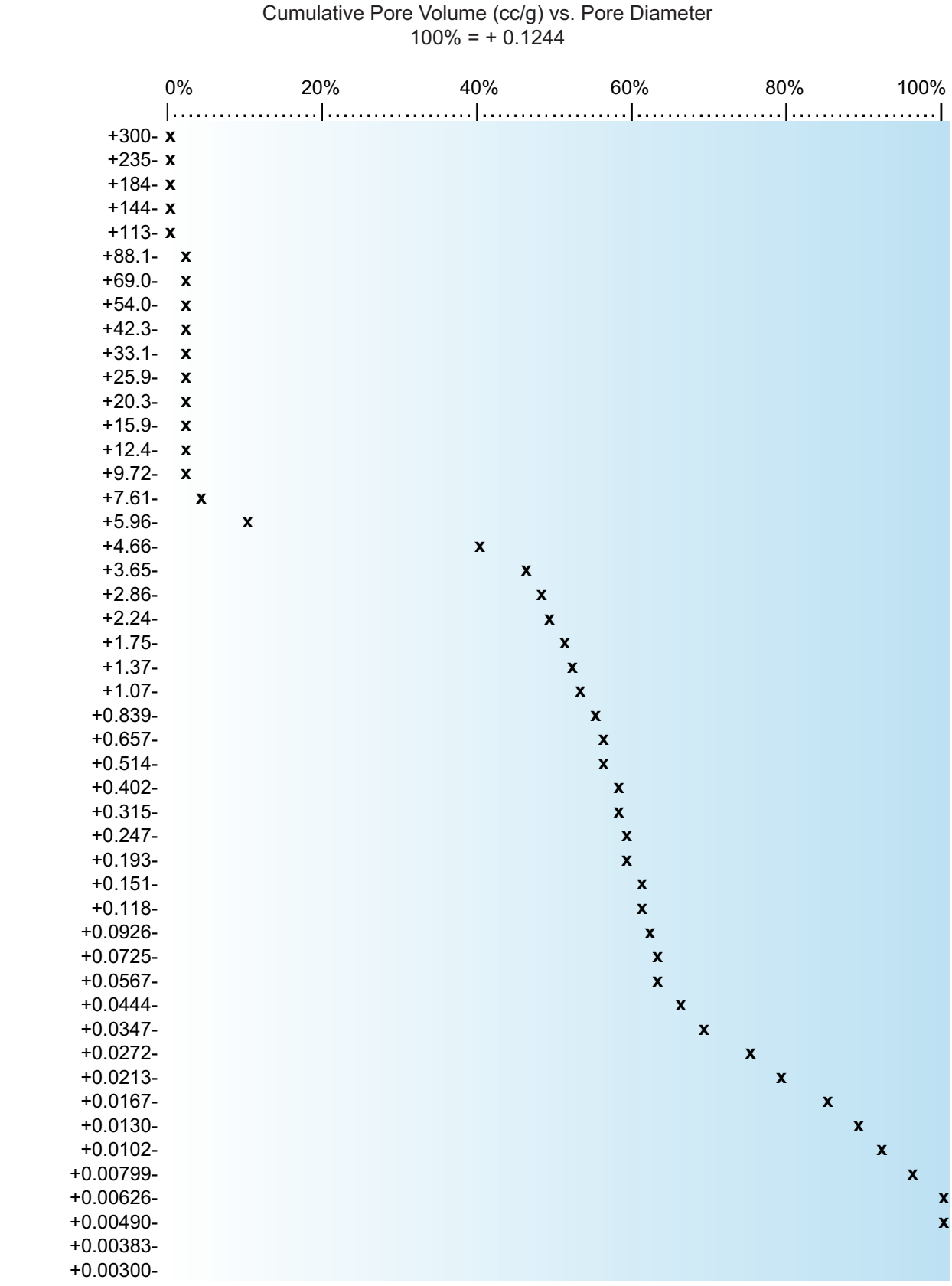


Figure 3.13 Porosimetry - UCAR® Grade ATJ



Coefficient of Static Friction

In certain applications, graphite is used in contact with other surfaces in such a manner that the static friction between the two surfaces becomes an important consideration. The actual degree of this friction is expressed by the coefficient of static friction which is defined by:

$$\mu = \frac{F}{P}$$

where F = the frictional force required to put one body in motion over another, P = the normal force with which the two bodies are pressed together, and μ = the coefficient of static friction. The value of the coefficient is dependent upon: (1) the composition of the two surfaces in contact; (2) the presence of any surface contaminants or oxides; (3) the temperature existent before sliding occurs; (4) condition of the surrounding atmosphere; and (5) the load on the surfaces.

In Table 3.9, average values of coefficients of static friction are given for monolithic graphite (UCAR® grade CS) in contact with a number of surfaces. The values presented are indicative of the range of frictional coefficients for graphite. Generally, the finer the grain size, the lower the coefficient. As temperatures increase, the coefficient also increases. Graphite to graphite static friction measurements on coarse grained electrode graphite run between 0.85 and 0.15 depending on the machined finish.

Table 3.9 Static Values of Coefficients of Friction of UCAR® Grade CS

Stationary Material	Sliding Material	Coefficient of Friction	Contact Surface Temp., °C	Atmosphere
Graphite (UCAR® CS)	Graphite (UCAR® CS)	0.15	25	Room Air
Graphite® (UCAR® CS)	18-8 Stainless Steel	0.35	204	Room Air
Graphite (UCAR® CS)	Cold-Rolled Steel	0.24	204	Carbon Dioxide

The important characteristics of graphite pipe slides are the following:

- Low Friction — Allow uniform thermal expansion in all directions
- Corrosion Resistance — Bearing surfaces that are permanently lubricated
- Long Life, Durable — Maintenance-free and lower costs
- Economical — Lower installation costs
- Temperature Stability — 1000°F air atmosphere; 3000°F inert atmosphere
- Thermal Stability — No cold flow deformation

Machinability

Graphite is machined with ease into almost any conceivable shape and contour. As an index of machinability, graphite is usually comparable to hardwood or aluminum. Graphite can be sawed, turned, milled, bored, drilled, planed, shaped, ground, tapped, reamed, honed and lapped. Tools will not seize or bind, and no buffs are left. Because of inherent characteristics, such operations as knurling and thread rolling cannot be performed.

Standard machining equipment generally used in woodworking and metal machining is easily adapted to graphite machining with only slight modification. Graphite may be cut rapidly, using cutting speeds employed with hardwood; cutting speeds are not critical and surface rates as high as 3000 ft/min have been successfully used. Generally, the limiting factors are the built-in range of the machine and tool and the ability to capture the dust and chips. Feeds and speeds normally associated with machining metals can be dramatically increased in machining graphite. It is not uncommon to run machinery at its rated capacity feed and speed in machining graphite.

Common, ordinary tooling can be used. The choice of economic tooling will depend on such factors as the number of pieces to be machined, tolerances, and surface finish. The most common and economical material is hardened tool steel. It is followed in cost by hardened high-speed steel, the “Stellite” alloy group (solid or tipped), tungsten carbide and diamond tools. Tungsten carbide and diamond are used in situations where long machining runs and minimum tool wear is desirable, and for the hard carbon and hard graphite such as the isomolded superfine and ultrafine grades.

With regard to tool design for machining graphite, the cutting and clearance angles agree closely with woodworking practice. A top-rake angle is of no particular advantage.

Depth of cut and feed rates are almost infinitely variable and are, generally, determined by tolerance, surface finish, and sturdiness of the machine tool. Feed rates may be as high as one-half inch per revolution, depending on the depth of cut. On the other hand, depths of cut may range up to two inches depending on feed rates. In general, it may be said that feed rates may vary over the entire range built into standard metal shaping or woodworking equipment; the depth of cut will depend upon the feed rate and the strength of the particular machine.

Since graphite is easily machinable, the use of lubricating oils to facilitate machining and reduce tool wear is not required. It is recommended that graphite be machined dry, since oils, greases, etc., can act as contaminants to the graphite.

Dust from synthetic graphite contains no significant amount of extraneous material and is classified as a nuisance dust by the American Conference of Governmental Industrial Hygenists (ACGIH). Because of the black color, it is more noticeable in the air and on the skin than most other industrial dusts. A threshold limit value (TLV) of 10 mg/m^3 has been established by the ACGIH. Prolonged or repeated breathing of dust can result in pneumoconiosis. For nuisance dust concentrations above the TLV, use a NIOSH-approved respirator.

Cold creams or soaps containing cold cream seem to be more effective in removing graphite from the skin than are the abrasive cleansers and soaps, and they are certainly gentler to the skin.

In the operation of graphite machining facilities, the following steps should be taken to insure economical and safe machining:

1 .Wear adequate eye protection.

2. Provide adequate dust collection. This may vary from the use of common shop vacuum canisters from single machine collection to highly sophisticated central vacuum systems for a series of machines. Graphite chips and dust should be collected as near the generating source as practical. This generally is accomplished through collection guides and cups directed at vacuum collection. Generation of chips are less of a nuisance and will generally not become airborne as will graphite dust or fines. Thus, machining feeds and speeds will dictate the type of collection needed. Good housekeeping practices are always recommended.
   Where dust collection systems are not available or not adequate to facilitate dust collection, graphite has been machined using a fine water mist to minimize airborne dust. This procedure has been effective on short-run or temporary machining situations. The graphite should be adequately dried after any wet machining.

3. Suitable hoods and guards prevent dust from getting into slides, gears, and feed screws. Since graphite acts as a lubricant, it generally will create little, if any, wear of machine parts. However, normal precautions in preventing excessive dust accumulations should be used.

4. All electrical equipment should be totally enclosed. Graphite dust is electrically conductive. Care should be taken to properly ground electrical equipment used in the machining area. Highly sensitive NC and CNC control systems should utilize positive pressure or vacuum systems to keep sensitive digital control panels dust-free.

Surface Finish
The surface of machined coarse grain graphite is made up in part of voids or valleys related to fundamental composition and plateaus carrying the tool marks where the peaks have been removed. A surface analyzer which measures vertical variations in microinches is meaningless as an expression of machining smoothness since it combines a variable basic stock factor along with a true machined surface factor. For the same reason, plus the fact that a graphite surface is not reflective (other than in a few exceptional cases), an optical method of surface flatness measurement is also not applicable.

It has been found practical to specify machined surface smoothness by another surface roughness method. This is a comparative method, made by vision and feel, in which the machined surface is judged against standard metal scales of roughness. Such scales are manufactured by a number of suppliers and are divided into gradations of average roughness expressed in terms of microinches. Bearing in mind the nature of the base graphite in terms of its inherent porosity, it is not difficult to distinguish by sight and feel the characteristic machine cut.
Different machining operations from rough sawing to lapping obviously create extremes in roughness. For any one operation, a range will be obtained, dependent on tool shape, feed, depth of cut, and sturdiness of machine. Under normal, good shop practice and tool maintenance, it is possible to predict the expectancy of the range to be realized in machined surface roughness.

Tolerances
It is not unusual to machine small graphite parts to tolerances as close as 0.0002 inch or less. This may require higher strength stock, since graphite is more elastic than most commonly used metals and will spring more under tool pressure. An example where such small tolerances are required is in the machining of EDM electrodes.

Hardness

The hardness of a substance may be defined in various ways, depending on the method of testing that is used and the use to which the results are to be put. In general, hardness is the ability to resist indentation, abrasion, and plastic deformation.

Resistance to indentation under static loads is measured by the Brinnel, Rockwell (ASTM C748 & E18), Vickers, Tukon, and Knoop methods, Rebound hardness or energy absorption under impact loads is measured by the Shore Scleroscope (ASTM C886). Resistance to scratching is measured by the Moh scratch hardness scale. The results of these methods are not translatable to one another and, to compare results, measurements must be made by the same apparatus.

The Rockwell with its fifteen scales (A to V) is in common use. Carbon and graphite are usually on the scale H (hardest, 1/8" ball, 60 kg load), L (1/4" ball, 60 kg load), or R (softest, 1/2" ball, 60 kg load). The Rockwell scales are so constructed that no linear correlation exists between the scales. Formulae have been proposed[12] which correlate measurements of hardness (N) using the 1/2", 1/4", and 1/8" balls, irrespective of the load, L (kg). The load L and the hardness N (irrespective of scale) are converted using the formulae in Table 3.10 to a Linear Scale Hardness, LSH, having the units of kg/m^2.

The Rockwell Hardness for typical carbon and graphites are shown in Table 3.1, and LSH in Table 3.2.

Table 3.10 Conversion Formula for Rockwell and Linear Scale Hardness

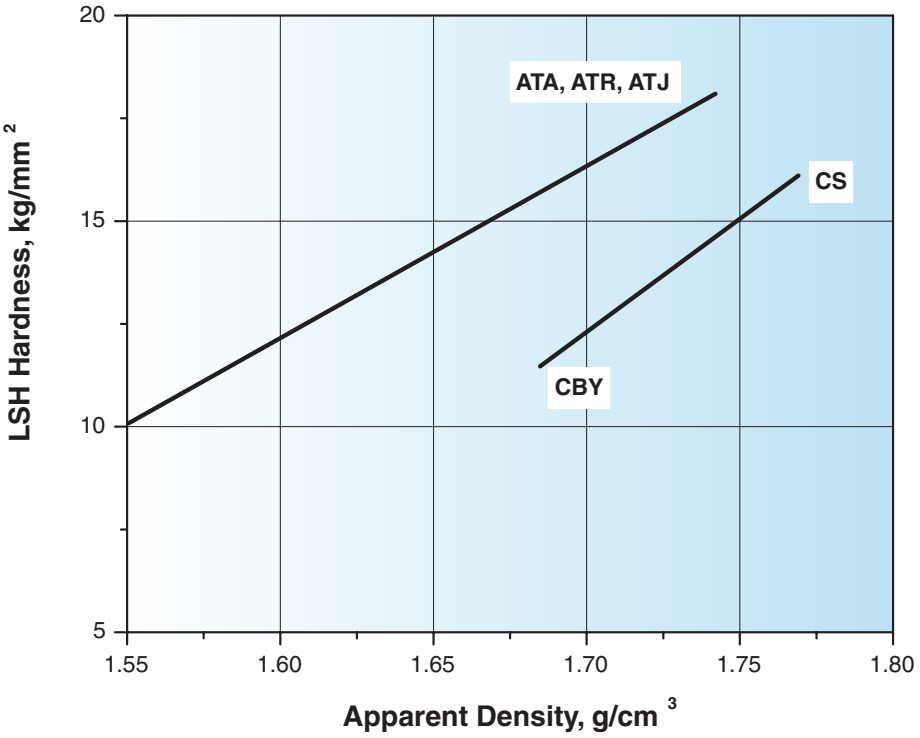
Ball Diameter, inches	Linear Scale Hardness
1/2	12.7(L)/(130 - N)
1/4	25.5 (L-4)/(130 - N)
1/8	52.8(L-8)/(130 - N)

L = Load, kg
N = Rockwell Hardness

Variation of Hardness with Other Properties

Within any one piece of stock, the hardness is always linearly dependent upon the apparent density. This correlation also is sometimes evident between different pieces of the same type and sometimes shows up between different types of stock. A graph illustrating this relationship is shown in Figure 3.14.

Figure 3.14 Static Values of Coefficients of Friction of UCAR® Grades



All properties which are significantly correlated with density will almost invariably be correlated with hardness as well. These include specific resistance, flexural strength, machinability, and crushing strength. Graphs illustrating the relationship between hardness and flexural strength and hardness and machinability are shown in Figures 3.15 and 3.16. Table 3.11 shows the correlation of hardness with resistivity, and thermal conductivity, of UCAR® grade ATJ.

Table 3.11 Resistivity, Hardness, and Thermal Conductivity for Grade ATJ[15]

Resistivity, μΩm	9	10.4	12.8
Hardness, Rockwell H	30	38	50
Thermal Conductivity, W/mK			
Room Temperature	131	119	102
300°C	102	93	82
600°C	80	71	63
1050°C	47	42	39
1325°C	40	38	4

Figure 3.15 Mean Flexural Strength vs. Linear Scale Hardness

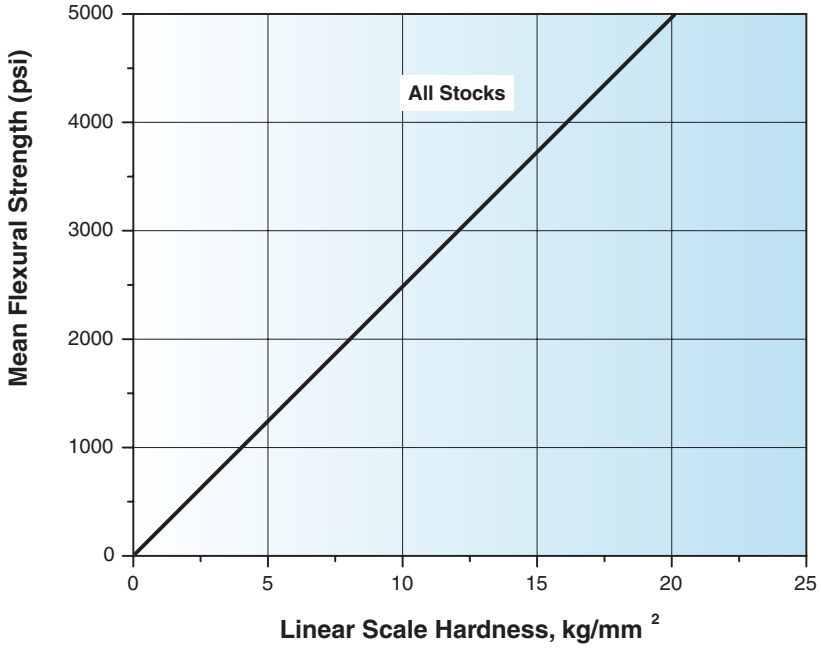
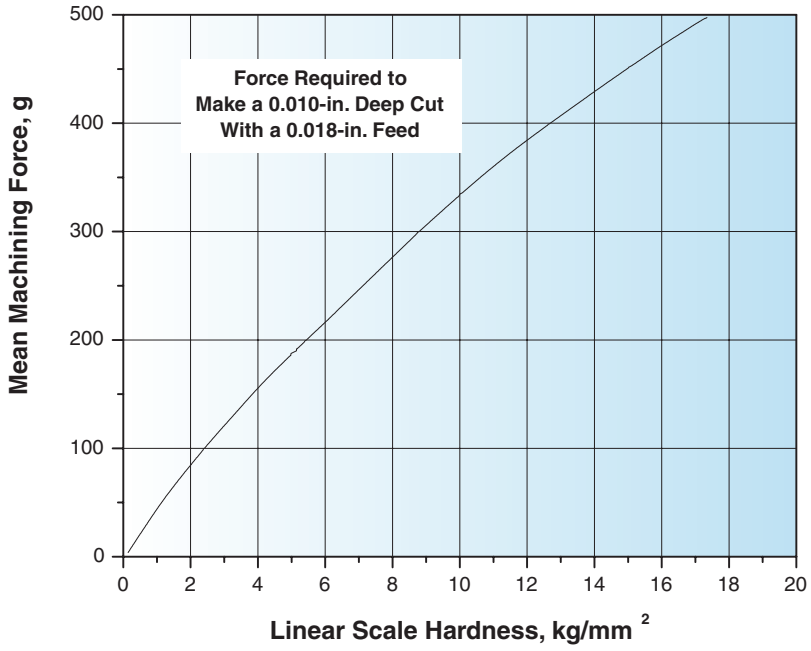


Figure 3.16 Mean Machining Force vs. Linear Scale Hardness

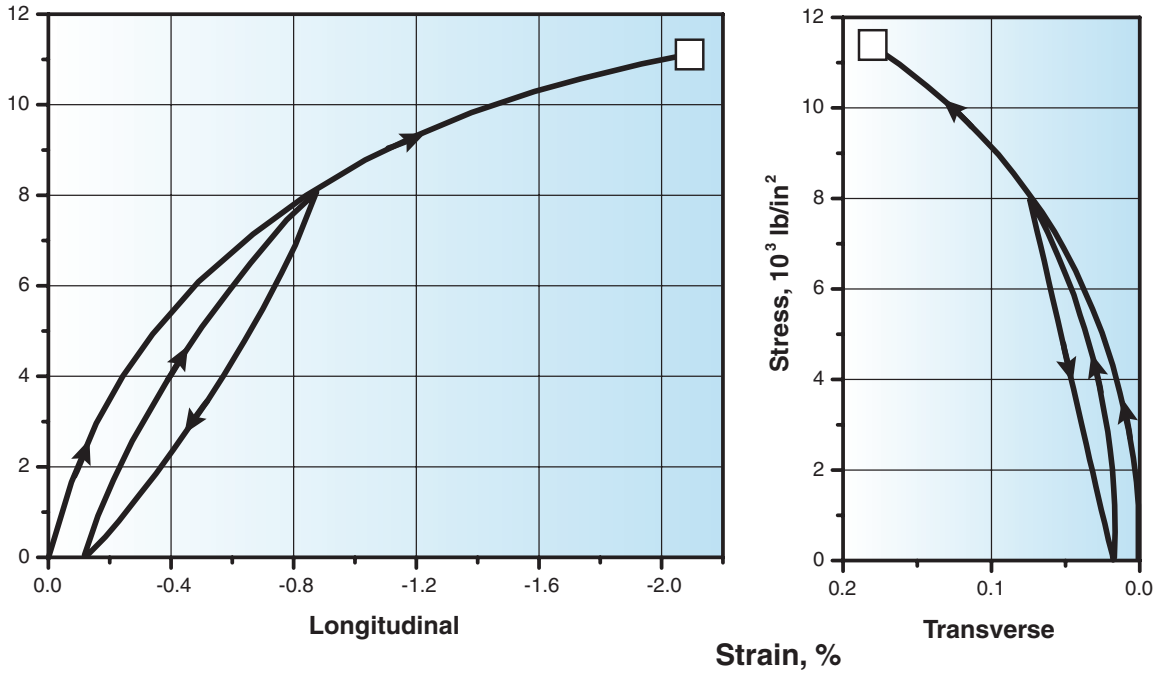


Stress-Strain Relationships

The stress-strain relationships of materials may be depicted graphically by plots of stress (load per unit area) versus strain (change in length per unit length). For elastic materials, the stress-strain curves are linear and reversible when the stress is cycled. The stress-strain curves of standard elastic materials do not deviate from linearity until the yield strength is reached and plastic deformation occurs; plastic deformation is defined as deformation which is not recovered when the stress is reduced to zero. The stress-strain relationships of polycrystalline graphites are different than those of standard elastic materials; the stress-strain curves of polycrystalline graphite are not linear, plastic deformation may be introduced at very low values of stress, and the stress-strain behavior differs in tension and compression[13].

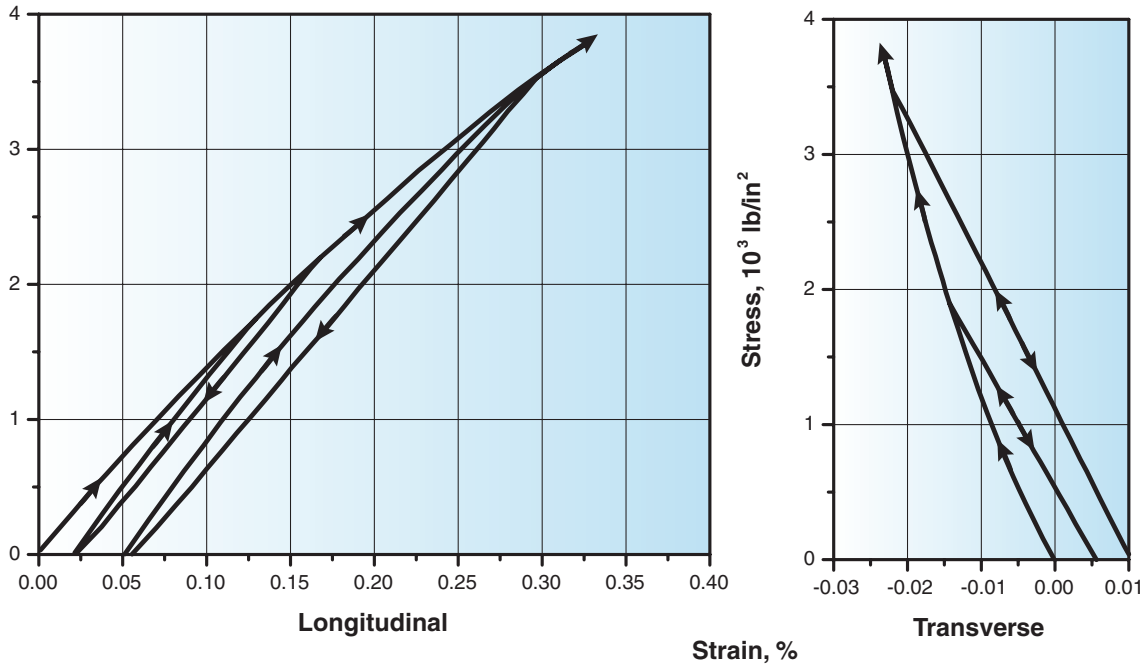
Representative of the manufactured polycrystalline graphites, some stress-strain curves for UCAR® grade ATJ stressed in compression at room temperature, are shown in Figure 3.17. The two plots in Figure 3.17 correspond to strains in two different directions: the longitudinal strain is the strain in the direction parallel to the direction of the applied stress and the transverse strain is the strain in a direction perpendicular to the applied stress. The compressive stress was applied in the with-grain direction, and the transverse strain is the strain in the across-grain direction. The stress-strain curves which result from an increasing stress are nonlinear. After the applied stress is reduced to zero, the graphite retains residual strains, and new curves are obtained when the stress is reapplied. The stress-strain curves were obtained by cyclically increasing and decreasing the applied stress. This procedure led to the observed hysteresis loops and showed that the residual strain in each direction increased with the applied stress. The ratio of the transverse strain to the longitudinal strain (Poisson's ratio) of the envelope curves in Figure 3.17 is constant over the entire range of stress.

Figure 3.17 Longitudinal and Transverse Stress-Strain Curves for UCAR® Grade ATJ With-the-Grain. Compression at Room Temperature



Stress-strain curves for UCAR® grade ATJ stressed in tension at room temperature are shown in Figure 3.18. These curves also show the longitudinal strain in the with-the-grain direction and transverse strain in the against-the-grain direction. The stress-strain curves for the longitudinal strains are qualitatively similar in both tension and compression. However, the stress-strain curve for the transverse strain in tension is quite different from its counterpart in compression. The residual strains introduced by an applied tensile stress are positive in all directions, i.e., the dimensions of the graphite are increased in all directions after a tensile stress has been applied and removed.

Figure 3.18 Longitudinal and Transverse Stress-Strain Curves for UCAR® Grade ATJ, With-the-Grain. Tension Test at Room Temperature



Unlike the stress-strain curves of polycrystalline graphites, the stress-strain curves at room temperature for tensile or compressive stresses applied parallel to the major axes of a good graphite crystal are linear and reversible. However, the stress-strain curves for the graphite crystal which are obtained by shear parallel to the basal planes are nonlinear and are qualitatively similar to the stress-strain curves of polycrystalline graphites in tension and compression.

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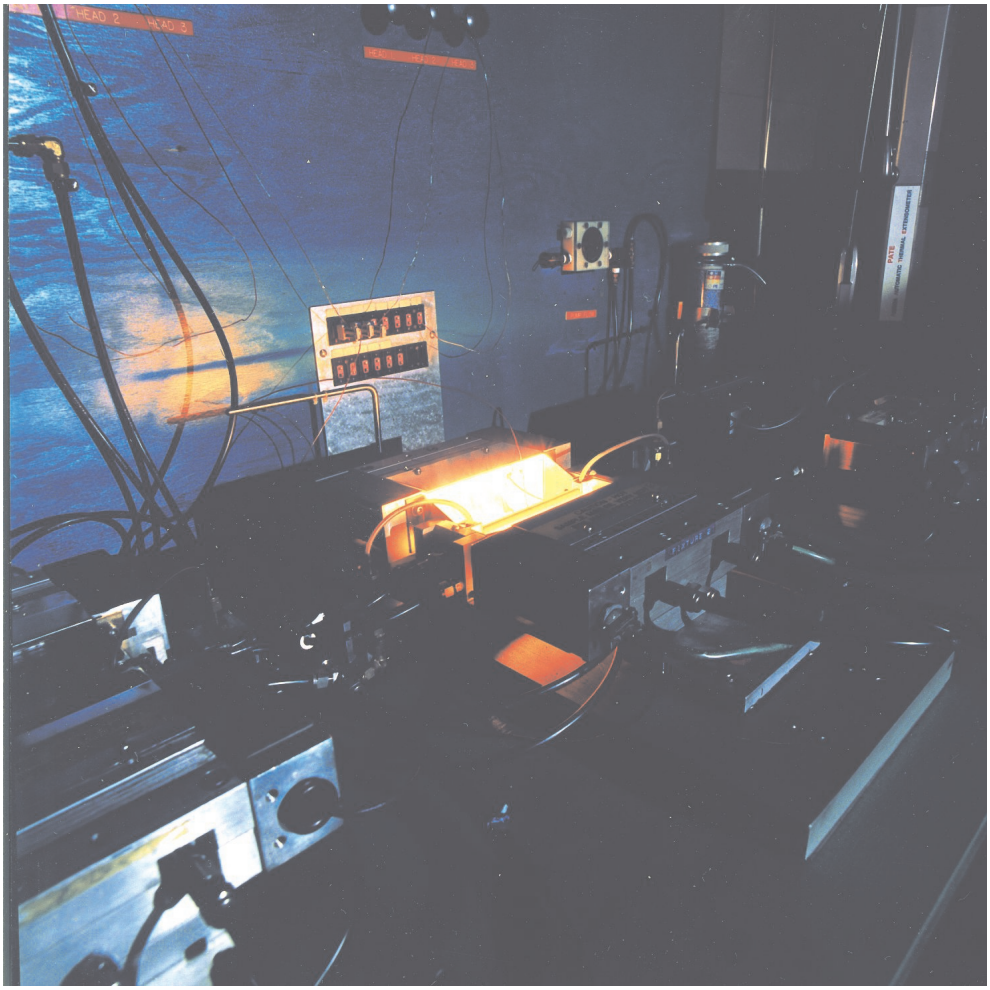
Chapter Four

THERMAL PROPERTIES

Industrial Graphite Engineering Handbook



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Thermal Properties

Thermal Conductivity

The thermal conductivity of a material is defined as the quantity of heat transferred by that material per unit area per unit temperature gradient. This is written:

$$K = \frac{WL}{A\Delta T}$$

where K is the thermal conductivity, W is the heat transferred per unit time, A is the area at right angles to the direction of heat flow, L is the length of sample along which the heat flows, and ΔT is the temperature differential along that length. Usually, the laboratory units of thermal conductivity are cal-cm/sec cm^2K or watt/mK and the engineering units are Btu-ft/hr $ft^2°F$. The conversion factors in Table 4.1 apply.

Measurements between room temperature and 1000°C are typically made using a laser flash thermal diffusivity method such as ASTM C714.

Table 4.1 Thermal Conductivity Conversion Factors

	cal-cm/sec cm^2K	W/mK	Btu-ft/hr $ft^2°F$
cal-cm/sec cm^2K	1	418.6	242
W/mK	0.002389	1	0.578
Btu-ft/hr $ft^2°F$	0.004135	1.73	1

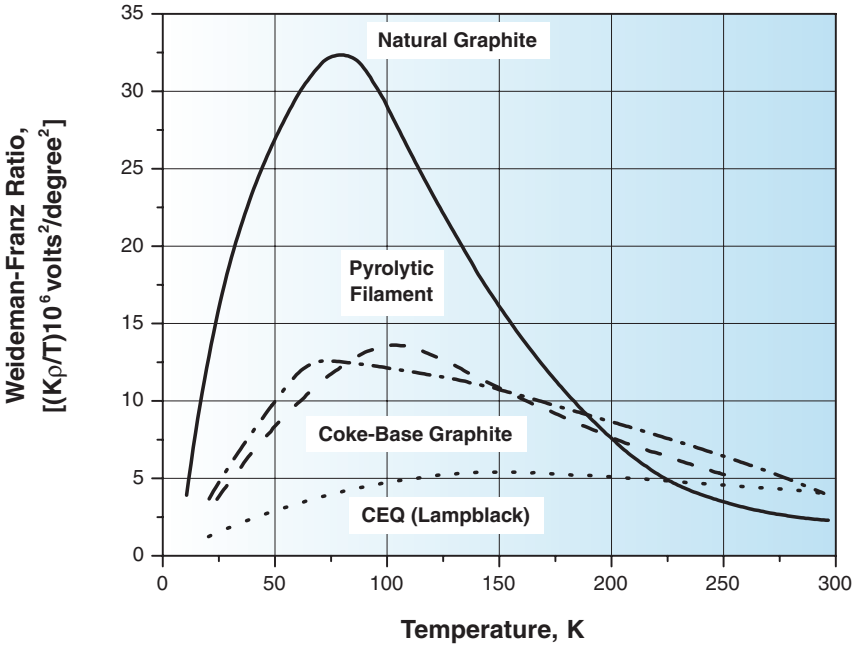
Manufactured graphite has an unusually high thermal conductivity over a considerable temperature range near room temperature. With-the-grain, it is comparable to aluminum, and against-the-grain, it is comparable to brass. Unlike metals, however, graphite behaves as a dielectric solid in that the principal mechanism for heat transfer is lattice vibrations or phonon waves. The electronic component so important with metals is much less than one percent at ordinary temperatures with graphite. Table 4.2 shows typical thermal conductivities of various materials.

Table 4.2 Typical Thermal Conductivities Measured At Room Temperature

Material	W/mK
Aluminum	210
Brick	0.6
Bronze (High Copper Content)	65
Carbon	4-30
Copper (Electrolytic)	350
Firebrick	0.1
Graphite	70-200
Iron	75
Molybdenum	125
Nickel	90
Silicon Carbide	15
Silica Glass	2
Silver	420
Tungsten	165

The Weidemann-Franz ratio is useful as a means of relating the thermal conductivity with the more easily determined electrical conductivity for metals. Figure 4.1 shows the measurements of this ratio for a number of graphites made by several investigators[2,17]. It is apparent that the W-F ratio is not constant with graphite.

Figure 4.1 Weidemann-Franz Ratio vs. Temperature for Typical Graphites



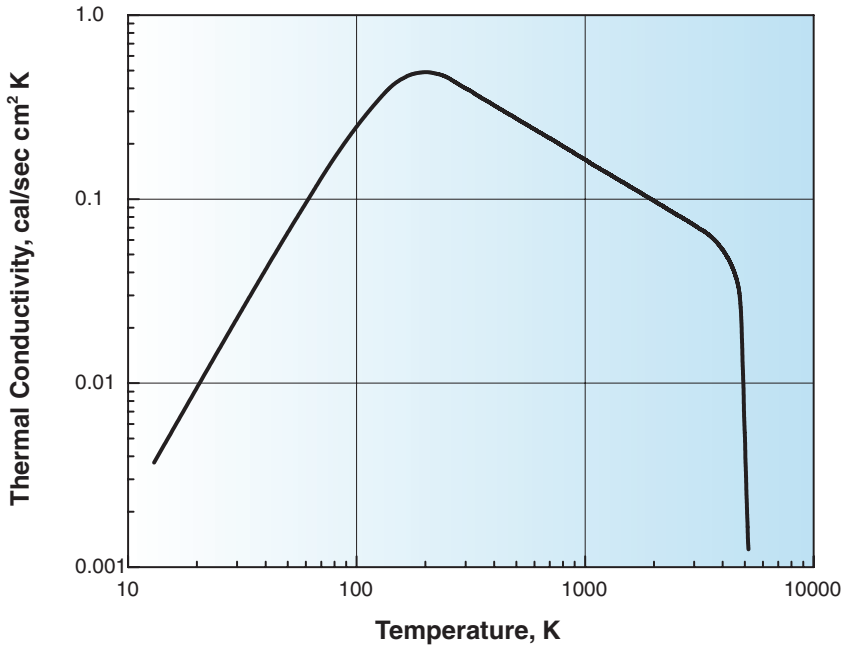
The W-F ratio for graphite is several hundred times the free-electron value. In general, samples that have the highest electrical conductivities show the greatest variations from the free-electron value. Additional evidence for phonon conductivity is that by the addition of the acceptor impurity, boron, to graphite, the electrical properties can be varied over a wide range without measurable effect on the thermal conductivity[1].

The general features of the thermal conductivity variation with temperature are shown in Figure 4.2. This is a composite curve drawn for illustration and does not represent an actual sample. The average crystallite size can be computed roughly from the position of the maximum conductivity using the relationship:

$$K = (1/3\ C_p) \cdot V_g \cdot \lambda$$

where C_p is the specific heat, V_g is an average group velocity, and λ is the average phonon wavelength. At the maximum, λ approximates the average crystallite diameter. Such calculation can be roughly confirmed in the crystallite size range where X-ray techniques can be applied.

Figure 4.2 General Nature of the Temperature Dependence of the Thermal Conductivity of Graphite



At temperatures below the maximum conductivity, where the mean-free path is limited by the crystallite size, the thermal conductivity of many manufactured graphites has a greater temperature dependence than does the specific heat[2,3]. The temperature dependence of the thermal conductivity in manufactured graphite can be interpreted[4] by considering the interconnections between crystallites as having a T^3 variation, while the crystallites are considered to have a T^2 dependence. Although this model is over-simplified and has defects in its details[28], the amount of nongraphitic material necessary for its success is roughly in agreement with that found to be present by measurements of magnetic susceptibility.

Thermal conductivity variation with temperature has been measured by a number of investigators[5,7]. The general shape of the thermal conductivity curve with temperature is shown in Figure 4.3. Throughout this temperature range, above the maximum shown on Figure 4.2, the thermal conductivity falls off roughly as T^{-1}. As the vaporization temperature of graphite is approached, the thermal conductivity drops sharply[6,7]. Figure 4.4 shows the general shape of the thermal conductivity curve for the UCAR® isomolded grades ATJ, DM302, DM332, and DM362 to 1000ºC.

Figure 4.3 Thermal Conductivity - Percent of Room Temperature Value with Temperature Increase

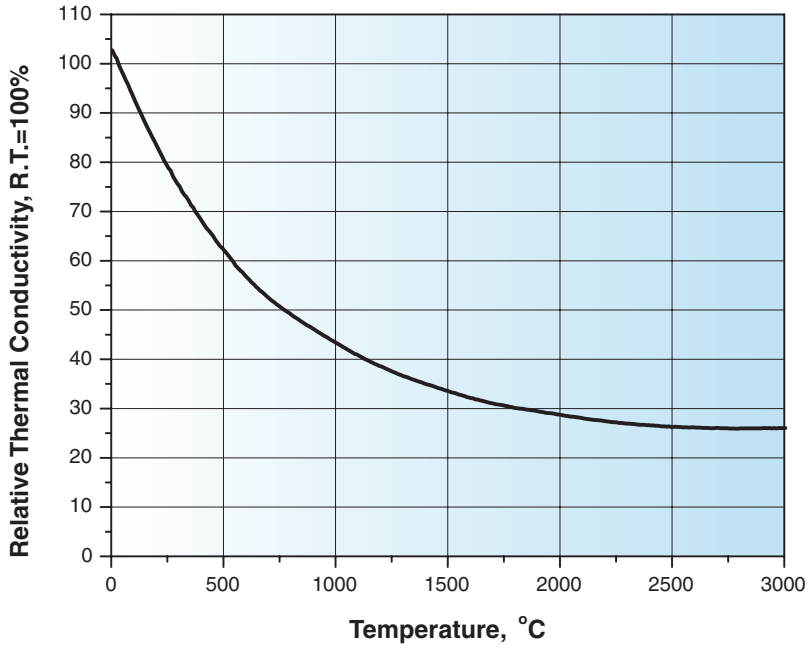
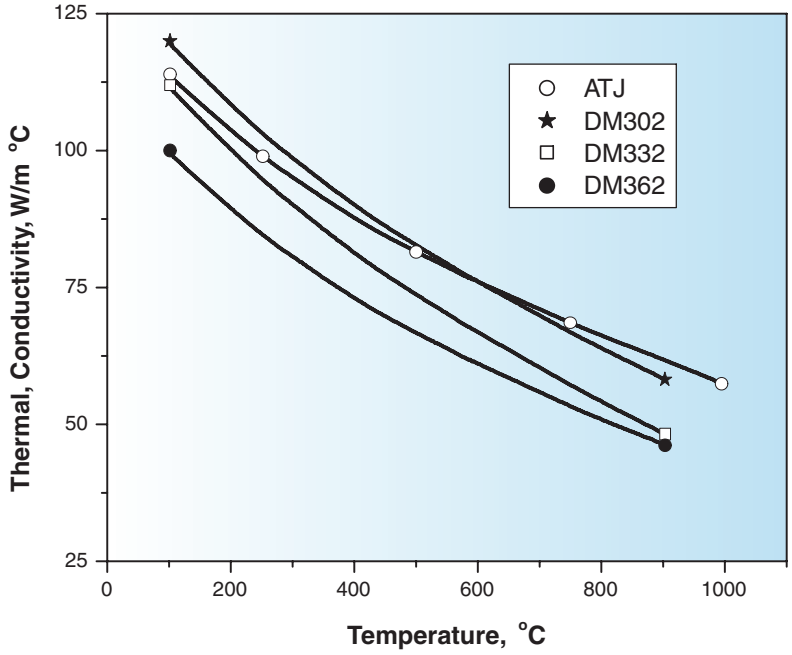


Figure 4.4 Thermal Conductivity of Isomolded UCAR® Grades ATJ, DM302, DM332, and DM362



Although a number of thermal conductivities at room temperature and above for manufactured graphite are reported in the literature, few of the references give sufficient information on the other properties of the graphite to characterize it properly. The thermal and electrical conductivities of a number of identified graphite grades are given in Table 4.3 where it is noted that within the accuracy of the measurements, the ratio between the two is fairly constant. In spite of the wide variation in stock size, mix particles, and pitch treatment represented by these graphites, the ratios of the two conductivities shown in the last column remain remarkably constant. A recalculation[8] of the data in Table 4.3 converting all the thermal conductivities to 25°C, yields a value of 0.00030 for the ratio of the thermal conductivity in cal-cm/sec cm^2 K to the electrical conductivity in mho/cm. This product has no relation to the Weidemann-Franz law, because the thermal conductivity is not electronic in nature.

Table 4.3 Thermal Conductivity (K) of Graphite (With-the-Grain)

Source	Diameter of Original Extruded Piece, inches	Maximum Size Particle in Mix, inches	Pitch Impregnated	Mean Temperature of K Determination, °C	Thermal Conductivity, K cal-cm/sec cm²K	Electrical Conductivity,σ mho/cm	(K/σ) x 10³
A	18	18	No	58.8	0.268	917	0.292
A	18	18	No	58.0	0.258	850	0.304
A	12	1/16	No	57.4	0.274	897	0.305
A	3	1/32	No	54.0	0.324	1098	0.295
A	3	1/32	Yes	47.3	0.428	1508	0.284
A	4	1/32	No	59.3	0.257	894	0.287
A	4	1/32	No	50.8	0.382	1337	0.286
A	2	Flour Only	No	45.7	0.357	1209	0.295
A	2	Flour Only	Yes	46.2	0.37	1269	0.292
B	18	1/8	No	53.3	0.341	1185	0.288
B	12	1/16	No	52.7	0.355	1264	0.281
B	4	1/32	No	51.7	0.364	1241	0.293
B	2	Flour Only	No	46.1	0.400	1462	0.274
B	2	Flour Only	Yes	42.0	0.510	1832	0.278
C	12	1/16	No	56.8	0.353	1236	0.286
C	4	1/32	No	49.5	0.415	1453	0.286
C	2	Flour Only	No	45.1	0.397	1385	0.287
C	2	Flour Only	Yes	44.9	0.375	1205	0.311
D	2	Flour Only	No	42.6	0.419	1462	0.286
D	2	Flour Only	Yes	46.9	0.409	1393	0.294
E	2	Flour Only	Five Times	40.1	0.451	1529	0.295

The anisotropy of the thermal and electrical conductivities has also been reported by others[9]. Some general measurements of thermal conductivity so reported are given in Table 4.4, but the accurate determination of the constancy of the ratio of thermal and electrical conductivities was accomplished by measuring both properties on a number of individual samples of widely varying conductivity. The ratio of conductivities so determined is 0.00032 when both are measured at room temperature. The ratio is general enough in validity to justify the use of the relationship in calculating the room temperature thermal conductivity of graphite from the room temperature electrical conductivity. The latter property is easy to measure accurately. If the relation:

$$K \text{ (cal-cm/sec cm}^2\text{K)} = 0.00031\ \sigma \text{ (mho/cm)}$$

is used for measurements at 25°C, the thermal conductivity so calculated will be accurate within ±5 percent.

The thermal conductivity data for several typical carbon and graphite grades is shown in Tables 4.5 and 4.6.

Table 4.4 Thermal Conductivity of Reactor Graphite Measured at Room Temperature

	Thermal Conductivity, Btu-ft/hr ft²°F	
Coke Source	Parallel to Grain	Perpendicular to Grain
B	0.56	0.29
C	0.4	0.3
D	0.44	0.32

Table 4.5 Typical Room Temperature Properties of UCAR® Grades, English Units

Group	UCAR® Grade	Largest Cross Section inches	Grain Size(1) inches	CTE(2) (RT-212°F) 10^{-6}/°F		Thermal Conductivity Btu • ft / hr • ft² • °F	
				WG	AG	WG	AG
Extruded Medium Grain							
	CBY	24	0.03	1.5	1.9	85	75
	CS	30	0.03	1.5	2.0	90	85
	SLWX	24	0.03	2.0	2.2	100	85
	CHV	24 x 48	0.06	0.6	1.3	80	60
Molded Medium Grain							
	PGW	52	0.03	---	2.0	---	75
	PGX	73	0.03	1.2	1.4	72	60
Extruded Fine Grain							
	FGE	8.5 x 8.5	0.006	1.0	2.3	100	59
	FGEX	8.5 x 8.5	0.006	1.2	2.2	88	59
Isomolded Fine Grain							
	ATJ	24	0.001	1.4		68	
	CGW	16 x 16	0.001	1.4		71	
Isomolded Superfine Grain							
	DM306	8 x 16	0.0002	3.1		12	
	DM300	8 x 16	0.0002	2.5		49	
	DM301	8 x 16	0.0002	2.2		60	
	DM302	8 x 16	0.0002	2.4		70	
	DM303	8 x 16	0.0002	2.6		50	
	DM342	8 x 16	0.0002	2.3		64	
	DM332	24	0.0002	2.0		64	
Isomolded Ultrafine Grain							
	DB362	6 x 12	0.00008	2.8		47	

(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.
(2) CTE = Coefficient of Thermal Expansion

Table 4.6 Typical Room Temperature Properties of UCAR® Grades, SI Units

Group	UCAR® Grade	Largest Cross Section mm	Grain Size(1) mm	CTE(2) (RT-100°C) 10^{-6}/°C		Thermal Conductivity W/mK	
				WG	AG	WG	AG
Extruded Medium Grain							
	CBY	600	0.8	2.5	3.4	145	120
	CS	600	0.8	2.5	3.6	175	145
	SLWX	600	0.8	3.4	3.9	170	150
	CHV	600 x 1200	1.7	1.5	2.6	140	105
Molded Medium Grain							
	PGW	1300	0.8	---	2.5	---	130
	PGX	1825	0.8	2.2	2.6	125	104
Extruded Fine Grain							
	FGE	215 x 215	0.150	1.8	4.1	170	100
	FGEX	215 x 215	0.150	2.1	4.0	150	100
Isomolded Fine Grain							
	ATJ	600	0.025		2.5	115	
	CGW	400 x 400	0.025		2.5	120	
Isomolded Superfine Grain							
	DM306	200 x 400	0.005		5.5	20	
	DM300	200 x 400	0.005		4.5	84	
	DM301	200 x 400	0.005		4.2	105	
	DM302	200 x 400	0.005		4.3	120	
	DM303	200 x 400	0.005		4.8	90	
	DM342	200 x 400	0.005		4.3	110	
	DM332	600	0.005		3.7	112	
Isomolded Ultrafine Grain							
	DM362	150 x 300	0.002		5.1	80	

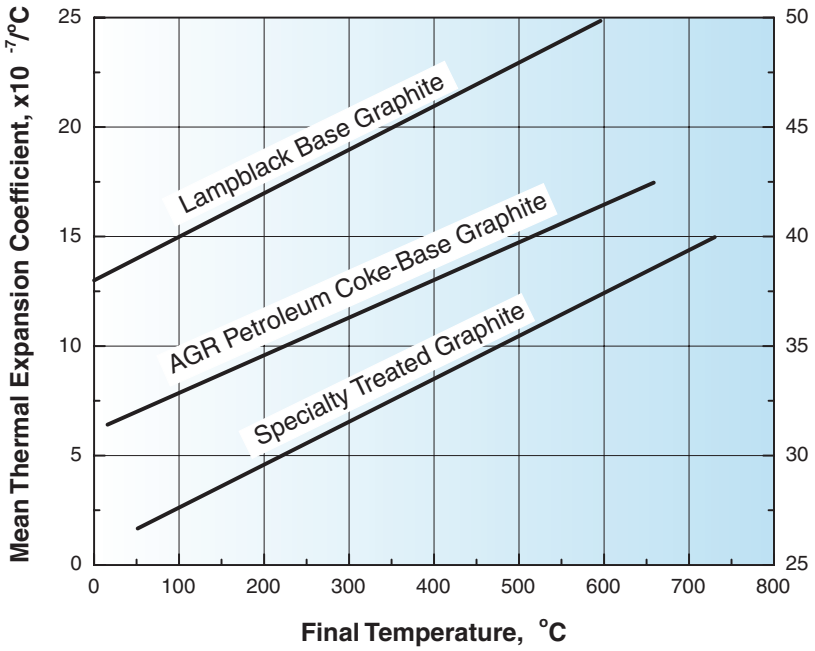
(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.
(2) CTE = Coefficient of Thermal Expansion

Thermal Expansion

The coefficient of thermal expansion of (CTE) a material is the increase in length per unit length per degree rise in temperature. For methodologies see ASTM E228 and References 110 and 111. Typical coefficients for some common graphites are listed in Tables 4.5 and 4.6.

The variation of the thermal expansion coefficient of graphite with temperature is most accurately measured by the interferometer method in the 0 to 600°C range[10,11]. The results of these interferometer mean-measurements for three very different types of graphite, between room temperature and the final temperature indicated, are shown in Figure 4.5.

Figure 4.5 Mean Thermal Expansion Coefficient vs. Temperature (Interferometer Method)

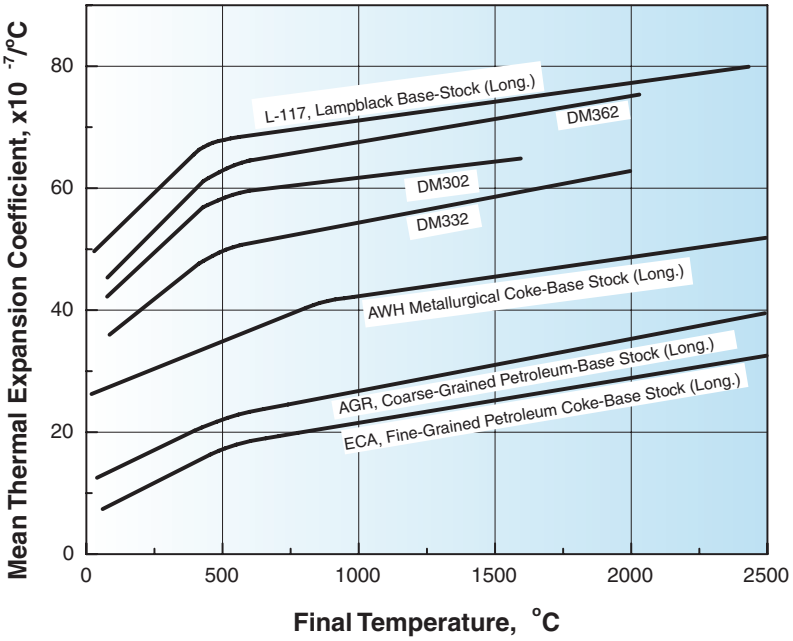


The points at the higher temperatures are more accurate since their calculation involves a wider temperature interval. They have been weighed more heavily than the low temperature points in drawing the curves. The slope of the curves is approximately the same for all three graphites, as well as many others tested.

Thermal expansion measurements at temperatures up to 2500ºC have been made by the optical dilatometer method on the UCAR® extruded graphites[12] L117, AWH, AGR, and ECA, and more recently on the UCAR® isomolded graphites DM302, DM332, and DB362. This was done by mounting 5 to 12 inches long samples in a 4 inch I.D. graphite resistance tube furnace. Length changes were determined by the telemicroscopic observation of pointed tungsten probes mounted 10 inches apart in the sample. An apparatus of considerable physical and thermal stability was required, giving measurements with a precision of +0.002 inches expansion and ±5°C.

Figure 4.6 shows the results measured parallel to the axis of the extruded grades. Small obvious irregularities of expansion occur in individual samples, probably as a result of irregular release of internal stresses. Nevertheless, the parallel nature of the curves is consistent with the idea that the change of the coefficient of thermal expansion with temperature is approximately the same for all graphites at high temperatures.

Figure 4.6 Mean Coefficient of Thermal Expansion of UCAR® Graphite vs. Temperature (Optical Dilatometer Method).



This leads to a simplification, for engineering purposes, in the calculation of high temperature thermal expansion. The mean linear coefficient of thermal expansion between room temperature and any final temperature can be obtained from the value for the temperature interval RT-100ºC by adding to it the numbers shown in Table 4.7. This method is valid for stock of any grain orientation. Using the table, the expansion curves of individual graphite samples can be computed to obtain good agreement with the measured curve.

The added factor indicated is to be combined with a coefficient determined in the temperature range RT-100ºC[45], to give the mean coefficient of thermal expansion over the range from RT to the final temperature of interest.

These factors have been compared with some very accurately measured coefficients of thermal expansion for graphite, independently determined and published[13], to check their accuracy. Up to the 1000ºC limit of these measurements, the agreement between actual and calculated values is very close. More recent studies[46] confirm the efficacy of combining CTE measured between RT and 100ºC with a factor from Table 4.7 to estimate CTE at temperatures up to 2500ºC.

Since graphite has a low coefficient of thermal expansion (CTE), manufactured parts being cast can be designed for an essentially finished size. Raw material selection, particle size, and impregnation will affect CTE.

Table 4.7 Factors for Calculation of Mean Coefficient of Thermal Expansion

Final Temperature, ºC	Added Factor, x 10^{-7}/ºC
100	0
200	2
300	4
400	6
500	7.7
600	9.2
700	10.4
800	11.4
900	12.3
1000	13.2
1500	17.2
2000	21.2
2500	25.2

The effect of raw material selection on thermal expansion is significant in that the lower the CTE of the calcined coke, the lower the CTE of the finished graphite. The addition of coarser particles to the base formula also helps decrease CTE.

The CTE of graphite, compared to other structural materials, is illustrated in Table 4.8.

Table 4.8 Thermal Expansion of Various Structural Materials

Material	x 10^{-6}/ºC
Copper	17
Carbon	4
Glass	0.5-10
Graphite	1-7
Iron	12
Porcelain	1.6-5.5
Molybdenum	5
Silicon Carbide	3.6
Silver	19
Tungsten	4.5

Thermal Shock Resistance

Resistance to thermal shock is a measure of the ability of a material to resist weakening or fracture when subjected to sudden heating or cooling. No standard tests exist that will accurately evaluate this property since shape is an important factor, as in the case of impact strength. Several practical tests are in use for specific applications and, in the case of graphite, temperature gradients and times on the order of those encountered in the striking of an arc are necessary, even for relative comparisons.

The exceptional thermal shock resistance of graphite arises from a unique combination of properties. The properties which appear to be necessary are high tensile strength, high thermal conductivity, low modulus of elasticity at failure, and low coefficient of thermal expansion. More sophisticated analyses take into account the temperature dependence of these properties, but for relative comparison of materials, simpler analyses may be used.

A test for the thermal shock resistance of refractory materials[18] consists of heating 2 inch diameter by 1/4 inch thick discs in a furnace, followed by sudden quenching in an air stream having a velocity of 250 feet per second past the specimen. The test determined the thermal shock resistance of only refractory oxides and carbides and expressed it as a parameter $Ks/\alpha E$ for these materials[19]. Table 4.9 shows a comparison of the value for graphite and the other refractory materials using this formula.

Table 4.9 Relative Evaluation of Thermal Shock Resistance

Order of Merit of Materials Evaluated in Thermal Shock	980°C	1100°C	1200°C	1320°C	$\frac{Ks}{\alpha E}$
UCAR® ATJ Graphite	(a)	(a)	(a)	(a)	3,080,000
Titanium Carbide	25	25	25	21	18,200
Beryllium Oxide	25	3			6,390
Zircon	1				2,345
Magnesium Oxide	1/2				641-1,840
Stabilized Zirconia	0				345

(a) Specimens at 1650°C have been quenched in water without damage. Spalling has been obtained by quenching in liquid sodium.

The table shows the marked superiority of graphite over ceramic materials in resistance to thermal shock. For a qualitative prediction of the resistance of materials to fracture by thermal shock, the material should have a high value of the parameter $Ks/\alpha E$ where:

K = Thermal Conductivity;
s = Tensile Strength;
α = Coefficient of Thermal Expansion; and
E = Modulus of Elasticity.

Within the carbon-graphite family of materials, the following are listed in descending order (i.e. best to worst) by their ability to resist thermal shock.

1. Graphite with large grain size is more resistant than fine grain graphite.
2. Semi-Graphite.
3. Coal-Base Carbon.
4. Coke-Base Carbon.
5. Lampblack-Base Carbon.

For any grade, a thin section is more resistant to thermal shock than a thick section. Carbon or graphite should be dried thoroughly to prevent internal steam explosions causing cracking. This is sometimes mistaken for thermal shock spalling.

Emissivity and Spectral Characteristics

Emissivity is a measure of the heat-radiating ability of a surface. It may be expressed as total emissivity or spectral emissivity. The total emissivity[14] of a surface of a radiator is the ratio of its radiant flux density (radiancy) to that of a black body at the same temperature. A representative emissivity value for graphite is 0.77[15], but is dependent upon the specific grade and surface finish. A true black body has an emissivity of one.

The spectral emissivity of the surface of a radiator is the ratio of its radiant flux density per unit wavelength interval (spectral radiancy) at that wavelength to the spectral radiancy of a black body at the same temperature. The spectral emissivity of graphite at 6500Å has been measured in connection with optical pyrometry applications. Values of 0.90 for dulled-surface graphite and 0.77 for polished graphite are typical[16]. The emissivity of graphite varies slightly with temperature, having a temperature coefficient of 1.9×10^{-5}/K. In the wavelength range 2000 to 7500Å , graphite emissivity is nearly constant.

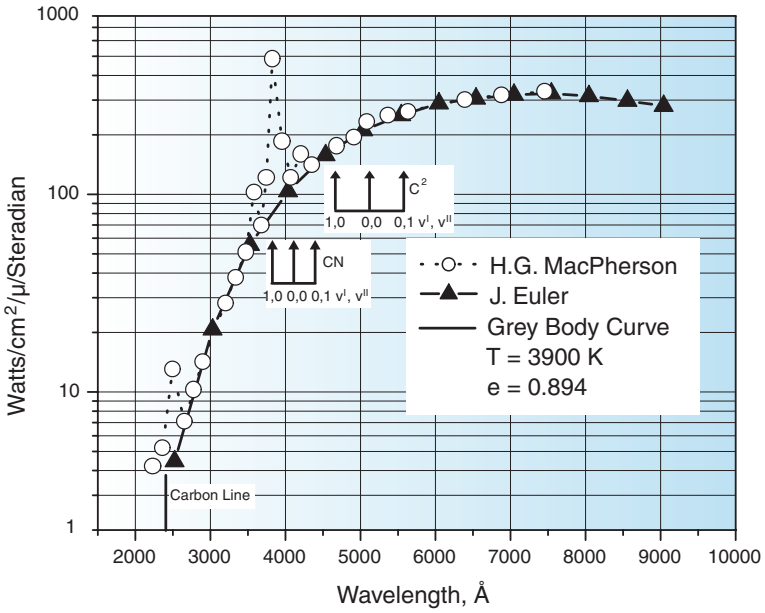
The crater of a carbon arc using a pure graphite anode is a useful high-temperature radiation standard. The brightness of the anode seen through an optical pyrometer (6530Å) matches that of a black body at 3790K (1948 I.T.S.) within the experimental error[15,20,21,22].

The crater radiation is also useful for spectral radiance calibration throughout the wavelength region 0.25 to 1.8µ. The measurements of MacPherson[15] and Euler[21,22] using a graphite anode agree within six percent above 3000Å and are shown plotted in Figure 4.7. Although different temperatures and emissivities were used to specify their data, for practical purposes, both sets of data may be described as the spectral radiance of a grey body at 3900K with an emissivity of 0.894. The resultant calculated curve (drawn in Figure 4.7) fits the observed data within six percent throughout the wavelength range 2500 to 18,000Å. (The spectral radiance using a carbon anode is 10 to 15 percent less than with a graphite anode.)

A value of 0.98 was obtained for the emissivity of the crater of an operating arc based on reflectance measurements[20]. Subsequent data[23] on the arc and on samples at temperatures in the range 500 to 2600K confirm the low reflectance of graphite and suggest emissivities in the range 0.96 to 0.98. Why reflectance measurements yield higher emissivity values than are obtained with an optical pyrometer is not understood.

A carbon arc operating in air produces a strong band system due to the CN molecule in the arc plasma and a C_2 system, as shown in Figure 4.7. A strong atomic line due to carbon occurs at 2478Å. Additional weaker bands due to CN, C_2, and C_3 molecules are observed[24-27].

Figure 4.7 Spectral Energy Distribution



In general terms, emissivity can be characterized in Table 4.10, as follows:

Table 4.10 Total Emissivity

Material	Total Emissivity
Carbon	0.60-0.80
Graphite	0.70-0.95
Lampblack	0.90-0.99
Silver	0.04

Specific Heat

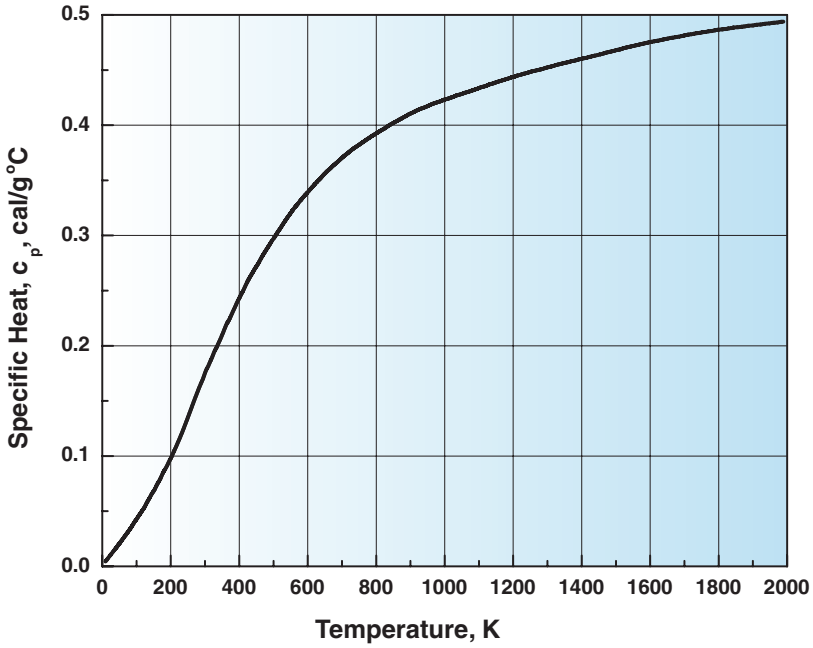
Specific heat is the quantity of heat required to raise the temperature of a unit mass of material one degree. The common units of measure are related as follows:

$$1 \frac{BTU}{lb\ ^\circ F} = 1 \frac{cal}{g^\circ C}$$

Except at temperatures near absolute zero, the values of the specific heat of all types of natural and manufactured graphites are essentially the same. From an extensive survey[30] of the measurements of the specific heat reported in the literature, it appears that the difference between different measurements of the same grade of graphite is as large as the difference between measurements of different grades. Below 20K, the specific heat of a graphite single crystal is slightly less than that of petroleum coke-base manufactured graphite; on the other hand, below 20K, lampblack-base manufactured graphite and unannealed pyrolytic graphite have slightly higher specific heats than petroleum coke-base manufactured graphite[30,31]. De Sorbo[32] found that the specific heat of Ceylon natural graphite is up to nine percent less than that of manufactured graphite in the temperature interval 120 to 300K. At higher temperatures, the difference between the values of all types of graphite is less than the experimental error.

A number of individual measurements[29] have been made of the specific heat at constant pressure of natural and manufactured graphites. Figure 4.8 represents the best fit of the values for manufactured graphite with proper consideration given to the accuracy of the individual measurements. Below 300K, the average curve essentially coincides with several individual measurements.

Figure 4.8 Specific Heat of Manufactured Graphite



The specific heat at constant pressure, c_p, may be written as the sum of several terms,

$c_p = c_v + c_w + c_e$ where:

c_v = specific heat at constant volume of the lattice vibrations;
c_w = difference between the total specific heat at constant pressure and at constant volume;
c_e = specific heat of the electrons at constant volume.

Table 4.11 gives the numerical values of c_p taken from the average curve for manufactured graphite and the corresponding values of Δc_p, c_w and c_v. The quantity Δc_p represents a probable error such that most of the reliable individual measurements fall within the range $c_p \pm \Delta c_p$. The table also extends the range of values to 3800K. The difference, c_w, has been calculated[28] from the density, coefficient of thermal expansion, and compressibility. The compressibility at high temperatures has to be roughly estimated from the room temperature value, so the values of c_w at higher temperatures may be in error by a factor of two or three. The values of the electronic specific heat, c_e, have been computed from formulas[29] based on McClure's calculation of the electronic band structure of graphite.

The data in Table 4.11 indicate that at about 2200K, the lattice specific heat c_v at constant volume, from $c_v = c_p - c_w - c_e$, becomes greater than the classical Dulong and Petit value, which for graphite is 0.496 cal/g°C. Values of c_v greater than 0.496 would indicate the existence of anharmonic atomic forces, but the possible errors in c_p and c_w are large enough that this point is not at all certain. The value of c_p increases sharply as the temperature is increased above 3200K. This result is based on the work of Rasor and McClelland[33] and, according to these authors, is due to the creation of vacancies at temperatures near the sublimation temperature.

Table 4.11 Specific Heat of Manufactured Graphite

Temperature, K	Specific Heat, c_p, cal/g°C	Δc_p	c_w	c_e
0	0	0	0	0
50	0.0101	0.0002	0.00002	0.00001
100	0.0335	0.0003	0.00014	0.00002
150	0.0643	0.0007	0.00027	0.00004
200	0.0995	0.001	0.00039	0.00005
250	0.1357	0.0013	0.00054	0.00007
300	0.1723	0.0017	0.00069	0.00009
350	0.209	0.003	0.0009	0.0001
400	0.245	0.004	0.001	0.0001
450	0.276	0.005	0.0012	0.0002
500	0.303	0.006	0.0014	0.0002
550	0.323	0.007	0.0016	0.0002
600	0.34	0.008	0.0017	0.0002
650	0.356	0.009	0.0019	0.0003
700	0.37	0.009	0.0021	0.0003
750	0.382	0.01	0.0023	0.0004
800	0.393	0.01	0.0025	0.0004
850	0.402	0.01	0.0027	0.0004
900	0.409	0.01	0.0029	0.0005
950	0.415	0.01	0.0031	0.0005
1000	0.421	0.011	0.0033	0.0006
1100	0.432	0.011	0.004	0.001
1200	0.443	0.011	0.004	0.001
1300	0.452	0.012	0.004	0.001
1400	0.46	0.012	0.005	0.001
1500	0.468	0.013	0.005	0.001
1600	0.474	0.014	0.005	0.002
1800	0.486	0.015	0.006	0.002
2000	0.496	0.016	0.007	0.002
2200	0.504	0.018	0.007	0.003
2400	0.511	0.02	0.007	0.004
2600	0.516	0.022	0.008	0.004
2800	0.521	0.024	0.008	0.005
3000	0.527	0.026	0.008	0.006
3200	0.536	0.03	0.008	0.006
3400	0.548	0.03	0.01	0.01
3600	0.58	0.04	0.01	0.01
3800	0.69	0.05	0.01	0.01

The average experimental values of c_p can be represented by simple polynomial functions of the absolute temperature T, with T in K to give c_p in cal/g°C.

Within the temperature interval 0K to 300K:

$$c_p = 7.2774 \times 10^{-12}T^4 - 8.5781 \times 10^{-9}T^3 + 3.6890 \times 10^{-6}T^2 + 4.4112 \times 10^{-5}T - 7.9099 \times 10^{-5}$$

Above 40K, the calculated values are within 2.2 percent of the average experimental values. This equation is not reliable below 40K

.

Within the temperature interval 300K to 3200K:

$$c_p = 0.44391 + 0.30795 \times 10^{-4}T - 0.61257 \times 10^{5}T^{-2} + 0.10795 \times 10^{8}T^{-3}.$$

This equation reproduces the experimental values within 1.5 percent over the entire interval.

These functions can be easily integrated to obtain the heat energy added or given off on changing the temperature of graphite. Rasor and McClelland[33] give a formula for the exponential increases in c_p due to the creation of vacancies at temperatures above 3200K.

Vapor Pressure

Vapor pressure is defined as the pressure exerted when a solid or liquid is in equilibrium with its own vapor. The vapor pressure is a function of the substance and of the temperature. Graphite is unique in exhibiting a vapor pressure that is negligible up to about 2500°C under normal conditions and, for that reason, the vapor pressure of graphite is not a limiting property in most applications. Even under high vacuum, it is practical to use graphite up to 2000°C. Other factors may have a dominant influence on weight loss, namely loss of carbon due to chemical reaction of graphite with container material or with residual gases, notably oxygen. These reactions will predominate unless the graphite is protected. Also, when graphite is heated to temperatures above 1000ºC, outgassing occurs, and in order of increasing temperatures, gases H_2O, CO, CO_2, and H_2 are released. Depending on the previous history of the graphite, these gases may vary from minute quantities to several hundred times the volume of the graphite. Finally, most graphites, unless purified, contain small amounts of ash, the constituents of which volatilize at various temperatures, causing a weight loss.

Standard techniques for vapor pressure determination are based on the classical Clapeyron equation:

$$\frac{dp}{dT} = \frac{\Delta H}{T\Delta V}$$

where:

p is the vapor pressure;
ΔH is the heat of vaporization;
ΔV is the increase in volume from solid or liquid to vapor; and
T is the absolute temperature.

When ΔH and ΔV are not constant, it is necessary that the deviations be known as functions of temperature in order to integrate this equation. In cases where several species are produced during vaporization, e.g., C, C_2, C_3.... it is further necessary that the above equation be applied separately to each constituent. Because the required information on the variation of ΔH and ΔV with temperature has not been determined experimentally outside the temperature range of 2500K to 3000K, the following relation is employed to obtain vapor pressures outside this range:

$$-R \log.p = \frac{\Delta F_\tau^\circ}{T} = fef\,(\text{gas}) - fef\,(\text{solid}) + \frac{\Delta H_o^\circ}{T}$$

where:

R is the Universal Gas Constant = 1.987 cal/°C/g mol;
ΔF_τ° is the free energy change at T K;
ΔH_o° is the heat of vaporization at 0 K; and

$fef = \frac{\Delta F_\tau^\circ - \Delta H_o^\circ}{T}$ is the free energy function, a quantity that is related to the entropy by the equation:

$$fef = S_\tau + \frac{1}{T}(H_\tau - H_o)$$

As with entropy, free-energy functions in liquid or solid phases are determined from heat capacity data, whereas in the gaseous phase, they are determined from known or assumed structures, vibrational frequencies, and multiplicities of ground and low-lying excited states of the various molecular species present.

Both monatomic and polyatomic species of carbon result from the vaporization of graphite; molecules of C through C_5 have been observed[34-36]. These species were identified by effusion studies and further characterized by spectroscopic measurements[37,38]. A theoretical treatment, based upon the assumption that the bonding can be described by molecular orbitals, predicts that the larger molecules become relatively more abundant with temperature and that the resulting molecules are linear chains (ring-shaped molecules are expected to be considerably less abundant), of which those containing an odd number of atoms are more abundant than the even numbered ones[39,40]. At 2500K, the saturated vapor contains about 60% C_3, 10% C_2, 30% C, and small amounts of larger molecules. For these species, free-energy functions were obtained and combined with existing vapor pressure data[34-36] to yield the vapor pressure information listed in Table 4.12 and depicted in Figure 4.9. The equivalent pressure is the pressure which would be observed if the vapor consisted entirely of monatomic carbon. This quantity was obtained as a weighted sum of partial pressures for the different species. In turn, the weight loss of material under free vaporization was determined from the equivalent pressure assuming that Knudsen flow conditions prevailed (essentially ideal gas behavior). This information is presented in Table 4.12 and Figure 4.10.

Table 4.12 Vapor Pressure of Graphite

Temperature		Partial Pressures of Species in Atmospheres					Equivalent Vapor Pressure,	Free Vaporization Rate,
K	ºC	C	C_2	C_3	C_5	C_5	atm[1]	g cm^2 sec
1200	927	1.01×10^{-23}					1.17×10^{-23}	5.19×10^{-23}
1300	1027	2.81×10^{-21}		2.38×10^{-22}			3.54×10^{-21}	1.51×10^{-20}
1400	1127	3.21×10^{-19}	1.015×10^{-21}	4.37×10^{-20}			4.54×10^{-19}	1.87×10^{-18}
1500	1227	1.95×10^{-17}	1.34×10^{-19}	3.96×10^{-18}			3.16×10^{-17}	1.26×10^{-16}
1600	1327	7.08×10^{-16}	8.55×10^{-18}	2.04×10^{-16}			1.34×10^{-15}	5.15×10^{-15}
1700	1427	1.68×10^{-14}	3.34×10^{-16}	6.55×10^{-15}	3.11×10^{-20}	4.47×10^{-20}	3.72×10^{-14}	1.39×10^{-13}
1800	1527	2.81×10^{-13}	8.65×10^{-15}	1.42×10^{-13}	1.62×10^{-18}	2.33×10^{-18}	7.25×10^{-13}	2.64×10^{-12}
1900	1627	3.47×10^{-12}	1.59×10^{-13}	2.45×10^{-12}	5.52×10^{-17}	8.00×10^{-17}	1.08×10^{-11}	3.81×10^{-11}
2000	1727	3.34×10^{-11}	2.21×10^{-12}	3.10×10^{-11}	1.03×10^{-15}	1.92×10^{-15}	1.31×10^{-10}	4.61×10^{-10}
2500	2227	1.79×10^{-7}	4.30×10^{-8}	2.62×10^{-7}	2.15×10^{-10}	3.17×10^{-18}	1.05×10^{-6}	3.23×10^{-6}
3000	2727	5.43×10^{-5}	3.03×10^{-5}	1.41×10^{-4}	6.05×10^{-7}	8.97×10^{-7}	5.38×10^{-4}	1.51×10^{-3}
3500	3227	3.18×10^{-3}	3.20×10^{-3}	1.38×10^{-2}	1.68×10^{-4}	2.49×10^{-4}	5.30×10^{-2}	0.138
4000	3727	6.67×10^{-2}	0.103	0.47	1.10×10^{-2}	1.64×10^{-2}	1.83	5.29
4500	4227	0.708	1.52	7.13	0.276	0.408	28.3	65
5000	4727	4.66	12.9	72.5	3.54	5.2	288	627
5500	5227	21.6	72.8	839	27.8	40.6	3000	6240
6000	5727	77.6	307	6390	152	220	21,600	43,000

(1) The equivalent pressure is that pressure which would be observed if the vapor were monatomic carbon.

Figure 4.9 Vapor Pressure of Graphite

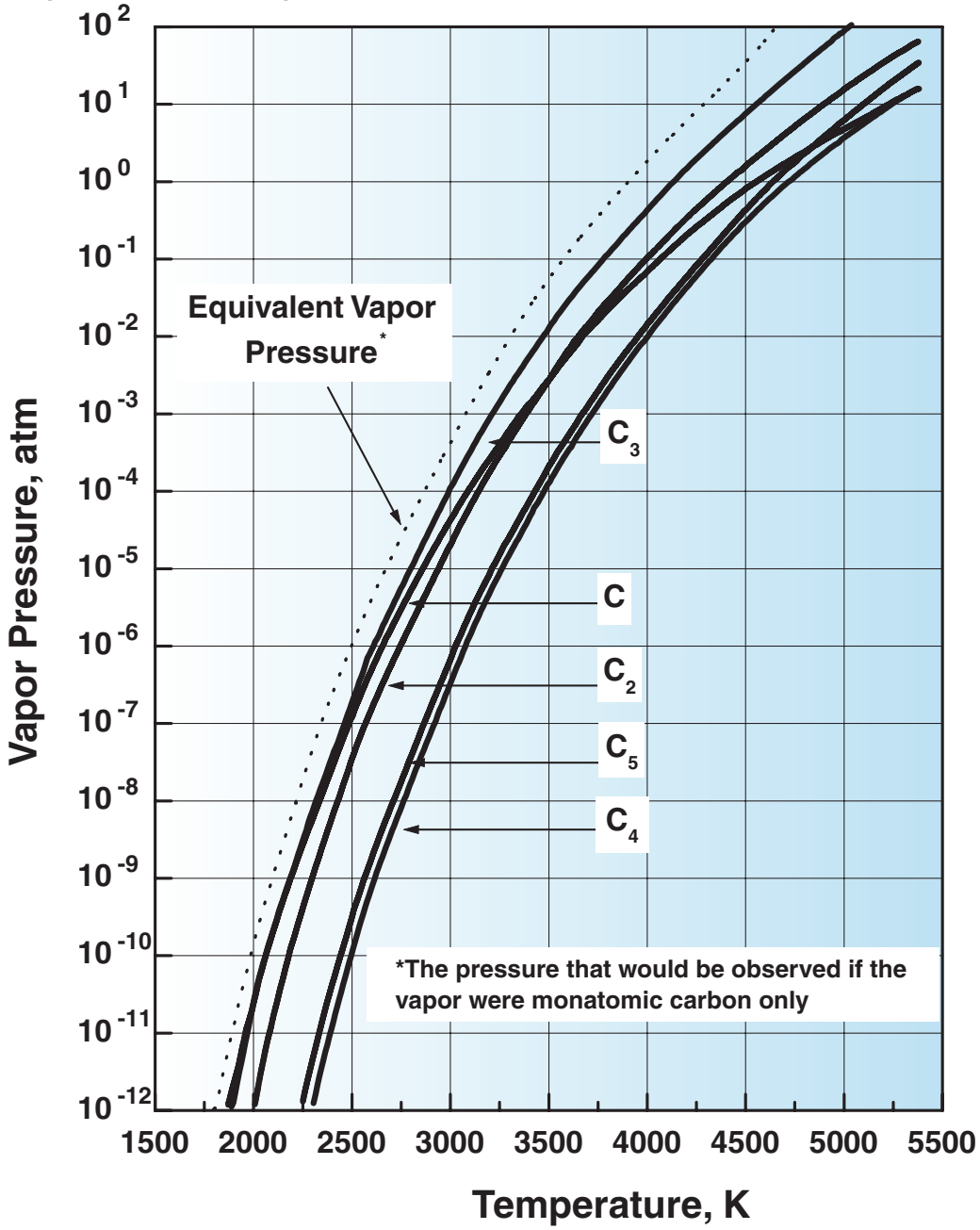
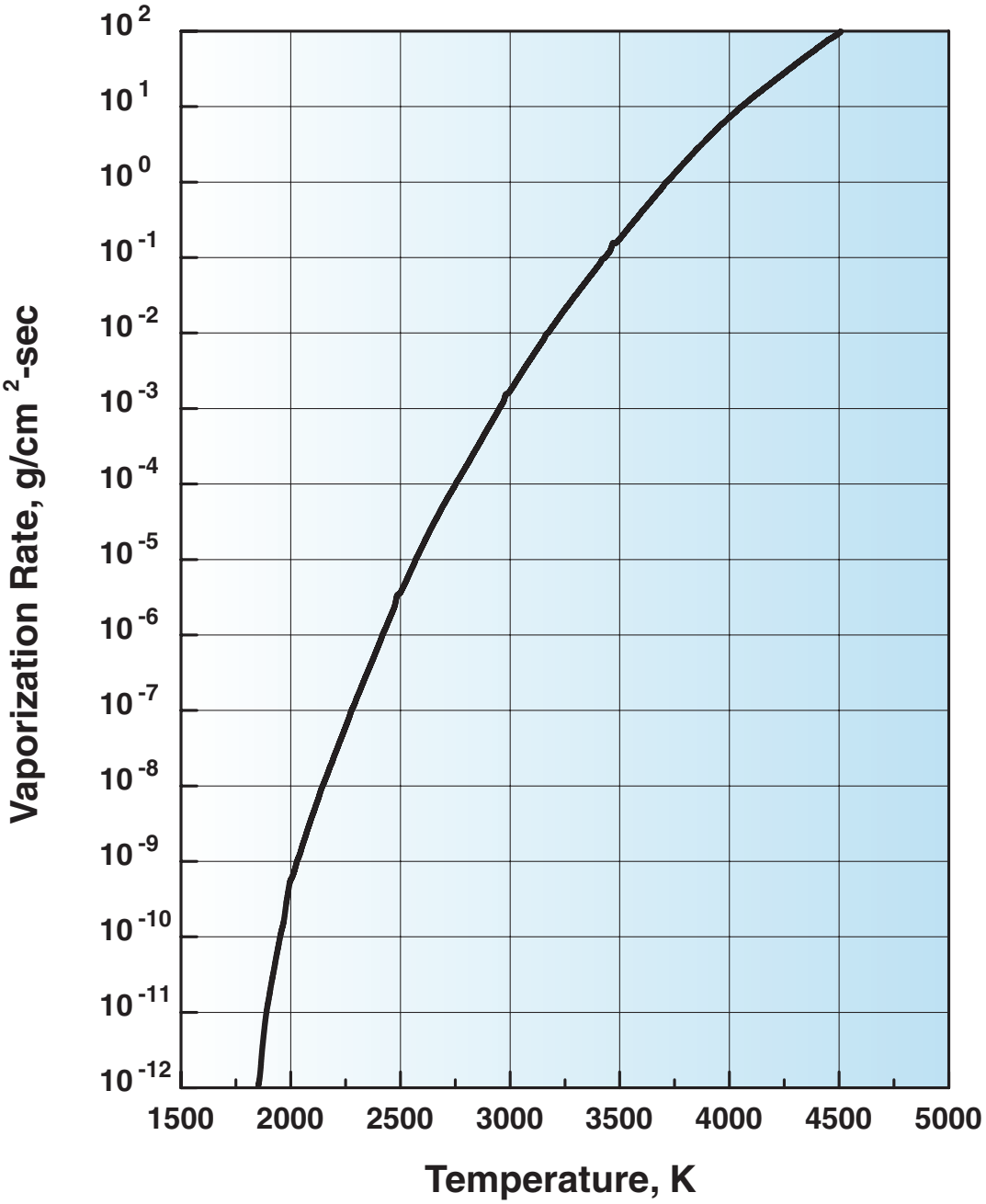


Figure 4.10 Free Vaporization Rate of Graphite



All of the measurements used in the above determinations were made under conditions of equilibrium between solid and vapor phases. Unpublished work on accommodation coefficients (a factor taking into account the resistance to vaporization due to energy exchange effects at the surface) shows that for high-density graphites, free vaporization and equilibrium are essentially equivalent[41]. In similar work on single crystals of pyrolytic graphite[42], the accommodation coefficients for the C and C_2 species were found to be normal (approximately 0.3), whereas the value for C_3 was surprisingly low (approximately 0.4). This implies that the free vaporization results given here should be adjusted by reducing the contribution due to the C_3 species by a factor of about 10. In general, therefore, the present free vaporization values should be regarded as upper limits; they will apply to graphites with normal accommodation coefficients and they will be too high when this quantity is smaller. The equilibrium pressures given here are valid in all cases.

The free-energy functions used for vapor pressure calculation are based on the following considerations.

Graphite
Since pressures of the various constituents in the vapor are obtained by difference of the free energy functions of gaseous and solid materials, data concerning graphite are necessary. There are two sources of heat capacity data from which free energy functions of solid may be obtained: either the JANAF tables[43], which are employed here, or the specific heat. Both these sets of values were obtained by averaging experimental data for different graphites; although the results differ slightly, sometimes one set is larger and sometimes the other, so that these effects compensate each other upon integration and the free energy functions differ by a maximum of two percent.

C and C_2
The physical properties of these molecules are well defined; hence, the values of the free energy functions given in the JANAF tables[43] are regarded as reliable and were used.

C_3
At present, the reported physical properties of C_3 lead to conflicting results and, unfortunately, this species is the most important constituent of the vapor. Mass spectrometric effusion studies by several investigators[34-36,39] are in good agreement with each other in yielding heats of formation, of about 188 kcal/mole. Recent spectroscopic work[37,44] indicates, however, that there is a low bending vibrational frequency of about 70 cm^{-1}. The resulting free-energy functions in conjunction with the mass spectrometric results lead to heats of formation of about 230 kcal/mole; moreover, these values are not self-consistent because they increase with the temperature of the mass spectral measurements. Because of the conflicting data, the spectroscopy-based free energy functions are employed here in connection with a set of arbitrary temperature dependent heats of formation, ΔH, obtained by extrapolation of Drowart's data[34]. The resulting vapor pressures appear plausible and are somewhat higher than the values derived from the free energy functions in the JANAF tables, which are based on earlier work. Hence, the data reported for temperatures exceeding 3000K should be regarded as tentative and subject to revision. The values given for temperatures of 2500 to 3000K are experimental quantities and considered to be exact.

C_4 and C_5
The free energy functions are based upon theoretical estimates of heat capacities and are regarded as only approximate. Between 2500 and 3000K, as with C_3, the data are experimental quantities.

C_6 and C..., Ring Molecules, etc.
Theoretical estimates concerning these molecules have been given by Pitzer[39,40]. They have not yet been observed experimentally, but may appear at temperatures above 3000K. Their partial pressures, however, will probably not be significant, relative to those of the other species reported here.

Graphite has a measurable vapor pressure at 2400K of 1.6×10^{-7}. The normal pressure is only obtained at 3491°C, the sublimation point. The sublimation rate increases rapidly with an increase in temperature. It does not liquefy until approximately 3700°C and 100 atmospheres.

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Chapter Five

ELECTRICAL PROPERTIES

Industrial Graphite Engineering Handbook

HIGH TECH HIGH TEMP

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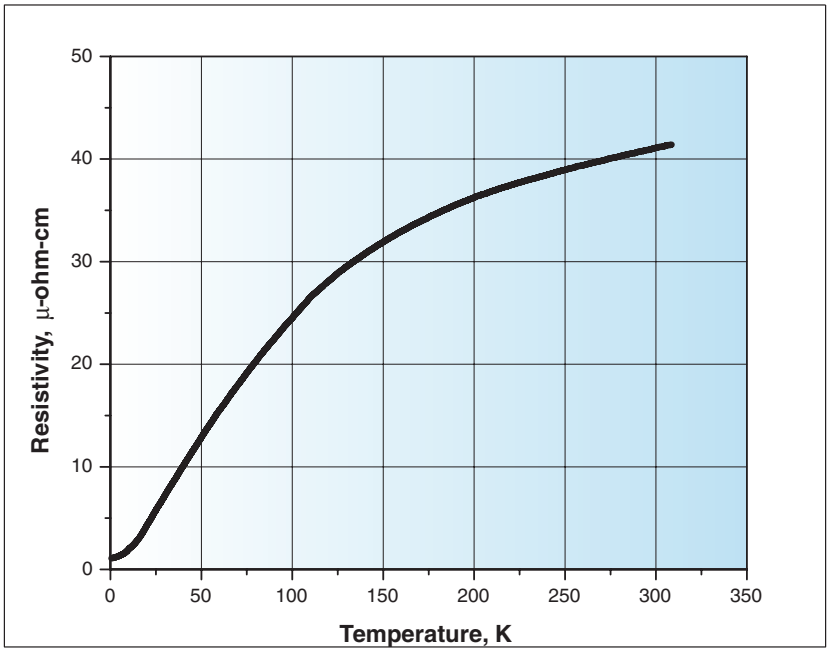
Electrical Properties

Electrical Resistivity

Graphite Single Crystals

The resistivity of a pure single crystal of graphite is an intrinsic property and represents the lower limit that can be achieved in any pure, formed, graphite body. Since graphite crystal resistivity is intermediate between that of a metal and a semiconductor, it is a strong function of temperature and impurity content. In addition, the resistivity is anisotropic, being about two orders of magnitude higher in the C-axis direction at room temperature. Measurements on highly-purified, good, single crystals of graphite[1] represent the lowest values obtained in the temperature range below 300K. The fact that these values are the lowest of all measurements reported, that they approach zero resistivity at absolute zero, that they were obtained by trustworthy experimental techniques, is interpreted to mean that they most truly represent our knowledge of graphite single crystal resistivity. These data for resistivity in the A-axis direction are shown in Figure 5.1.

Figure 5.1 Graphite Single Crystal Resistivity vs. Temperature (A-axis Direction)



Manufactured Graphites

The values of resistance for polycrystalline manufactured graphites are not expected to have any simple relation to those for single crystals, although the anisotropy of resistance of manufactured graphite shown in most stocks is, of course, related to the single crystal anisotropy. For instance, the room-temperature resistivities of commercial graphites are of the order of 20 times that for a single crystal, while the nature of the temperature dependence of resistivity of commercial materials is widely different as well. For lampblack-base, poorly crystalline graphite, the temperature coefficient of resistance is continuously negative to 1000°C, with evidence of flattening suggesting a reversal at some higher temperature. With coke-base graphites, the coefficient is negative only to about 500°C, where a minimum resistivity occurs; at higher temperatures, the coefficient becomes positive. A physical basis for explaining this behavior is readily seen in the combined effect of the two types of mechanisms contributing to the resistivity of formed graphite. First, the number of electrons available to carry the current increases with increasing temperature, which should lower the resistivity; but second, the freedom of these electrons to move in a given direction is restricted by scattering

effects of various sorts, with different temperature dependences. Thus, while the increasing electron density always tends to reduce the resistivity as the temperature rises, scattering effects may dominate at any temperature to modify or even to reverse this trend. With an assembly of disoriented crystallites bonded together in a matrix with 10%-30% porosity, as in manufactured graphite, it is easily seen that scattering will be a very important, as well as a very complex factor, in the determination of electrical resistivity.

Typical resistivity data are shown in Figure 5.2 for a coke[2,3,5] and coal-base and lampblack-base graphites. Independent measurements on a coke-base graphite to about 3700ºC[4] show a smooth continuation of the upward trend in resistivity to this very high temperature. Table 5.1 references normal resistivities for various graphites. Figure 5.3 illustrates the temperature resistivity curves for the UCAR® Isomolded Grades ATJ, DM302, DM332, and DM362.

Figure 5.2 Electrical Resistivity vs. Temperature

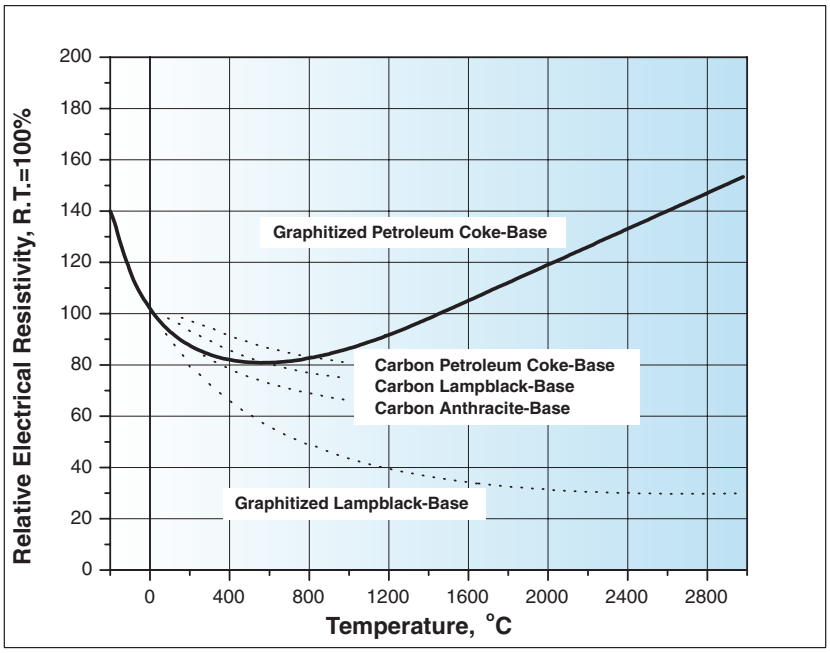
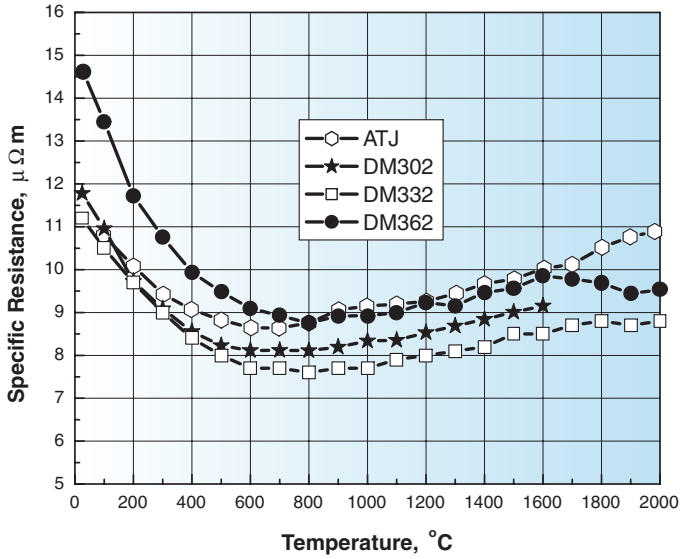


Table 5.1 • Normal Resistivity of Various Graphites

Normal Resistivity at Room Temperature	μΩm
Carbon Anthracite-Base	38
Carbon Petroleum Coke-Base	38
Carbon Lampblack-Base	51
Graphitized Petroleum Coke-Base	9.1
Graphitized Lampblack-Base	47

Figure 5.3 Specific Resistance vs. Temperature for the UCAR® Isomolded Grades ATJ, DM302, DM332, and DB58



In graphite, which has a defined crystalline structure, numerous electrons are free, providing high conductivity. Carbon, with undefined monoplanes and relatively few available electrons has high resistance. See Table 5.2 for the specific resistivity of common materials. The room temperature resistivity of several typical carbon and graphite grades is shown in Tables 5.3 and 5.4.

Table 5.2 Room Temperature Electrical Resistivity

Material	μΩm
Aluminum	0.028
Carbon	50
Copper	0.017
Graphite	7
Iron	0.1

Table 5.3 Typical Room Temperature Properties of UCAR® Grades, English Units

Group	UCAR® Grade	Largest Cross Section inches	Grain Size(1) inches	Specific Resistance 10^{-4} Ωinches	
				WG	AG
Extruded Medium Grain					
	CBY	24	0.03	3.2	4.1
	CS	30	0.03	3.0	4.4
	SLWX	24	0.03	2.8	3.8
	CHV	24 x 48	0.06	3.2	4.7
Molded Medium Grain					
	PGW	52	0.03	---	3.7
	PGX	73	0.03	4.0	5.6
Extruded Fine Grain					
	FGE	8.5 x 8.5	0.006	2.8	4.7
	FGEX	8.5 x 8.5	0.006	3.2	4.7
Isomolded Fine Grain					
	ATJ	24	0.001	4.6	
	CGW	16 x 16	0.001	4.6	
Isomolded Superfine Grain					
	DM306	8 x 16	0.0002	14.2	
	DM300	8 x 16	0.0002	5.7	
	DM301	8 x 16	0.0002	4.9	
	DM302	8 x 16	0.0002	4.5	
	DM303	8 x 16	0.0002	5.5	
	DM342	8 x 16	0.0002	4.7	
	DM332	24	0.0002	4.4	
Isomolded Ultrafine Grain					
	DM362	6 x 12	0.00008	7.3	

(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.

Table 5.4 Typical Room Temperature Properties of UCAR® Grades, SI Units

Group	UCAR® Grade	Largest Cross Section mm	Grain Size[1] mm	Specific Resistance μΩm	
				WG	AG
Extruded Medium Grain					
	CBY	600	0.8	8.5	10.0
	CS	600	0.8	7.5	9.0
	SLWX	600	0.8	7.0	9.6
	CHV	600 x 1200	1.7	8.0	12.0
Molded Medium Grain					
	PGW	1300	0.8	---	9.6
	PGX	1825	0.8	10.2	13.0
Extruded Fine Grain					
	FGE	215 x 215	0.150	7.0	12
	FGEX	215 x 215	0.150	8.0	12
Isomolded Fine Grain					
	ATJ	600	0.025	11.7	
	CGW	400 x 400	0.025	11.7	
Isomolded Superfine Grain					
	DM306	200 x 400	0.005	36	
	DM300	200 x 400	0.005	14.6	
	DM301	200 x 400	0.005	12.4	
	DM302	200 x 400	0.005	11.4	
	DM303	200 x 400	0.005	14	
	DM342	200 x 400	0.005	12	
	DM332	600	0.005	11.2	
Isomolded Ultrafine Grain					
	DM362	150 x 300	0.002	15.5	

(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.

Electrical Contact Resistance

The term electrical contact resistance refers to the resistance occurring between any two contact points in electrical applications. A resistance characteristic of the two materials making the contact is established. This contact resistance is dependent upon a number of variables, the most important being the nature of the materials involved, the pressure at the contact, the nature of the surfaces making the contact, and the quality of the contact surfaces.

In Figure 5.4, the contact resistance of graphite to graphite and to a number of other frequently encountered metals is plotted against contact pressure. In making these determinations, all surfaces were smoothed to a plane and the metal surfaces were cleaned. Table 5.5 shows similar data.

Figure 5.4 Typical Electrical Contact Resistance of Graphite to Various Surfaces

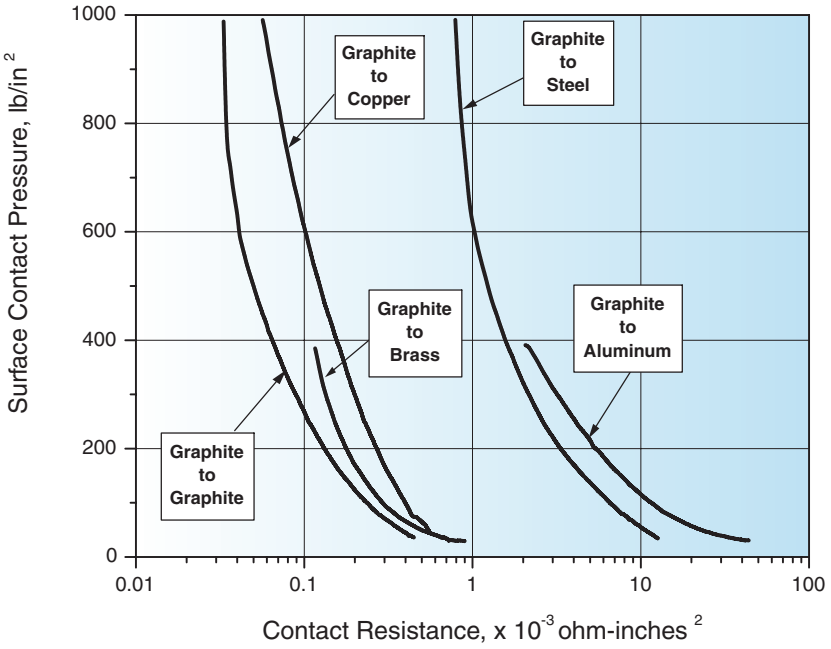


Table 5.5 Typical Contact Resistance of Graphite

	Contact Resistance, ohms-inches^2				
Pressure, psi	Graphite to Graphite	Graphite to Copper	Graphite to Steel	Graphite to Brass	Graphite to Aluminum
25	0.000473	0.000704	0.01309	0.00092	0.0448
75	0.000261	0.000424	0.00694	0.000345	0.0159
150	0.000175	0.000315	0.00438	0.000214	0.0067
250	0.000101	0.000237	0.00282	0.000162	0.0043
400	0.000064	0.000162	0.00177	0.000109	0.002
750	0.000036	0.0000755	0.000856		
1000	0.000031	0.0000555	0.000737		

The values must be considered as typical only, since the absolute electrical contact resistance for any given situation will depend upon the individual pieces and their exact surface condition. The contact resistance of carbon to metals is higher than that of graphite because carbon has a higher specific resistance and because it is harder. Table 5.6 shows some typical contact resistance of carbon-metal junctions.

Table 5.6 Typical Contact Resistance of Carbon

	Contact Resistance, ohms-inches^2			
Pressure, psi	Carbon to Carbon	Carbon to Copper	Carbon to Brass	Carbon to Aluminum
5	0.0058	0.0133	0.038	0.138
25	0.0026	0.0093	0.012	0.078
75	0.0017	0.0042	0.0052	0.015

Electrical Resistance - Designing Elements

In considering the design or operation of a furnace provided with graphite resistor elements, the following comments may be helpful:

Limiting Current Density

The maximum current density that can be put through a graphite resistor in an inert atmosphere is determined by two factors: (1) the rate of surface evaporation to be allowed for a reasonable life of the resistor; and (2) the interior temperature of the resistor should not reach such a value that sublimation of the graphite will be so rapid as to cause the rod to explode. Calculation of the maximum current in cylindrical graphite rods for the two cases in which each of these two factors, respectively, is limiting is given below.

Limited by Surface Evaporation
The following formula is calculated on the basis of allowing a maximum surface temperature of 2200°C, which should give a reasonably long life for a graphite resistor in an inert atmosphere:

$$(1) \quad I = \sqrt{\frac{A}{RD}}$$

where R is the specific resistance of the rod at 2200°C in ohm-in., D is the diameter of the rod in inches, and I is the limiting current density in amp/in.^{2}. Table 5.7 lists the factor A for various furnace temperatures.

Table 5.7 Temperature and Corresponding Affect

Temperature of Surrounding Environment °C	"A" Factor
1000	63.9
1200	61.9
1400	58.9
1600	54.1
1800	46
2000	35.4
2100	26.3
2200	0

Limited by Interior Temperature
A conservative calculation for the maximum current density based on the limitation of internal vapor pressure is given by the formula:

$$(2) \quad I = \frac{96.7}{D\sqrt{R}}$$

where R is the specific resistance of the rod at 2200°C in ohm-in., D is the diameter of the rod in inches, and I is the limiting current density in amp/in.^{2}.

The limiting current density used in any case should be the smaller of the calculations (1) and (2). For rods smaller than two inches in diameter, (1) will always be the limiting condition. For rods greater in diameter than two inches, the temperature of the surroundings will determine whether (1) or (2) is the determining factor.

Power Input

The wattage radiated by the resistor elements, of course, will just equal the rate of energy loss from the furnace by radiation, conduction, and sensible heat removed in the product tapped minus any heat contained in the charge or developed by chemical reactions. The relationship between the temperature within the furnace and the rate of radiation of energy from the resistors is given by the following equation derived from the Stefan-Boltzmann Law:

$$W = (36.6 \times 10^{-12})\ (.8)\ (T_2^4 - T_1^4)$$

where:

W = power radiated per unit area of resistor surface in watts/in.^{2}
Stefan-Boltzmann constant in watts/in.^{2}/K^4 = 36.6×10^{-12}
Emissivity ratio for graphite = 0.8
T_1 = Furnace temperature in K
T_2 = Resistor temperature in K

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Chapter Six

CHEMICAL PROPERTIES

Industrial Graphite Engineering Handbook

HIGHTECH HIGHTEMP

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Chemical Properties

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Chemical Properties

Ash Content

The ash content of all graphite is relatively low since most of the impurities originally present in the raw materials are volatilized and driven off at the temperatures encountered in the graphitization of the product.

In Table 6.1, the total ash contents of several typical carbon and graphite grades are presented.

The total ash content of graphite can have an important influence on oxidation characteristics (see the section on oxidation of graphite later in this chapter). Low-ash graphites, such as CBY and CS, are examples of graphites exhibiting good oxidation characteristics due to low ash.

Table 6.1 Typical Ash Content of UCAR® Grades

Group	UCAR® Grades	Largest Cross Section inches	Largest Cross Section mm	Grain Size,(1) inches	Grain Size,(1) mm	Ash ppm
Extruded Medium Grain						
	CBY	24	600	0.03	0.8	800
	CS	30	600	0.03	0.8	800
	SLWX	24	600	0.03	0.8	<5
	CHV	24 x 48	600 x 1200	0.06	1.7	1.50%
Molded Medium Grain						
	PGW	52	1300	0.03	0.8	1000
	PGX	73	1825	0.03	0.8	700
Extruded Fine Grain						
	FGE	8.5 x 8.5	215 x 215	0.006	0.150	1000
	FGEX	8.5 x 8.5	215 x 215	0.006	0.150	1000
Isomolded Fine Grain						
	ATJ	24	600	0.001	0.025	900
	CGW	16 x 16	400 x 400	0.001	0.025	2000
Isomolded Superfine Grain						
	DM306	8 x 16	200 x 400	0.0002	0.005	2000
	DM300	8 x 16	200 x 400	0.0002	0.005	1600
	DM301	8 x 16	200 x 400	0.0002	0.005	1300
	DM302	8 x 16	200 x 400	0.0002	0.005	1100
	DM303	8 x 16	200 x 400	0.0002	0.005	1500
	DM342	8 x 16	200 x 400	0.0002	0.005	1500
	DM332	24	600	0.0002	0.005	600
Isomolded Ultrafine Grain						
	DM362	6 x 12	150 x 300	0.00008	0.002	1500

(1) For the extruded medium and fine grain and molded medium grain, the grain size is the maximum; for all the isomolded, the grain size is the average.

Gas Evolution from Graphite

The many fine pores which characterize the structure of industrial graphite usually contain chemically- and physically-absorbed gases, some of which are retained despite prolonged evacuation of room temperature. Moreover, very small amounts of certain metallic carbide impurities will react with moisture to produce gases not originally present in the graphite. Such impurity centers, trapped inside the graphite structure, can operate to evolve gas for a long time. The use of graphite to form the plates and grids of high-power radio and rectifier tubes first called attention to the fact that removal of these gases and gas-forming impurities was necessary to ensure long-term maintenance of a good vacuum and to prevent filament poisoning.

The "gas content" of graphite has come to be defined as the volume of gas evolved from graphite when it is heated to a specified high temperature after it has been evacuated at a lower temperature, i.e., room temperature. In order to determine this value, the graphite sample is mounted inside a quartz bulb in a vacuum system and pumped out at room temperature for at least one hour, reaching a pressure well below 0.001 mm of Hg. Then the system is sealed off and the graphite heated by induction to the specified high temperature. The gas evolved increases the pressure. The volume of gas evolved is calculated from the pressure rise. Thus, the term "gas content" includes the gas which cannot be removed by prolonged pumping at the lower temperature as well as the gas which is produced by the decomposition of chemical impurities in the graphite at the higher temperature. Table 6.2 shows typical results with two standard grades, not intended for use where gas content is critical. Large variations in these values, dependent upon the particular sample and the environmental conditions, can occur with these grades.

Table 6.2 Gas Content of UCAR® Medium Grain Extruded Graphite

UCAR®	Gas Liberated, mm^3/g		
Grade	1400°C	2000°C	Total
AGSR	400	90	490
AGSX	484	80	564

Typical composition of gas liberated by UCAR® grade AGSR is shown in Table 6.3

Table 6.3 Percentage Composition of Gases from Medium Grain Extruded AGSR Graphite

	Composition, %	
Gas	1400°C	2000°C
O_2	0	0
H_2O	0	0
CO_2	13	10
CO	34	52
H_2	Trace	0
N_2	52	38

The nature of the impurities in graphite is known to have a marked effect on the gas content. Calcium is an especially critical material[1]. Initially present as the carbide, it hydrolyzes to $Ca(OH)_2$ on standing in air, and then partially converts to $CaCO_3$ in contact with CO_2. With even a fraction of one percent of calcium present, graphite heated under vacuum will evolve water vapor, CO_2 and CO for a long time, making a stable "clean-up" very difficult. In most graphite applications, this is of no concern, but where low gas content is required, special grades in which particular attention has been paid to the choice of materials and to the processing conditions which influence gas content should be used.

The gas content of graphite may also be of importance in nuclear reactors. If the graphite is to be canned in metal containers, an excessive evolution of gas could create certain pressure and embrittlement problems. Low gas content graphites are obviously recommended for such applications. Figure 6.1 shows gas evolution versus temperature data on two samples of nuclear UCAR® grade AGOT graphite[2]. In a similar study of purified grade TSP graphite, considerably less gas was evolved. The composition of the gas for UCAR® AGOT graphite was found to be predominantly CO with a lesser amount of H_2, while that for TSP was primarily H_2 with a lesser amount of CO (see Table 6.5)[3].

Figure 6.1 Total Gas Desorbed as a Function of Temperature for Various Types of Graphite

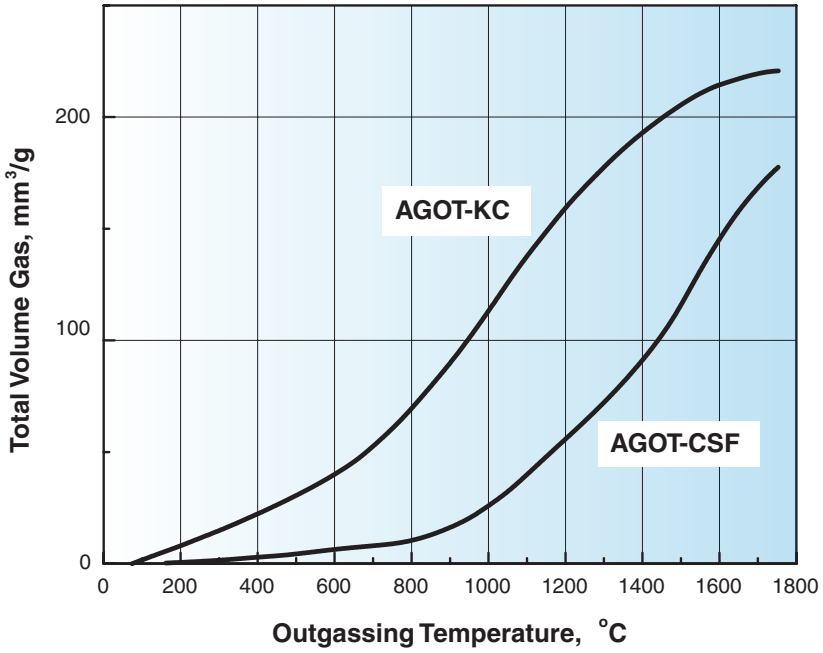


Table 6.4 Percentage Composition of Gases from UCAR® Grades AGOT and TSP Graphites at 1750ºC

	Composition, %	
Gas	UCAR® AGOT	UCAR® TSP
H_2	30.1	81.5
CO	~67	16.9
N_2	~3	1.4
CO_2	0.2	0.1
CH_4	0.1	0.1

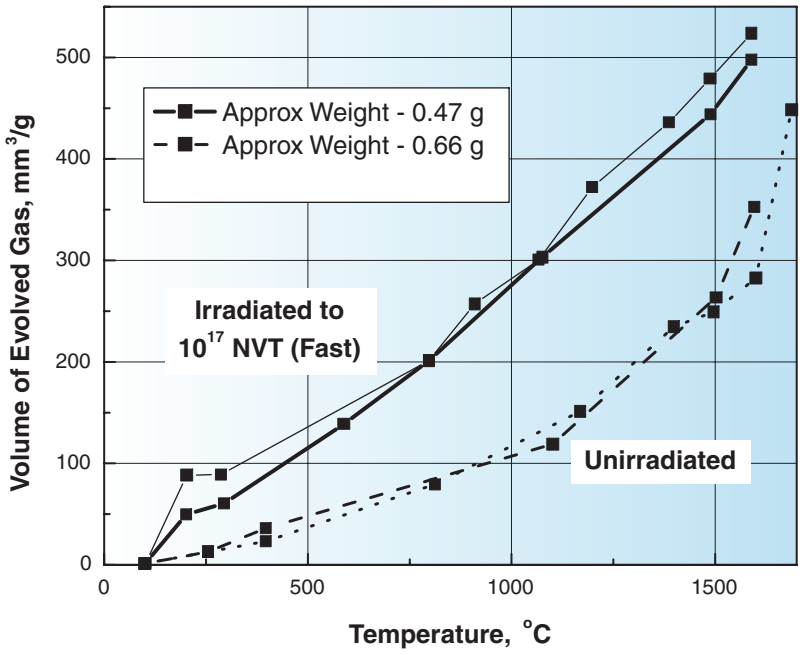
While the origin of the reported H_2 and CO is not positively known, it may result from the absorption of water vapor from the atmosphere after the stock is exposed to air. For example, the gas content of grade TSP graphite has been found to be very much less when stored in a desiccator with P_2O_5 than when exposed to moist air[4]. In this experiment, a given sample of UCAR® grade TSP graphite was exposed in turn to a series of wet and dry storage conditions, the gas content being measured at the conclusion of each exposure. The sample thus entered each new exposure in a relatively gas-free condition (see Table 6.5).

Table 6.5 Gas Content of TSP Graphite

Exposure No.	Type Atmosphere	Storage Period, Days	Gas Evolution, mm^3/g at 1800°C
0	As Received		62
1	Air + H_2O	8	104
2	Air + H_2O	5	85
3	Air + P_2O_5	23	27
4	Air + P_2O_5	25	23

Nuclear irradiation has also been shown to have a marked effect on the amount of gas evolved for reactor graphite[5]. Typical data, such as are plotted in Figure 6.2, show that prior irradiation almost tripled the amount of gas evolved at 1000ºC and almost doubled it at 1400ºC. The tendency for the curves to come closer at high temperatures suggests that both temperature and irradiation act to accelerate the processes which release gas for the same potential sources in the graphite.

Figure 6.2 Outgassing of Graphite vs. Temperature



Oxidation

The fact that graphite burns to form CO and CO_2 is very well known. There is a tremendous quantity of literature dealing with this apparently simple phenomenon. The reason relates directly to the form in which industrial graphite exists - a porous mass of solid particles of assorted kinds and sizes, held together with a bond carbon formed during fabrication of the piece. Relatively little agreement exists among the findings of many investigators who have studied the graphite-oxygen (or carbon-oxygen) reactions, because the rate, and perhaps even the mechanism of the reaction, depends strongly on the nature of the carbon, the impurities present, degree of graphitization, particle size, etc. These vary widely from one graphite grade to another.

Graphite is actually less reactive to oxygen at low temperatures than many metals, but since the oxide is volatile, no protective oxide film is formed. Still, the rate of oxidation of graphite is slow enough to permit its very effective use in oxidizing atmospheres at extremely high temperatures where a modest consumption rate can be tolerated. For example, a one-half inch diameter all-carbon arc lamp electrode operates in the open air with the tip at about 4000K and a consumption rate of no more than three inches per hour - half of which is the result of direct volatilization from the arcing tip and half is due to oxidation. When the arc current is interrupted, the electrode promptly cools, showing that ignition or self-sustained combustion does not occur even after heating to this very high temperature.

The differences in oxidation behavior of the various grades of graphite are widest at the lowest temperatures, tending to disappear as the temperature increases. If “threshold oxidation temperature” is defined as that at which a sample loses approximately one percent of its weight in 24 hours, pure graphite has a threshold oxidation temperature of 520 to 560ºC, depending on coke source and processing variables. Small amounts of catalysts are very effective in increasing the oxidation rate at low temperature, and there are enough of these catalysts in commercial graphite to lower the threshold oxidation temperature by as much as 100ºC. Tests of individual catalysts as additives to the graphite have shown, for example, that 20 to 40 parts per million of sodium, potassium, vanadium, or copper will increase the rate of oxidation by a factor of six at 550ºC, thereby reducing the time for a five percent oxidation from 120 hours to 20 hours.

In the temperature range 400 to 650ºC, just above the threshold oxidation temperature, the oxidation is primarily penetrative in character, such that the weight loss is greater than the volume loss. At still higher temperatures, the oxidation is diffusion-controlled, with virtually all oxidation taking place at the outside surface. In this diffusion-controlled temperature region, the oxidation rate is extremely sensitive to the velocity of air movement over the surface.
The oxidation of graphite has been measured between 425 and 575ºC in pure oxygen at pressures of 0.15 to 9.8 cm Hg[13]. It was found that the rate at which the samples lost weight with time could be expressed according to the equation:

$$W = Kt + Ct^2$$

where t is the time and K and C are constants. The activation energy was found to be 36.7 kcal/mol.

Oxidation tests in air have been carried out on a variety of purified graphite grades at 500, 600, and 700ºC[6]. The rate of oxidation was found to increase with time, but leveled off after the samples had undergone a weight loss of several percent. The oxidation rates during this latter period of oxidation varied with temperature in a manner consistent with an activation energy of 55 kcal /mol, averaged for ten different graphite grades. Since this value refers to a different part of the oxidation curve, it is not inconsistent with the value of 36.7 kcal/mol cited above.

To indicate the order of magnitude of oxidation rates in air, some of the average results on both purified and general-purpose graphites at 500, 600, and 700ºC are cited in Table 6.6. The samples were 1.25 inches diameter by 2 inches long, and were suspended in a furnace from an automatic recording balance. Individual values for different graphites deviated from the indicated averages by as much as 30 percent.

Table 6.6 Typical Values of Oxidation Rates of Graphite in Air

Graphite Sample	No. of Varieties in Average	Oxidation Temperature, °C	Time for 1% Weight Loss, hr	Time for 6% Weight Loss, hr
Pure	13	500	200	-
Pure	13	600	7	-
Pure	13	700	0.6	1.8
Commercial	1	500	6	-
Commercial	6	600	0.75	2.5
Commercial	6	700	0.2	0.7

A critical study of the graphite oxidation reaction has been published in a series of papers by Eyring and his associates[7-9]. After studying the heterogeneous gas solid reaction in the range of 1-100 micron pressure and 600 to 1300°C, they report that below 800ºC, the surface reaction is of zero order with an activation energy of 80 kcal/mol. On samples thicker than 0.1 mm, the diffusion of oxygen into the pores of the graphite results in an observed half order reaction with 42 kcal/mol activation energy. Above 1200°C, the reaction is controlled by the frequency of oxygen collisions with the surface. Between 800 and 1200ºC, a continuous transition region exists. It is suggested that whenever an O_2 molecule is absorbed on a C surface, an activated complex is formed which may break up into C (surface) + O_2 or into 2 CO, and that the latter reaction is favored by the potential energy situation prevailing under most circumstances. Eyring et al. point out that a great deal of the literature on carbon oxidation deals with materials which have been carbonized but not graphitized. These materials are usually oxidized in first order reactions with an activation energy of 20-30 kcal/mol.

Oxidation of graphite is also produced by steam and carbon dioxide atmospheres, although the threshold oxidation temperature is higher for these gases. Typical manufactured graphite will have a threshold oxidation temperature of about 700°C in steam and 900ºC in carbon dioxide, see Table 6.7. Graphite consumption by hydrogen and nitrogen is significant at temperatures above 1000°C for the former and 1500°C for the latter.

Table 6.7 Threshold Oxidation Temperatures of Graphite in Various Atmospheres

Atmosphere	Temperature, °C
Air	400
Steam	700
CO2	900

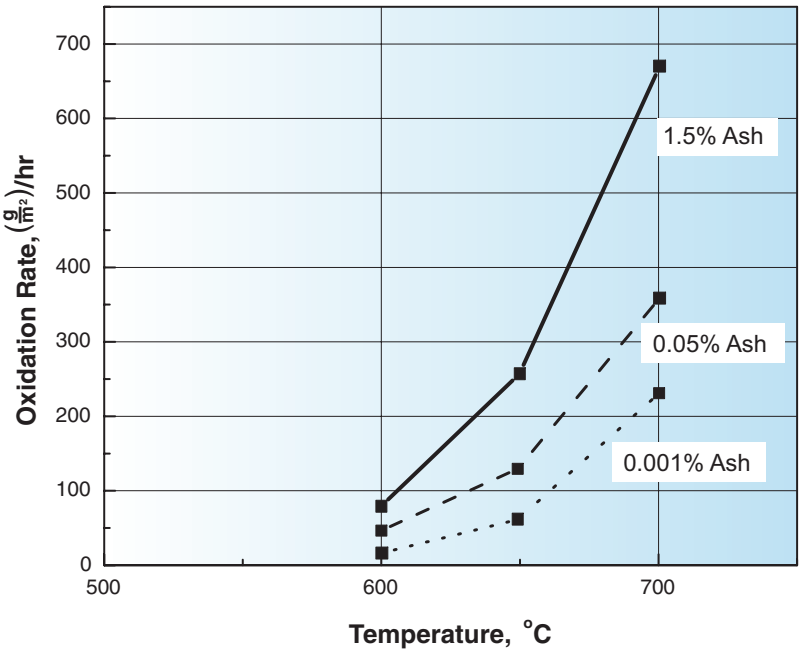
The amount of total ash and impurities in graphite plays a significant part in determining oxidation characteristics. Low-ash graphite, such as CBY and CS, have significantly lower oxidation rates than do commercial graphites having 1.5 percent ash. Similarly, purified graphite having ash levels less than 10 ppm have even lower oxidation levels. Table 6.8 and Figure 6.3 show this comparison[19].

Table 6.8 Oxidation of Graphite

Grade Grouping	High-Ash Commercial	CBY-CS	Purified CBY-CS
Ash, typical, %	1.5	0.05	0.001
Grain Size, max, inches	0.03	0.03	0.03
Oxidation Rates[1]			
600°C	66	38	12
650°C	243	126	53
700°C	651	343	210
Threshold Oxidation Temperature[2] °C	470	560	550

(1) $(g/m^2)/hr$
(2) 1% weight loss in 24 hr

Figure 6.3 Oxidation Rates of Graphite



Oxidation Retardation

Various forms of UCAR® graphite are used in applications where resistance to oxidation is a primary consideration. Research on the oxidation characteristics of UCAR® graphite has shown the following:

(1) Untreated graphite obeys an Arrhenius activation energy equation within certain temperature ranges up to 700ºC. Above 700ºC, oxidation is not governed solely by temperature.

(2) The treatment of UCAR graphite materials with an oxidation retardant materially enhances oxidation resistance. In general practice, a typical commercial oxidation retardant maintains a high degree of efficiency and is superior to untreated graphite at temperatures up to 750ºC.

Tests on graphite tubes summarized in Table 6.9 show the effectiveness of a commercial oxidation retardant treatment. The tests were conducted over a 24-hr period in an air atmosphere. The specimens were thick-walled pipes 2 inches outside diameter, 0.5 inches inside diameter, 5.5 inches long, and weighing approximately 500 g. The anti-oxidation treated samples are about ten times more resistant to oxidation at 750ºC than untreated specimens.

Table 6.9 Effect of Air Oxidation on Untreated and Treated Graphite

	Oxidation Rate, $(g/m^{2)})/$ hr	
Graphite	at 650ºC	at 750ºC
Untreated		
UCAR® CBY-Type	230	1800
UCAR® CS-Type	340	1900
Treated	23	200

Resistance to Erosion and Attrition

The ability of graphite to resist erosion or abrasion, either mechanical or chemical, is important in its applications as structural components in guided missiles and as containers for liquid metals in nuclear reactor high-temperature heat transfer systems.

The erosive effect of an inert hot gas on graphite has been shown to be negligible, e.g., a helium jet at 2000ºC and a velocity of 900 ft/sec produced, in 24 hours, no more than an initial change in surface quality[10] attributable to a blasting away of particles previously loosened in machining. However, any oxidizing gas will react with graphite (even at moderate temperatures) to chemically erode it. A sharp distinction must, therefore, be made between chemical erosion and mechanical erosion. Molecular nitrogen, which is inert, will act as the helium did. Nascent nitrogen, on the other hand, will readily react with graphite at high temperatures to produce cyanogen, and thus erode the graphite.

An example of chemical erosion caused by oxidation is seen in Table 6.10[11]. A supersonic kerosene-oxygen jet torch operating at a chamber pressure of 160 psig, an exhaust gas velocity of 6000 ft/sec, and a temperature of 2760ºC, was used. At lower chamber pressures, the exhaust velocity and temperature were somewhat reduced.

Table 6.10 Oxidation Erosion of Graphite Nozzle in Supersonic Kerosene - Oxygen Torch[11]

UCAR® Grade	Chamber Pressure, psi	Number of Tests	Life, seconds		
			Maximum	Minimum	Average
ATJ	100	5	43	36	40
	125	5	18	11	14
CS	100	5	33	15	20
	125	5	5	2	4
	135	4	4.5	2	3
AGSX	100	3	27	12	19
AGSR	100	5	22	12	17.5

The graphite grade to be tested formed the nozzle for the torch, simulating the role of the rocket nozzle. The life indicated in seconds is arbitrarily taken as the time required for a five-percent decay in the original peak chamber pressure of the torch, which is a direct measure of the throat enlargement (due to wear) of the nozzle.

Similar results are reported in measurements of erosion caused by directing the exhaust jet of a rocket motor against graphite test specimens[12]. The test samples, in this case, were graphite plates 4 1/8 inches square by 7/16 inches thick. The 2700K exhaust jet was adjusted to cover the entire face of the test plate. Dimensional, weight, and appearance changes following a 15-second exposure to the jet were recorded. These results are presented in Table 6.11.

Table 6.11 Oxidation Erosion of Graphite by Rocket Exhaust Jet Directed Against Graphite Surface

UCAR® Grade	Erosion Rate			Surface Appearance
	Temperature, °C	By Thickness, inches/second	By Weight, g/second	
AGSR	850	0.004	1	Rough, Pitted
CS	750	0.003	0.7	Slightly Rough
ATJ	465	0.002	0.2	Slightly Rough

While densification (impregnation) and other processing modifications effect small increases in abrasion resistance, the largest single influence for mechanical abrasion is raw materials. Petroleum coke type is the biggest factor. Table 6.12 shows the influence of raw materials on the mechanical abrasion of various graphites. Coke Type A is characterized as isotropic and shows the least weight loss on abrasion. Coke types B and C are anisotropic cokes with relatively low CTE and SR values, where abrasion is simulated in an in-house test by rotating a cylindrical specimen in a bed of sized aluminum particles, and measuring the weight loss with respect to time.

Table 6.12 Effect of Raw Materials on Graphite Abrasion Properties

Coke Type	Densification	BD, g/cc	SR, μΩm	Hardness, Rockwell "R"	Abrasion Resistance, % Weight Loss after		
					10 minutes	30 minutes	50 minutes
A	No	1.6	12.7	91	2	6	9
A	Yes	1.7	11.7	99	2	5	8
B	No	1.64	8.9	62	8	18	24
B	Yes	1.72	8.6	67	7	18	23
C	No	1.7	8.6	65	7	17	22
C	Yes	1.77	9.5	79	6	15	20

Heat of Combustion

The heat of combustion of graphite is a reproducible thermochemical quantity that has been determined with high precision. It measures the heat that is liberated when graphite is burned with oxygen to form CO_2. A U.S. Bureau of Standards[14] compilation gives the "standard heat of formation" of CO_2 as 94.0518 kcal/mole. More precisely, this refers to the "standard heat of formation" and "standard states" as defined and used in thermochemistry[14]. Practically, it agrees within a fraction of a percent with the heat of combustion at 10 atmospheres pressure and 25ºC.

Several high-precision studies have been made at the National Bureau of Standards. Dewey and Harper[15] found, in 1938, that several low-ash natural graphites and spectroscopic-grade artificial graphite gave the same heat of combustion. Jessup[16] confirmed this and found that diamond has approximately 0.5 percent larger heat of combustion than graphite. Prosen and Rossini[17] made refined measurements in 1944 on some of the same material used earlier by Jessup[16]. These men[18] used these results along with a recalculation of some of the earlier data to obtain the currently accepted value[14]. Measurements on anthracite cokes indicated that the heat of combustion of these depended on the hydrogen content, but in the hydrogen-free form, it was about 2.5 percent higher than for graphite[15].

Heat of combustion measurements made on reactor-irradiated graphite samples are affected by the amount of stored energy.

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Chapter Seven

INDUSTRIAL APPLICATIONS OF BULK CARBON AND GRAPHITE

Industrial Graphite Engineering Handbook

HIGHTECH HIGHTEMP

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Introduction

This chapter is concerned with industrial applications of bulk carbon and graphite. This is not an exhaustive treatment, but rather a brief survey of metallurgical, chemical process equipment, electrical, electronic, electrolytic, aerospace, and nuclear applications. For greater detail, and for information about applications which are not mentioned here, please contact a UCAR Carbon representative.

Metallurgical

Metallurgical applications are defined as those in which graphite comes in contact with metal, normally in the presence of high temperatures. Since the processing of metals generally involves high temperatures, the term metallurgical applications has, in many cases, become synonymous with high-temperature applications. The properties of graphite that make it particularly useful in a broad range of metallurgical applications are listed below.

- Unreactive with many molten metals (e.g., Zn, Al, Mg, Cu and Cu Alloys) and metal-producing slags.
- Does not adhere to most castings, billets, and ingots.
- Strength properties increase as temperature increases.
- Low thermal expansion; high dimensional stability.
- High thermal-shock resistance due to a combination of high thermal conductivity, low Young's modulus, low coefficient of thermal expansion, and high strength at temperature.
- Will not melt at high temperatures under ordinary pressures.
- Machinability.
- High strength-to-weight ratio.
- Performance at high temperatures.

Boats and Trays

Graphite is used extensively in the sintered carbide and powdered metallurgy fields for molds and crucibles, pre-sintering and sintering boats, and trays, as well as for hot pressing and other methods of carbide and sintered carbide production. Graphite is also used widely for trays and reference forms in furnace brazing.

Crucibles

Since graphite may be heated either by induction or electrical resistance, as well as by conduction or radiation from other sources, it is an excellent crucible material. Customary crucible sizes vary from the small, vacuum-fusion-type (less than 1-in. diameter) to the large type (up to 73-in. diameter) for melting beryllium and uranium and other high-melting point metals. Most metals may be melted directly in graphite since graphite is not attacked, and the metal not contaminated.

Fluxing, Injection Tubes

Graphite fluxing tubes are used (1) for the introduction of gas into the gas purification of molten aluminum and other nonferrous metals, (2) for the desulfurization of blast furnace iron, and (3) for the injection of graphite powder into foundry iron. Graphite is well established as the standard fluxing tube material in many industries. Fluxing tubes can also be treated with oxidation retardants for resistance to oxidation.

Molds and Dies

Graphite may be adapted to molds for static castings of most metals or to molds or dies for the continuous casting or drawing of nonferrous metals. Such products vary in size from molds with cavities of one inch diameter by one inch depth to molds used in the centrifugal casting of bronze and bearing metals with a 73 inch diameter and 72 inch length. Other mold and die applications include molds for the hot pressing of powdered metals and ceramics, ingot molds, jominy bar molds, pig and over metal molds and precious metal molds. Graphite is also used in pressure and static casting of carbon steels as a mold to form precision cast parts or slabs. Graphite can be remachined to produce hundreds of castings.

Furnace Parts

Because of desirable refractory properties, in addition to relatively high resistance, graphite makes excellent resistor elements where they may be operated in a reducing or non-oxidizing atmosphere, or in a vacuum. Graphite can be used at temperatures up to 2800°C in inert atmospheres. The temperatures attainable with graphite are well above those of any other known resistor elements.

Graphite is also used extensively in atmospheric-controlled furnaces as a lining material. Its high thermal conductivity, low thermal expansion, and ease of machinability are characteristics conducive to lining furnaces in virtually any shape or size. Graphite slabs can be bolted together with a variety of jointing techniques. On any structural furnace part (hearth, wall, roof, door) where there is a non-oxidizing atmosphere, graphite should be considered as a material of construction.

Graphite and semi-graphites are also used in blast furnaces as hearth, bosh, and stack materials. Its high thermal conductivity, thermal shock resistance, high-temperature dimensional stability, and machinability are ideal characteristics for cost-efficient linings.

Foundry Accessories

In foundries, graphite plates and rods are used as molds, chills, and cores because of their rapid chill characteristics, resistance to metal attack and smooth surfaces. Graphite riser rods are used in conjunction with steel castings to add carbon to the riser metal in order to maintain a fluid liquid head of metal for a longer period of time to feed the casting during cooling and shrinking. In brass foundries, graphite skimmer floats are used in pouring crucibles to hold back the dross from the metal being cast. In nonferrous melting pots, graphite rods are used for stirring since this material does not contaminate the melt, does not crack, and does not have an affinity for the molten metal.

Aluminum Processing Parts

In the extrusion of small-section aluminum shapes, such as angles, counter edge- concave and convex beadings, window frame channels, etc., it is general practice to use multiple die openings in each cylinder. Where as many as four or six such shapes are being extruded simultaneously, the tendency to twist or curl upon leaving the die causes considerable tangling of the leading ends of the several extrusions and the subsequent loss of considerable material in removing the tangled ends. Tangling may be prevented by using graphite extrusion guides which also assure freedom from scratching of the aluminum between exit from the die and alignment on the runout table. The guide may be a simple graphite cylinder having rectangular or pie-shaped openings that run the length of the graphite tube between the bolster and runout table for whatever distance may be required to ensure satisfactory operation.

Chemical Process Equipment

As a construction material in chemical processes, manufactured graphite offers a unique combination of properties: resistance to practically all corrosives (except highly oxidizing conditions), inertness to thermal shock, high thermal conductivity, thermal stability and good machinability - all at moderate cost. Some of the major applications of graphite, illustrating its wide use in the chemical industry, follow.

Graphite vessels and reactors are used in applications, such as burning hydrogen and chlorine to form hydrogen chloride, and burning phosphorus and its subsequent hydration product to form phosphoric acid. Such vessels range in size from 12 inches to 20 feet in diameter by 10 to 30 feet in height and are constructed from either single graphite shapes or fabricated from premachined graphite blocks. Units can be arranged for water cooling to remove the heat of reaction. Because of the chemical inertness of graphite, impure reactants can be handled without concern.

When installed as a vessel lining, graphite provides inertness to chemical attack, resistance to abrasion, and a heat barrier to protect the vessel proper or its inner lining (either metallic or non-metallic). Graphite linings can be installed in vessels of any conceivable size and cross section. They can be erected with either a keyed construction to provide mechanical stability or with suitable chemical and heat-resisting cements. Graphite-lined vessels of this type are well suited for high-temperature processes, such as chlorination reactions for the production of boron trichloride and other metallic chlorides.

Because it offers resistance to corrosion, high temperature, and thermal shock, and can be easily fabricated, graphite has many internal uses in chemical processing equipment. Typical are graphite bubble caps and trays, sieve plates, and packing support plates. Graphite is employed for constructing submerged guide rolls used in the production of synthetic fibers and films because of its resistance to corrosive action and freedom from metallic contamination. Graphite rolls are also used as conductor rolls on electrolytic tin-plating lines where the electrical conductivity of graphite is an additional advantage.

A complete line of chemical processing equipment items can also be manufactured. In this field of application, graphite is outstanding in the construction of heat exchangers of all types. Here it offers unusually high heat transfer rates together with freedom from corrosion, metallic contamination and thermal shock. Because of its ease in machining and fabrication, graphite is an economic material of construction for centrifugal pumps, valves, pipe and fittings, towers, hydrochloric acid absorbers, and a variety of process accessories.

Electrical

In electrical and electronic applications, graphite is used primarily for component parts, such as vacuum power tubes and resistor elements. It is also used to some extent as electrical contacts, brazing tips, etc. The properties of graphite of particular utility for electrical applications are summarized in the list below:

- Low gas content.
- High thermal emissivity.
- High thermal conductivity.
- Low coefficient of thermal expansion.
- Low vapor pressure.
- Good electrical conductivity
- No electron emission at high temperatures.
- High-temperature strength.
- Machinability.

Electrical Discharge Machining (EDM)

Electrical discharge machining or EDM, as it is most commonly known, uses the energy contained in an electrical discharge to remove metal. These discharges are generated many thousands of times per second. Simple or complex operations, such as through-hole boring or cavity sinking in any electrically-conductive work material, including carbide, tool steels, high-temperature alloy steels, and metals of all types, are made possible with this process. The EDM process will machine any electrically-conductive metal, regardless of its hardness.

Electrical discharge machining is used for a variety of machining applications and is especially noted for its ability to produce intricate shapes, while holding extremely close tolerances. For many machining jobs, EDM is the only viable means to manufacture a particular shape. The technique may be used in the manufacture of forging dies, cavity inserts, plastic injection molds, and many tools.

UCAR Carbon superfine and ultrafine graphites are used in EDM applications for the following reasons:

- Relatively inexpensive
- Available in a variety of sizes
- Readily machinable to complex shapes with conventional tools
- May be used in “No-Wear” (positive electrode) machining
- Provide highest metal removal rates with lowest wear
- Are non-melting.

The EDM operation is a melting or vaporization process: the spark energy jumps across the gap to melt or vaporize a portion of the workpiece. This energy also affects the electrode. If the electrode is made of the same material as the workpiece, the electrode will be eroded at the same rate as the work piece - a “one-to-one” wear ratio. Graphite does not melt. Instead, it sublimes at a temperatures above 3000°C. Thus, graphite resists the spark action, making wear ratios of 100:1 and greater possible.

Resistance Heating Elements

Resistance heating elements may be fabricated in almost any shape and size from round rods to rectangular beams joined and fitted to obtain the desired resistance for a given application. Assemblies can be arranged in banks, series, or in series-parallel. Graphite tubes or slabs may also be machined to form a hairpin resistor or to form an extended path by alternate saw cuts resembling numerous hairpin resistors in series; or, the graphite cylinder may be machined into a single or double helix to provide the maximum resistance in a given space. A resistor suitable for three-phase operation may be machined from a single graphite tube, connections for such resistors may be made from water-cooled copper parts; or, the graphite may be joined directly by threading, pressed fit, or spring fit, which is provided by making a saw cut in the end of a graphite rod for insertion into a drilled hole a few thousandths of an inch smaller in diameter than the resistor rods.

Electronic

Fiber-Optic

High grade quartz is drawn at temperatures around 1700°C to form a cable with fiber-optic qualities. In one industrial process the furnace container and heater are fabricated from high purity isomolded superfine graphite such as UCAR® DM302. Graphite with its combination of thermal characteristics, chemical inertness, high purity, superfine grain texture, and structural integrity at elevated temperatures is eminently suitable.

Semiconductors

There are several major applications for graphite in the semiconductor sector with most requiring purification to at least 10ppm ash.

Single Crystal Growth - Silicon Carbide: Purified graphite with its enviable thermal properties and inertness provides a suitable container material in which single crystals of silicon carbide can be grown. The isomolded grades are ideal for this application.

Single Crystal Growth - Silicon: Silicon is the most widely used material for the manufacture of integrated circuits and other semiconductor devices. Silicon, the second most abundant element, comprising 28 percent of the earth's crust, is a good material for circuits because its electrical properties can be precisely altered by adding controlled amounts of impurities, called dopants. One of the most essential operations in the manufacture of an integrated circuit is the conversion of this silicon into single crystal form. The most common method of converting the polycrystalline silicon to a single crystal ingot is through growing the ingot from molten silicon by the Czochralski crystal-growing process. This process involves wetting a properly-oriented seed in a melt and withdrawing it vertically to grow a crystal which might be 12" (300 mm) in diameter.

Graphite is extensively used in this application for a number of reasons. First, graphite has electrical properties that are necessary for good heater and crucible- operation. Second, it has the required thermal properties. Material is needed that can withstand temperature recycling from room temperature to above 1400°C, maintaining strength at these high temperatures. Graphite has excellent thermal conductivity, enabling it to maintain thermal equilibrium in the growing process. Third, the combination of cost, availability, machinability, and life of the graphite parts has not been improved upon by other materials.

Purified medium-grain graphites, such as UCAR®Grades CS and PGW, are used as resistance heaters, which melt the silicon and maintain it at a temperature of approximately 1400°C. UCAR®Grades SLWX and purified CS and purified isomolded superfine are used as holders or crucibles to hold a quartz crucible liner containing the molten silicon. These same grades are used as a graphite pedestal which support the crucible, rotate it, and lift it as necessary. The graphite is subjected to the same thermal conditions as the crucible and also acts as a path for removal of excess heat of fusion from the puller. A graphite heat shield, surrounds the heater. This shield prevents the loss of heat outward from the crucible area. Lastly, graphite chucks are used in the polysilicon reactors. These hold filaments upon which the polysilicon is grown.

The choice of grade depends upon the size of the hot zone, customer preferences, and the graphite performance required.

Epitaxial Deposition: Another application for graphite is epitaxial deposition, a process utilized throughout this industry for doping silicon or III - V compound substrates. Epitaxy, in silicon semiconductor technology, refers to the oriented overgrowth of one crystalline material on to another. This process is known as "vapor phase epitaxy" and uses silicon carbide coated graphite susceptors to hold the silicon wafers. Controlled doping of "P" or "N" type impurities is also possible in silicon epitaxial technology.

Graphite is used in epitaxial deposition because of its high purity, electrical properties, thermal conductivity, low gas evolution, and coating characteristics. Graphites are generally superfine grain and possess CTE characteristics compatible with the coatings.

Glass-To-Metal Sealing: In glass-to-metal sealing, a "chip" mounted to a metal lead frame must be protected, quite often by glass encapsulation. The device is mounted in a graphite fixture, heated to a desired temperature and the molten glass applied. Graphites are generally used due to their low gas evolution, high thermal conductivity, ability to be machined to very close tolerances, excellent resistance to abrasion and wear, and superior oxidation resistance.

Coatable Graphites: Coating of graphite with silicon carbide extends the applications of graphite into high temperature oxidation regimes and eliminates interaction with potentially reactive materials such as silicon. The UCAR® superfine grain grades are eminently suitable substrates for CVD, CVI, PBN, and SiC coatings.

Electrolytic

Several basic chemicals are manufactured by electrolytically decomposing compounds into their elements. Graphite can be used for either the anode or cathode in electrolytic applications.

Graphite, in performing its function in an electrolytic cell, must meet seven fundamental requirements:

1. High electrical conductivity - high electrical resistance results in expensive power losses. As a conductor of electricity, graphite holds a position between the lesser conducting metals (iron and steel) and the best conducting metals (copper and aluminum).
2. Operate at low voltage. Graphite has a lower anode voltage than any other practical material except precious metals when used for the electrolysis of brine. Graphite possesses a remarkably low chlorine discharge voltage and a relatively high oxygen discharge voltage. Platinum exhibits these properties in some degree and lasts longer than graphite, but it operates at a higher voltage and is much more expensive.
3. High purity - the cell products must be protected from contamination. Graphite can be manufactured to almost any desired purity.
4. Satisfactory mechanical strength - the mechanical strength of graphite anodes has been found satisfactory in all electrolytic applications.
5. Easily machined and readily assembled - the remarkable ease of machining graphite permits its use in any type cell, no matter how intricate the design. Its comparatively light weight makes assembly relatively easy.
6. Low initial cost - graphite is an inexpensive material, particularly when compared with its nearest rival (platinum).
7. High durability - with an expected life of 200 to 800 days in modern cells, depending on the type and the severity of operation, graphite meets this basic practical requirement to a high degree. Purity of the graphite improves life since iron and other metallic contaminants can act as oxidation catalysts.

Electrolytic manufacturing of chemicals is conducted by two methods: solution electrolysis and fused-salt electrolysis. Fluorine, chlorine, chlorate, and manganese dioxide are liberated from water solutions while magnesium and sodium are generated from molten salt solutions.

Aerospace

Rockets and Missiles

The vast field of rockets and missiles takes advantage of many of the properties of graphite. Applications include tactical rockets, strategic rockets and missiles, and large, advance-launch systems. Rapid temperature rise and unusually high operating temperatures are encountered and unusual cone, nozzle and vane shapes are needed. Graphite is one of the few materials that can reasonably meet the demands encountered under these conditions.

Of particular importance in this type of application are the excellent thermal properties of graphite - high thermal shock resistance, high thermal stress resistance, and a strength increase with temperature increase. In addition, its excellent machinability makes it possible to maintain the required close tolerances.

For most short-time applications, firing times are normally less than one minute, graphite is very cost effective in the hot, reactive atmosphere.

With a proven application history involving high-temperature casting and shaping, monolithic graphite should be considered as a viable tooling material for forming composite parts.

While desirable tooling parameters depend on the circumstances of composite part manufacture, the following are generally agreed upon as being most influential: thermal expansion and thermal conductivity.

The thermal expansion of the tool must be compatible with the composite part. This characteristic becomes even more critical when the part manufacturer requires higher temperatures or as part sizes become larger. This compatibility of thermal expansion between tool and part allows the machining of precise dimensions and tolerance directly eliminating costly intermediate steps.

High thermal conductivities are also necessary. The ability to transmit or dissipate heat rapidly is necessary for rapid cycle turnaround times. Monolithic graphite thermal properties match these characteristics.

The structural characteristics of the tooling material are equally important. Integrity of the working tooling surface must be assured by providing a low porosity material void of defects and uniform in physical properties.

Composite parts require exact dimensions and tolerances, especially where assembly to related parts is required. Tooling must be machined to precision tolerances with resultant smooth working surfaces. Multiple axis machining is necessary with integrated CAD/CAM link-ups between the tooling designer, tooling manufacturer, and machiner. Lead times are significantly reduced if this linkage is available.

Monolithic graphite machines comparable to hard wood with ordinary hardened steel tooling. It must be machined dry. Stress relieving is not required.

Techniques are regularly employed to repair graphite. Also, since the material is easily machinable, design changes or contour modifications can be accommodated at a reasonable cost by remachining.

The high-temperature thermal stability and the machining innovation possible with monolithic graphite allow this material to be used for high-temperature composite resin systems for complex shapes and sizes.

Nuclear

Graphite finds widespread use in many areas of nuclear technology based on its excellent moderator and reflector qualities, which are combined almost uniquely with strength and high temperature stability. The function of a moderator is to slow fast neutrons to thermal velocities at which fission in Uranium-235 and Uranium-233 are most efficient. The reflector serves to reflect neutrons, which otherwise would escape, back into the active core region.

The UCAR Carbon Company has participated in the development and production of nuclear grade graphite for fission reactors since the beginnings around the Manhattan project in the 1940s. The low-boron moderator UCAR® grade AGOT was developed for the air-cooled X10 reactors at Oak Ridge, TN, and the first water-cooled reactors at Hanford, WA. Later water-cooled reactors built at Hanford were moderated with purified grades CSF, GBF, and TSF, respectively.

Several carbon dioxide cooled graphite moderated commercial reactors were built in the UK, France, and Japan. UCAR Carbon produced grade UCAR® grade PGA as the life-of-reactor moderator for use in the post-war UK Magnox design. This PGA is notable in that like its UCAR® grade AGOT predecessor, it was not purified, but produced from specific “clean” (low impurity) raw materials. Many of the Magnox reactors built in the 1950’s and 1960’s are now decommissioned.

The later, higher operating temperature, Advanced Gas Reactors (AGR) built in the 1970’s and 1980’s use a

UCAR® grade GCMB as the life-of-reactor moderator. The AGR design incorporates a replaceable graphite sleeve, UCAR® purified grade PPEA.

Two designs of the most advanced gas cooled reactor type, the so-called High Temperature Reactor (HTR), use helium as the coolant and graphite as the moderator. In the one design, demonstrated in small reactors in the UK (Winfrith), and USA (Peach Bottom and Fort St. Vrain), the moderator is the usual prismatic bed characteristic of the Magnox and AGR types, but it is unique in that it is replaceable. There is a demonstration unit currently critical in Fuji, Japan. The alternative Pebble Bed Reactor (PBR) of German design demonstrated in the Jülich and THTR-300 units uses a moving bed of fuel embedded in 50mm diameter spherical graphite “pebbles”.

Glossary

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Glossary

Terminology Relating to Manufactured Carbon and Graphite

Acheson graphite - a synthetic graphite made by the Acheson process.

across grain - synonym for against grain (AG).

activated carbon - a porous carbon material, a char, which has been subject to reaction with gases, sometimes with the addition of chemicals, e.g. $ZnCl_2$, before, during, or after carbonization in order to increase its adsorptive properties.

activated charcoal - a traditional term for activated carbon.

against grain - (AG) the direction in a carbon or graphite body with preferred orientation that has the maximum c-axis alignment (synonym across grain).

agglomerate - a composite particle containing a number of grains.

amorphous carbon - a carbon material without long-range crystalline order. Short-range order exists, but with deviations of the interatomic distances and interbonding angles with respect to the graphite lattice as well as to the diamond lattice.

anisotropic - properties are not equal in all directions.

artificial graphite - synonym for synthetic graphite.

ash - the residue remaining after the complete oxidation of a carbon or graphite.

baking - the process in which the carbonaceous binder, usually coal-tar pitch or petroleum pitch, as part of a shaped carbon mix, is converted to carbon yielding a rigid carbon body by the slow application of heat.

binder - a substance, usually an organic material such as coal tar pitch or petroleum pitch, used to bond the coke or other filler material prior to baking.

binder coke - a constituent of a carbon (or a ceramic) artifact resulting from carbonization of the binder during baking.

calcined coke - petroleum coke or coal-derived pitch coke obtained by heat treatment of green coke to about 1600 K. It will normally have a hydrogen content of less than 0.1% wt.

carbon - an element, number 6 of the periodic table of elements, electronic ground state $1s^2 2s^2 2p^2$.

carbon - an artifact consisting predominantly of the element carbon and possessing limited long range order. The presence of limited long range order is usually associated with low electrical and low thermal conductivity and difficult machinability when compared with graphite.

carbon artifact - an “artificially produced” solid body which consists mainly of carbonaceous material in a distinct shape.

carbon black - an industrially manufactured colloidal carbon material in the form of spheres and of their fused aggregates with sizes below 1,000 nm.

carbon electrode - an electrode for an electrical application. In its green state, it comprises granular carbon material bound with pitch.

carbon material - a solid high in content of the element carbon and structurally in a non-graphitic state.

carbon mix - a mixture of filler coke, e.g.. grains and powders of solid carbon materials, and a carbonaceous binder and selected additives, prepared in heated mixers at temperatures in the range of 410 K - 525 K as a preliminary step for the formation of shaped green bodies.

carbonization - a process by which solid residues with increasing content of the element carbon are formed from organic material usually by pyrolysis in an inert atmosphere.

char - a solid decomposition product of a natural or synthetic organic materials.

charcoal - a traditional term for a char obtained from wood, peat, or coal or some related natural organic materials.

coal-derived pitch coke - the primary industrial solid carbonization product obtained from coal-tar pitch, and is mainly produced in the delayed coking process.

coal-tar pitch - a residue produced by distillation or heat treatment of coal-tar. It is a solid at room temperature, consists of a complex mixture of numerous predominantly aromatic hydrocarbons and heterocyclics, and exhibits a broad softening range instead of a defined melting temperature.

coarse-grained - containing predominantly large grains that are greater than 2mm in size.

coke - a solid high in content of the element carbon and structurally in the non-graphitic state. It is produced by pyrolysis of organic material which has passed, at least in part, through a liquid or liquid-crystalline state during the carbonization process. Coke can contain mineral matter.

compressive strength - a property of solid material that indicates its ability to withstand a uniaxial compressive load.

delayed coke - a commonly used term for a primary carbonization product (green coke or raw coke) from high-boiling hydrocarbon fractions (heavy residues of petroleum or coal processing) produced by the delayed coking process.

delayed coking process - a thermal process which increases the molecular aggregation or association in petroleum-based residues or coal-tar pitches leading to extended mesophase domains.

diamond - an allotropic form of the element carbon with cubic structure (space group O_h^7 - Fd3m) which is thermodynamically stable at pressures above 6 GPa at room temperature and metastable at atmospheric pressure. At low pressures, diamond converts rapidly to graphite at temperatures above 1900 K in an inert atmosphere. The chemical bonding between the carbon atoms is covalent with sp^3 hybridization.

electrographite - a synonym for manufactured graphite.

extruded - formed by being forced through a shaping orifice as a continuous body.

filler - carbonaceous particles comprising the base aggregate in an unbaked green-mix formulation.

flaw - a defect sufficiently greater than those typical of the morphology of a carbon or graphite body to influence a property.

flexural strength - a property of solid material that indicates its ability to withstand a flexural or transverse load.

flow line - a defect induced by discontinuous flow velocities during forming of molded or extruded bodies.

furnace black - a type of carbon that is produced industrially in a furnace by incomplete combustion in an adjustable and controllable process that yields a wide variety of properties within the product.

glassy carbon - a non-graphitizable carbon with homogenous microstructure which does not exhibit any structural components distinguishable by optical microscopy.

grade - the designation given a material by a manufacturer such that it is always reproduced to the same specifications established by the manufacturer.

grain - a region in a carbon or graphite body that is identifiable as being derived from a particle of filler.

graphite - an allotropic crystalline form of the element carbon, occurring as a mineral, commonly consisting of a hexagonal array of carbon atoms (space group P 6_3/mmc) but also known in a rhombohedral form (space group R 3m).

graphite - an artifact consisting predominantly of the element carbon and possessing extensive long range order. The presence of extensive long range order is usually associated with high electrical and high thermal conductivity in the hexagonal plane and relatively easy machinability when compared with carbon.

graphitic carbons - all varieties of substances consisting of the element carbon in the allotropic form of graphite irrespective of the presence of structural defects.

graphitizable carbon - a non-graphitic carbon which upon graphitization heat treatment converts into graphitic carbon.

graphitization - a solid-state transformation of thermodynamically unstable non-graphitic carbon into graphite by means of heat treatment.

green carbon - a formed, but unfired carbon body.

green coke (raw coke) - the primary solid carbonization product from high boiling hydrocarbon fractions obtained at temperatures below 900 K. It contains a fraction of matter that can be released as volatiles during subsequent heat treatment at temperatures up to approximately 1600 K. This mass fraction, the so-called volatile matter, is typically between 4% and
15% by weight.

hardness - the resistance of a material to deformation, particularly permanent deformation, indentation, or scratching.

hexagonal graphite - the thermodynamically stable form of graphite with an ABAB stacking sequence of the graphene layers. The exact crystallographic description of this allotropic form is given by the space group D_6h^4 **-** $P6_3$/mmc (unit cell constants: a=245.6 pm, c = 670.8 pm). Hexagonal graphite is thermodynamically stable below approximately 2600 K and
6 GPa.

impervious graphite - manufactured graphite that has been impregnated with a resinous material to make the final article impervious to liquids in the recommended operating range.

impervious carbon - the same as impervious graphite with the exception that the base stock has not been graphitized.

impregnation - partial filling of the open pore structure with another material.

intercalation - the insertion of an element or compound between graphitic layers.

isostatic - equal pressure from every direction.

isotropic - equal properties in all directions.

lamination - line of demarcation or elongated void generally parallel to the principal grain direction of a carbon or graphite body.

lamp black - a special type of carbon black produced by incomplete combustion of a fuel rich in aromatics that is burned in flat pans. Lamp black is characterized by a relatively broad particle size distribution.

longitudinal, L - direction in a carbon or graphite body coinciding (usually) with the center cylindrical axis, or plane, in the case of a rectangle. (Same as with grain for an extruded body, maybe either with grain or against grain, for both mechanically molded and isostatically molded bodies depending upon the orientation of the body in the press). Transverse (T) is the perpendicular direction to longitudinal.

machinability - a measure of the ease with which a material can be shaped with the aid of cutting or abrasive tools.

manufactured carbon - a bonded granular carbon body whose matrix has been subjected to a temperature typically between 1150 and 2650K.

manufactured graphite - a bonded granular carbon body whose matrix has been subjected to a temperature typically in excess of 2650K and whose matrix is thermally stable below that temperature.

mesophase - see spherical carbonaceous mesophase.

metallurgical coke - produced by carbonization of coals or coal blends at temperatures up to 1400K to produce a macroporous carbon material of high strength and relatively large lump size.

molded - formed in a closed die by the application of external pressure.

natural graphite - a naturally occurring mineral of varying crystalline perfection always intimately mixed with inorganic mineral matter.

needle coke - the commonly used term for a special type of coke with extremely high graphitizability resulting from a strong preferred parallel orientation of its turbostratic layer structure and a particular physical shape of the grains.

non-graphitic carbons - all variations of solids consisting mainly of the element carbon with two-dimensional long-range order of the carbon atoms in planar hexagonal networks, but without any measurable crystallographic order in the third direction (c-direction) apart from more or less parallel stacking.

non-graphitizable carbon - a non-graphitic carbon which cannot be transformed into graphitic carbon solely by high-temperature treatment up to 3300 K under atmospheric pressure or lower pressure.

nuclear graphite - a polygranular graphite material for use in nuclear reactor cores consisting of graphitic carbon of very high chemical purity. High purity is needed to avoid absorption of low-energy neutrons and the production of undesirable radioactive species.

orientation of a crystal - the angular position of a crystal described by the angles which certain crystallographic axes make with the frame of reference.

orientation of a grain - the angular position of a grain described by the angles which a defined set of axes of the grain make with the frame of reference.

orientation of an object - the angular position of an object described by the angles which a defined set of axes or surfaces of the object make with the frame of reference.

oxidation of carbon - the chemical combination of carbon with oxygen or oxygen-containing compounds.

particle sizing - segregation of granular material into specified particle size ranges.

penetration - the depths to which one material extends into or penetrates another.

permeability - a property measured by the rate of passage of a fluid under a pressure gradient through a material.

petroleum coke - a carbonization product of high-boiling hydrocarbon fractions obtained in petroleum processing (heavy residues). It is the general term for all special petroleum coke products such as green, calcined and needle petroleum coke.

petroleum pitch - a residue from heat treatment and distillation of petroleum fractions. It is a solid at room temperature, consists of a complex mixture of numerous predominantly aromatic and alkyl-substituted aromatic hydrocarbons, and exhibits a broad softening range instead of a defined melting point.

pitch - a residue from pyrolysis of organic material or tar distillation which is solid at room temperature, consisting of a complex mixture of numerous, essentially aromatic hydrocarbons and heterocyclic compounds. It exhibits a broad softening range instead of a defined melting point. When cooled from the melt, pitches solidify without crystallization.

polycrystalline graphite - a graphite material with coherent crystallographic domains of limited size regardless of the perfection and preferred orientation (texture) of their crystalline structure.

polygranular carbon - a carbon material composed of grains, which can be clearly distinguished by means of optical microscopy.

polygranular graphite - a graphite material composed of grains, which can be clearly distinguished by means of optical microscopy.

porosity - the percentage of the total volume of a material occupied by both open and closed pores.

preferred orientation - an alignment in the crystal or defect structure of a body leading to variations in physical properties as a function of direction; normally referenced to an orthogonal system where one of the axes is the working direction.

premium coke - an extremely well graphitizing carbon with a high degree of optical anisotropy (isochromatic areas of optical texture above about
100 mm) and is characterized by a combination of the following properties which differ significantly from those of regular coke: high real density, low reversible thermal expansion, and low ash content combined, in most cases, with low sulfur content.

puffing - an irreversible expansion of some carbon artifacts during graphitization heat treatment between 1650 K and 2700 K.

puffing inhibitors - metals or metal compounds with a high chemical affinity for the heteroatoms in the carbons. They are distributed as fine particles within the carbon materials to be graphitized.

pyrolytic carbon - carbon material deposited from gaseous hydrocarbon compounds on suitable underlying substrates (carbon materials, metals, ceramics) at temperatures ranging from 1000 K to 25000 K (chemical vapor deposition).

pyrolytic graphite - graphite material with a high degree of preferred crystallographic orientation of the c-axes perpendicular to the surface of the substrate, obtained by graphitization heat treatment of pyrolytic carbon by chemical vapor deposition at temperatures above 2500 K.

reactivity - rate at which another material will form compounds with carbon or graphite.

regular coke - a petroleum coke with good graphitizability and is characterized by a combination of properties which differ significantly from those of metallurgical coke.

rhombohedral graphite - a thermodynamically unstable allotropic form of graphite with an ABCABC stacking sequence of the layers. The exact crystallographic description of this allotropic form is given by the space group D_3d^5 - $R^{-3}m$, (unit cell constants: a = 256.6 pm, c = 1006.2 pm).

semicoke - a carbonaceous material intermediate between fusible mesophase pitch and a non-deformable green coke produced by the incomplete carbonization at temperatures between the onset of fusion and complete devolatilization. Semicoke still contains volatile matter.

single crystal graphite - a graphite exhibiting the perfect crystallographic hexagonal structure, an infrequently occurring and rather special natural graphite.

soot - a randomly formed particulate carbon material and may be coarse, fine, or colloidal in proportions dependent on its origin. Soot consists of variable quantities of carbonaceous and inorganic solids together with absorbed and occluded tars and resins.

specialty carbon - a carbon manufactured with specific characteristics of low impurity, high density, uniform strength, and certain grain size to meet specific industry needs.

specialty graphite - graphite manufactured with specific characteristics of low impurity, high density, uniform strength, and certain grain size to meet specific industry needs (such as semiconductor fabrication process parts, metal coating dies, crucible and furnace parts, aerospace and nuclear).

spherical carbonaceous mesophase - describes the morphology of carbonaceous mesophase which is formed in the isotropic pitch matrix. The spherical carbonaceous mesophase usually has a lamellar structure consisting of flat aromatic molecules arranged in parallel layers which are perpendicular to the sphere/isotropic phase interface as described by Brooks and Taylor. On coalescence, this spherical mesophase loses its characteristic morphology and is converted to the bulk mesophases.

surface finish - the geometric irregularities in the surface of a solid material.

tensile strength - a property of solid material that indicates its ability to withstand a uniaxial tensile load.

transverse, T - direction perpendicular to longitudinal, L. (Same as against grain or an extruded body, maybe either against grain, or with grain, for both mechanically molded and isostatically molded bodies depending upon the orientation of the body in the press.)

turbostratic - a structure in which the stacked layered planes show no crystallographic stacking registry.

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